

The most important thing we build is trust



C-LAN Linux Administrator Guide

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Executive Summary

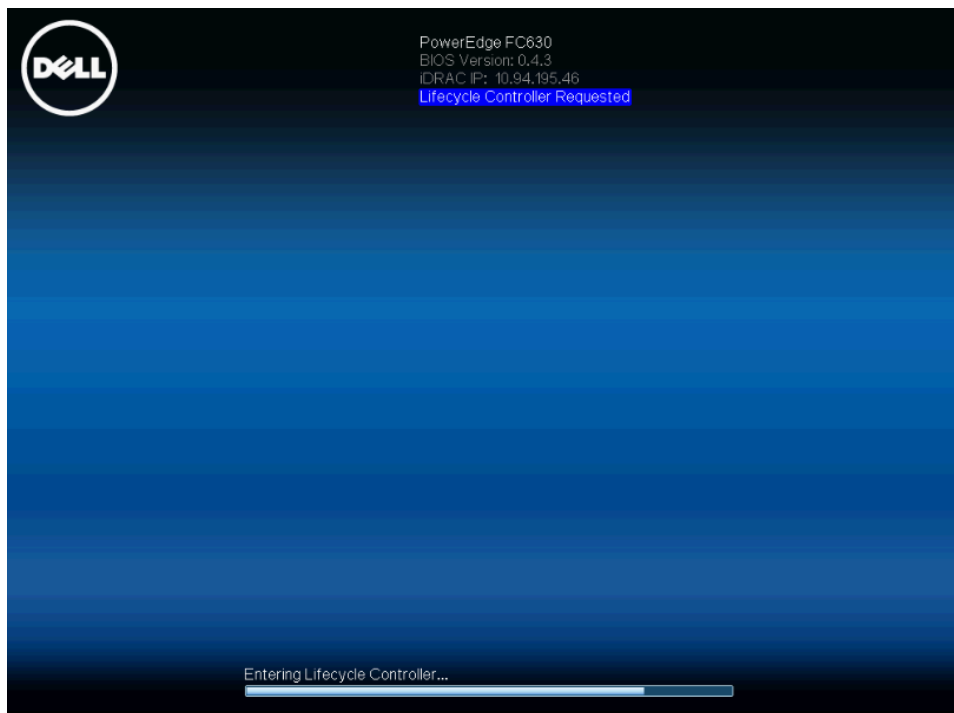
This document is written to explain the Windows Update process within the C-LANs.
Revisions

Revision	Name	Date	Changes Made
<i>Initial documentation</i>			
Ver 1.0	Dana MacAloney	2023.06.13	Initial Document
Ver 2.0	Derek Duchein	2024.03.21	Adding RedHat Installation and applicable chapters
Ver 3.0	Nadeem Sahial	2024.08.07	Changes made to most of the document. Added instructions

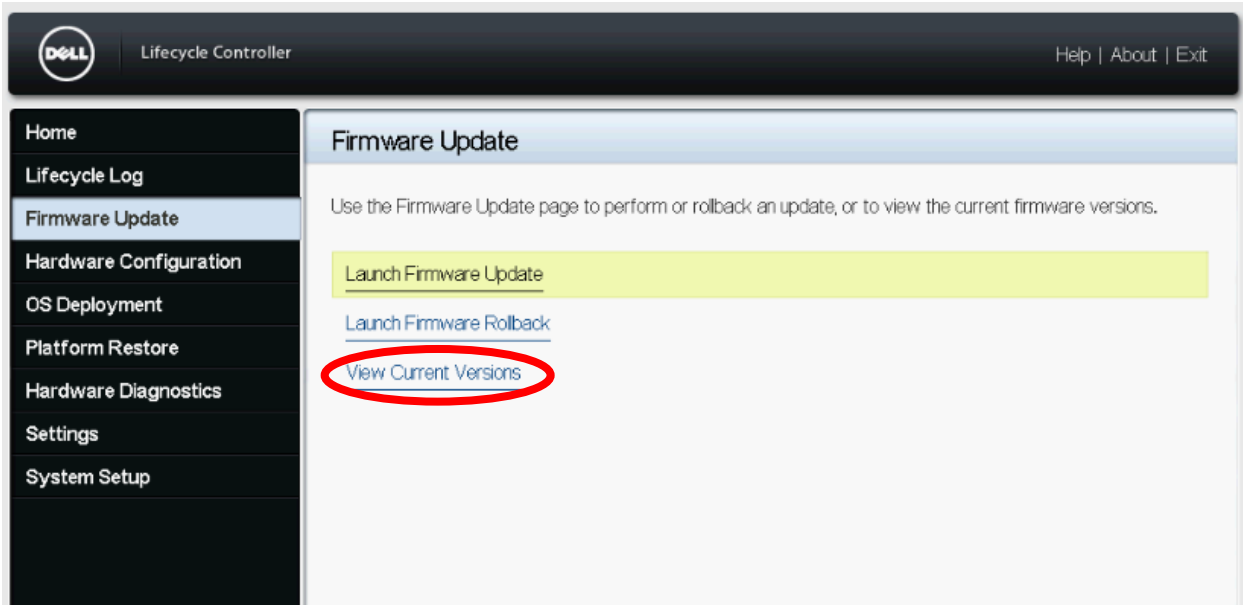
1. FIRMWARE UPDATE

Reboot the PC

Boot into Lifecycle Controller using F10 button



On the Lifecycle Controller page, head to “Firmware Update”, then “View Current Versions”. This will display the current versions and the dates they were updated.



Find the current versions

Asset – 39919

Tag – 81SRTQ2

BIOS – V2.21.2

Intel (R) Gigabit 4P I350-t rNDC – 19.0.12

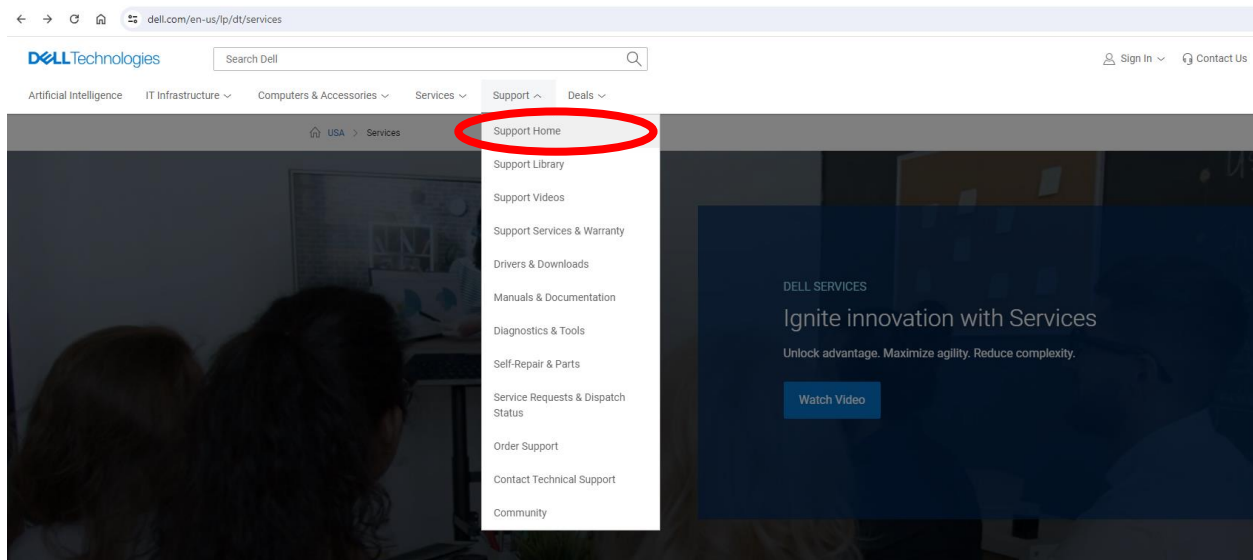
Lifecycle Controller – 7.00.00.172

PERC H330 Adapter – 25.5.9.0001

System CPLD – 1.1.4

To find the latest versions head to dell.com/support and input the service tag number to populate the different versions of updates needed after searching by keywords.

Go to Dell.com and click on “Support Home”



Scroll down and input your “Service Tag” number

Identify your product or search support

Enter a product identifier, model, service request or describe what you are looking for.

[How to find your product identifier](#)

 You could also [detect this PC automatically](#)

Go to “Drivers & Downloads” & go to “Expand”

[Overview](#)[Drivers & Downloads](#)[Diagnostics](#)[Documentation](#)[Service Events](#)[Parts & Repairs](#)

RECOMMENDED

 Let Dell automatically find driver updates for you [Check for Updates](#)

Access the complete list of drivers and downloads for Precision 7920 Rack [Expand](#)

Use keywords check the latest version availabilities

Access the complete list of drivers and downloads for Precision 7920 Rack [Collapse](#)

Keyword
Enter a driver name or keyword

Operating system [Don't see the one you need?](#)
Windows 10, 64-bit

Download Type
All

Category
All

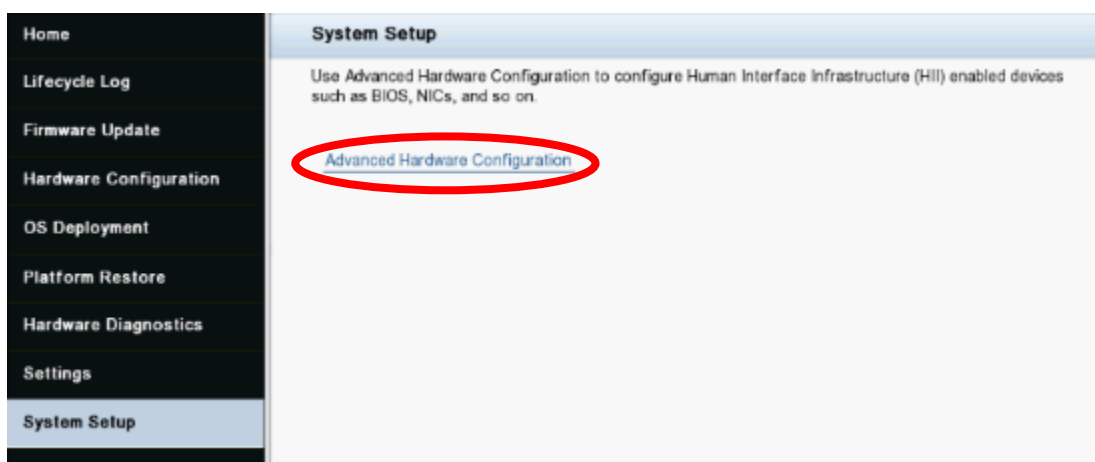
2. SYSTEM SETUP SETTINGS

If you want to check the different settings configured in the System BIOS, follow these step-by-step instructions.

On the Lifecycle Controller, go into “System Setup”



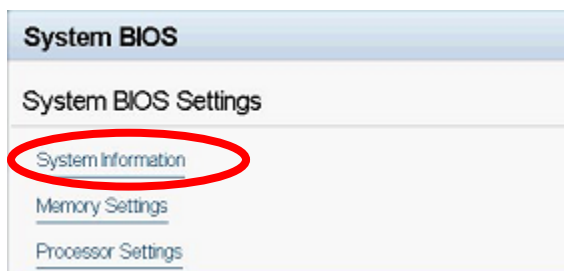
Then go into “Advanced Hardware Configuration”



Click into "System BIOS"

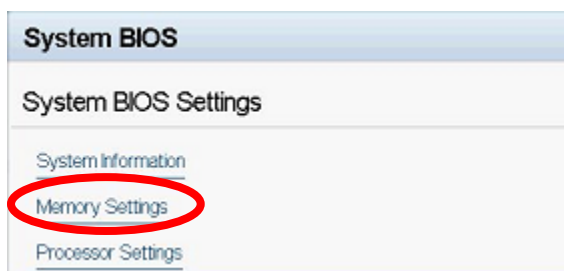


Go into "System Information"



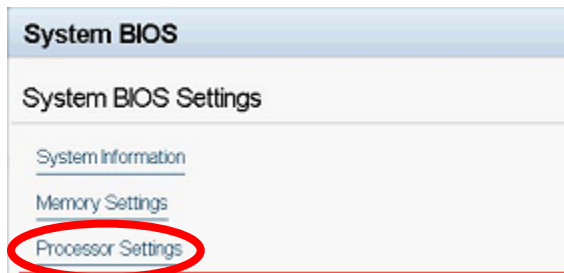
In the System Information make sure it is all up to date. Double check by accessing Dell's website for the latest firmware.

Next go into "Memory Settings"



Leave all of the settings at default and also take a look at the system memory size which is 64GB and the memory type which is ECC DDR4.

Next go into the “Processor Settings”



Leave all of the settings at default and also take a look at the processor “Brand” by scrolling all the way down. The brand is Intel(R) Xeon(R) Silver 4210 CPU @ 2.20GHz.

Next go into the “SATA Settings”

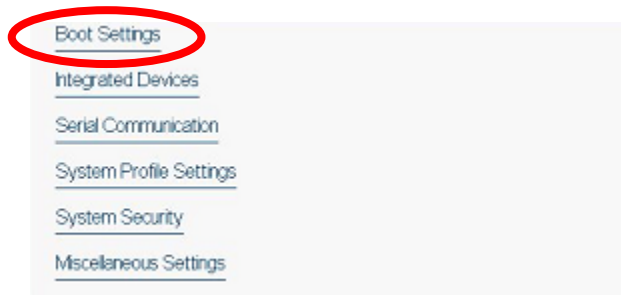


Leave all of the settings at default and also take a look at the “Embedded SATA” – make sure it is on “AHCI Mode”.

Next go into the “NVMe Settings”

Leave all of the settings at default.

Next go into the “Boot Settings”

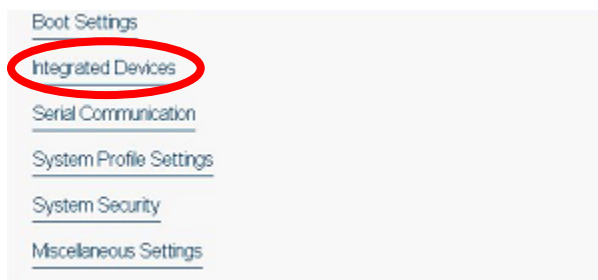


Leave all of the settings at default and also take a look at the “Boot Mode”. Make sure it is set at “UEFI” Mode.

Next go into the “Network Settings”

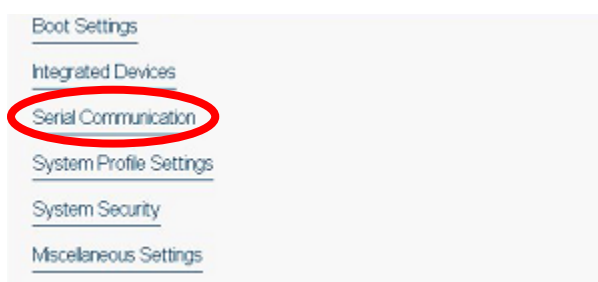
Leave all of the settings at default.

Next go into the “Integrated Devices”



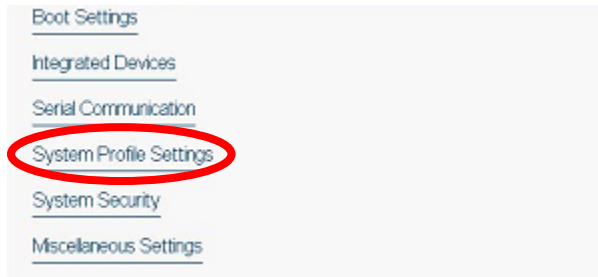
Leave all of the settings at default and also take a look at the “User Accessible USB Ports”. Make sure all the ports are on.

Next go into the “Serial Communication”



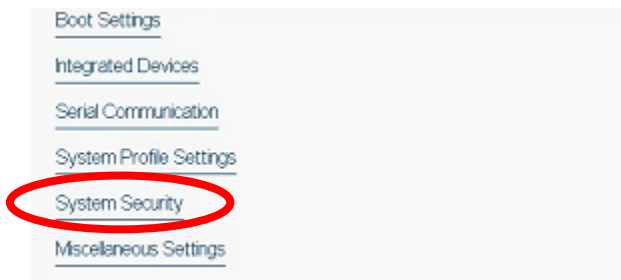
Leave all of the settings at default.

Next go into the “System Profile Settings”



Leave all of the settings at default and also take a look at the “System Profile”. Make sure it is set to “Workstation Performance” for best performance in work settings.

Next go into the “System Security”

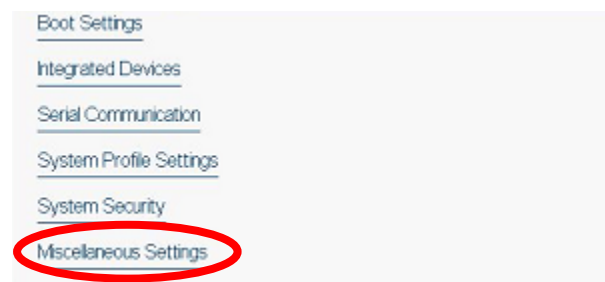


Leave all of the settings at default and also take a look at the “Setup Password”. Check toolroom safe to setup the password.

Next go into the “Redundant OS Control”

Leave all of the settings at default.

Next go into the “Miscellaneous Settings”



Leave all of the settings at default and also take a look at the “Asset Tag” and “System Time”. Input the asset tag and correct time.

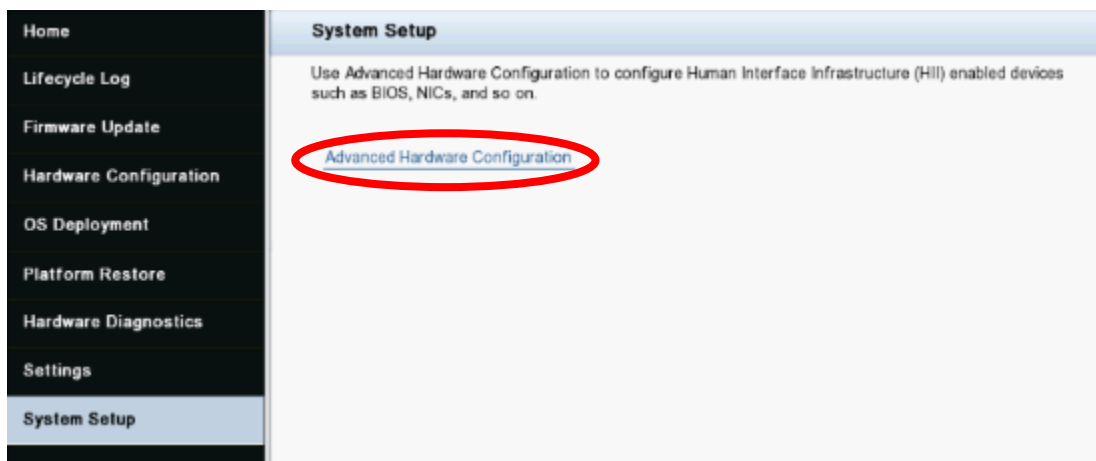
3. DEVICE SETTINGS –

Next, we will go into the device settings to check out our RAID controller settings.

First, on the Lifecycle Controller, go into “System Setup”



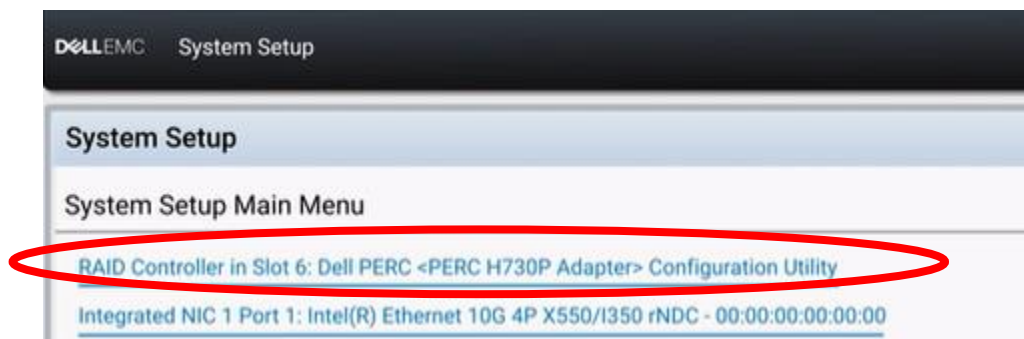
Then go into “Advanced Hardware Configuration”



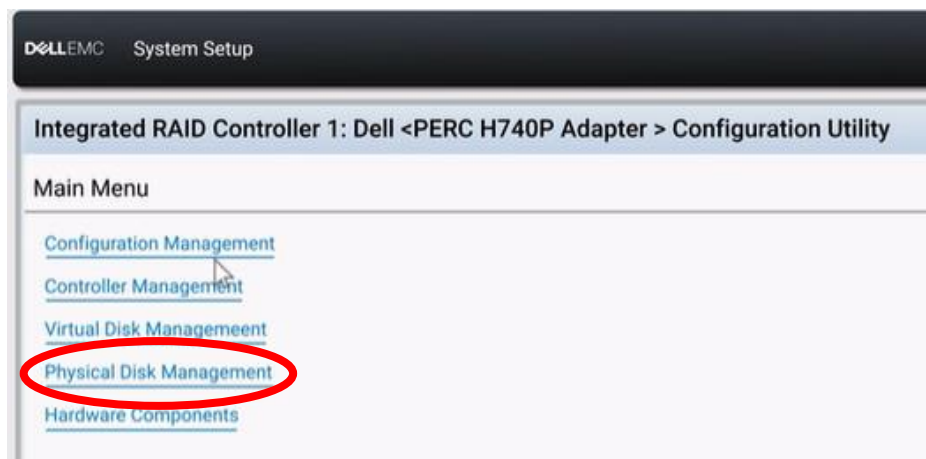
Click into “Device Settings”



Then click into the RAID controller



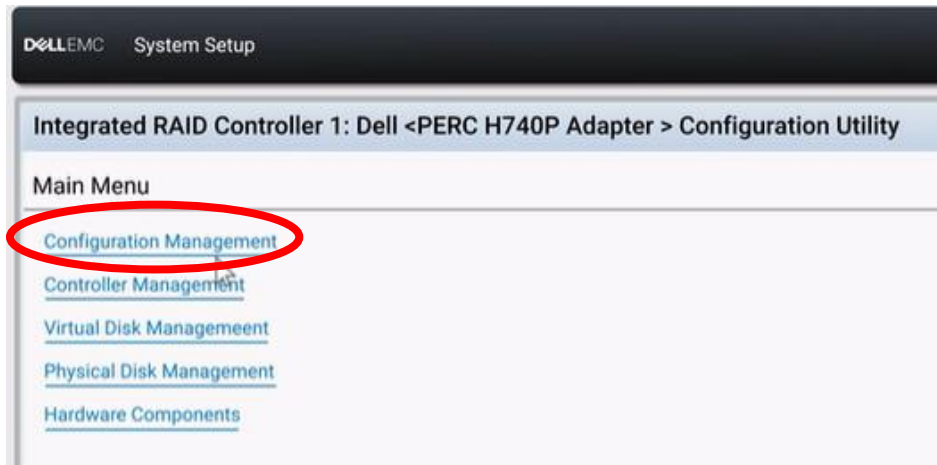
We will start with explaining the “Physical Disk Management” so click into it.



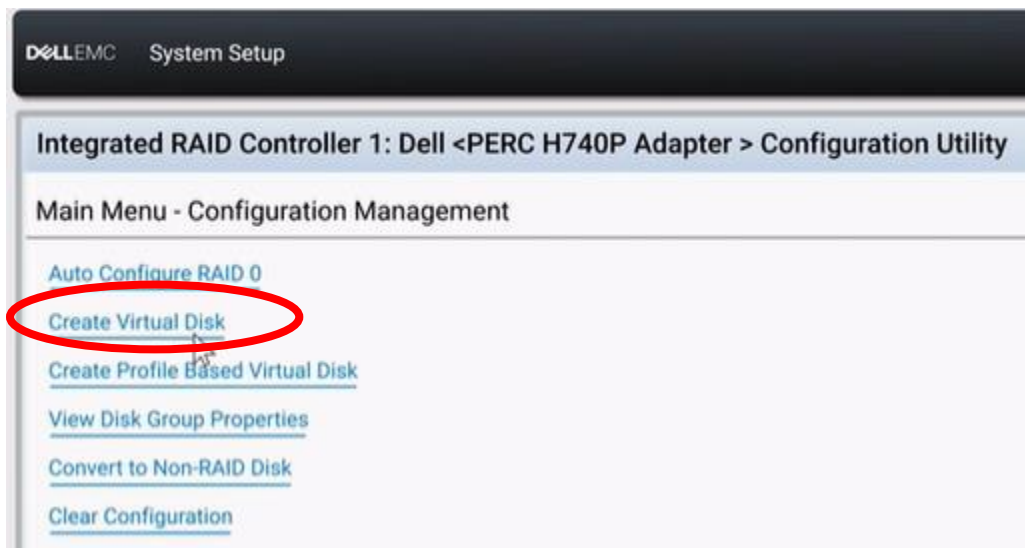
In the menu for “Physical Disk Management” depending on how many disks are installed, they should appear here. If not a great way to troubleshoot is by removing and reinstalling the disks in the slots with a soft reboot. This usually will have the disks appear again.

Next, we will configure virtual disks. In order to do so,

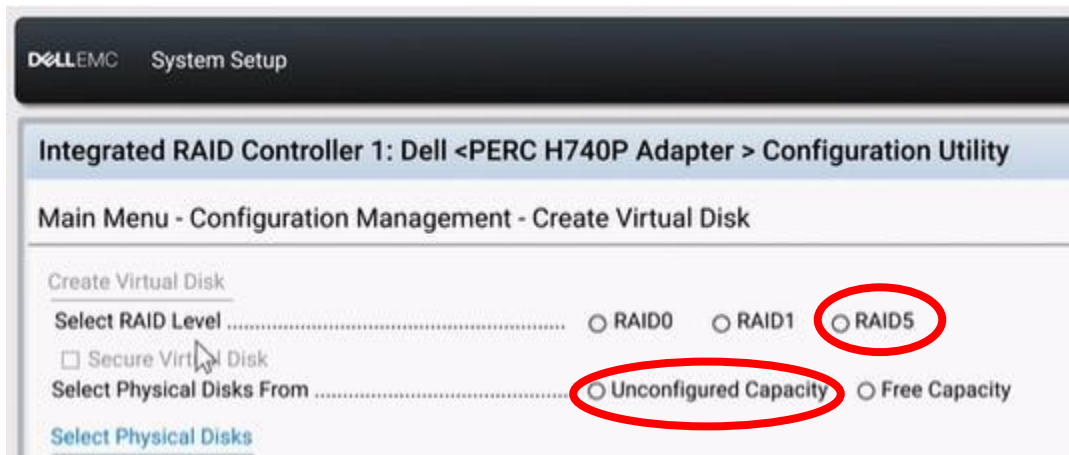
Click into “Configuration Management”



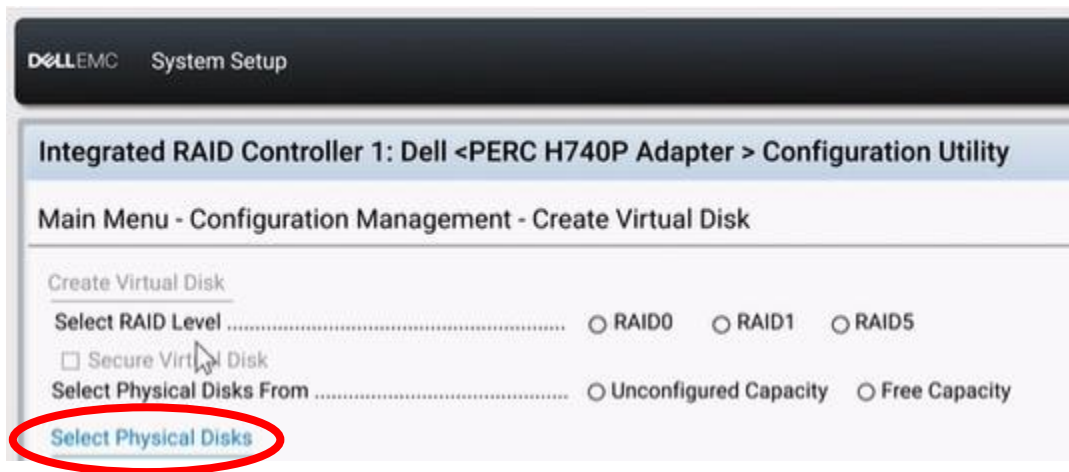
Then click into “Create Virtual Disk”



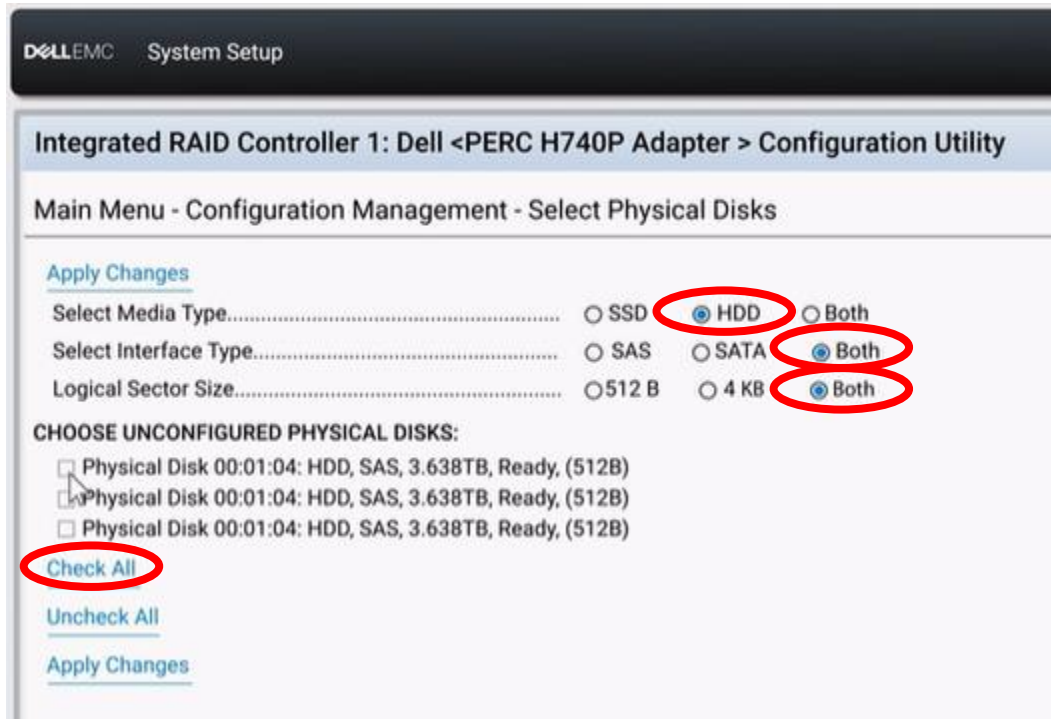
Then choose the options –



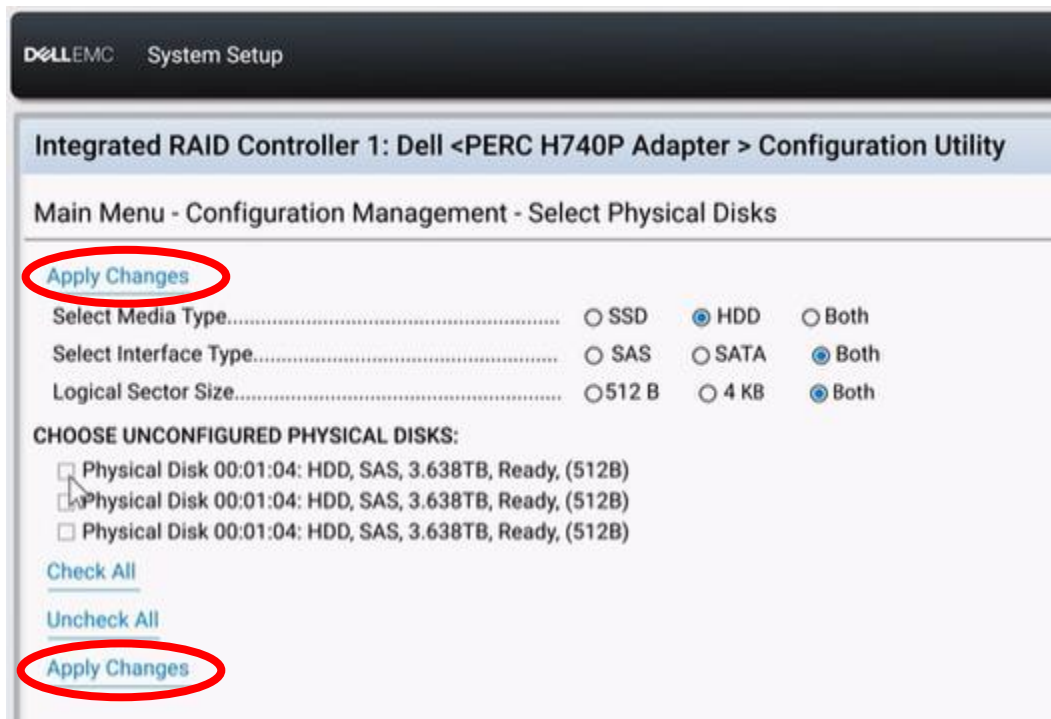
Then click “Select Physical Disks”



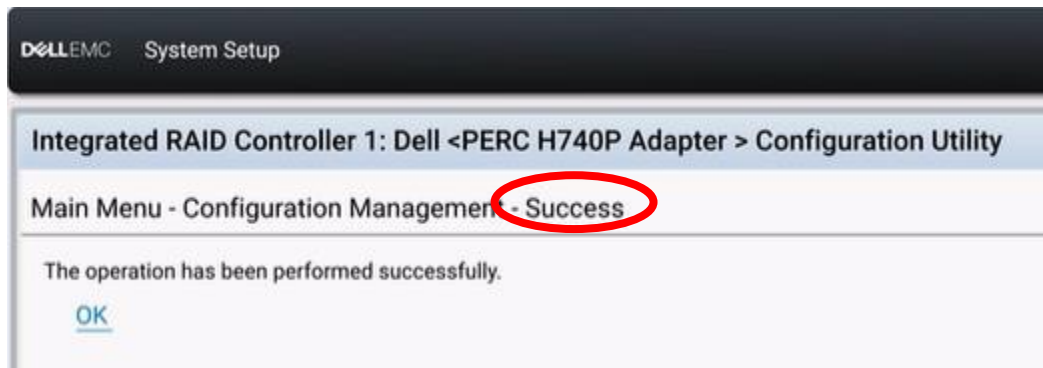
Then click the following options, and “Check All” Physical Disks if that applies to you.



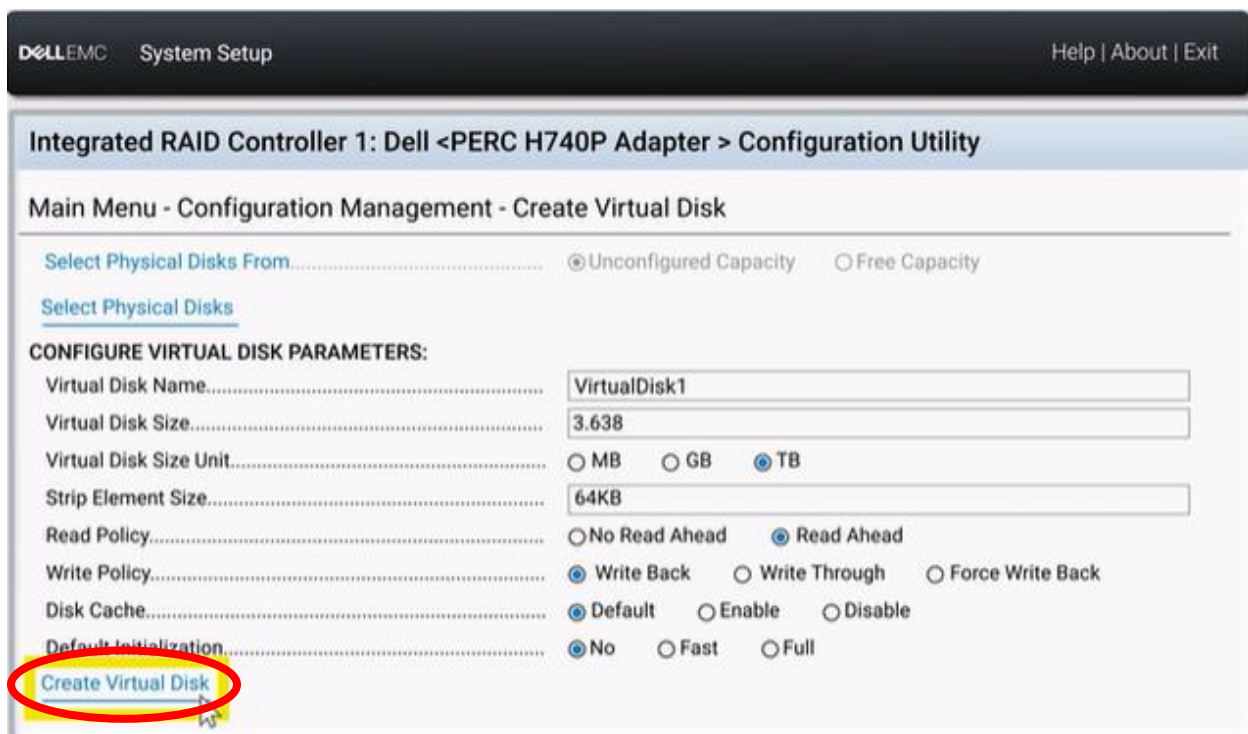
Then click “Apply Changes” on the top left and bottom left



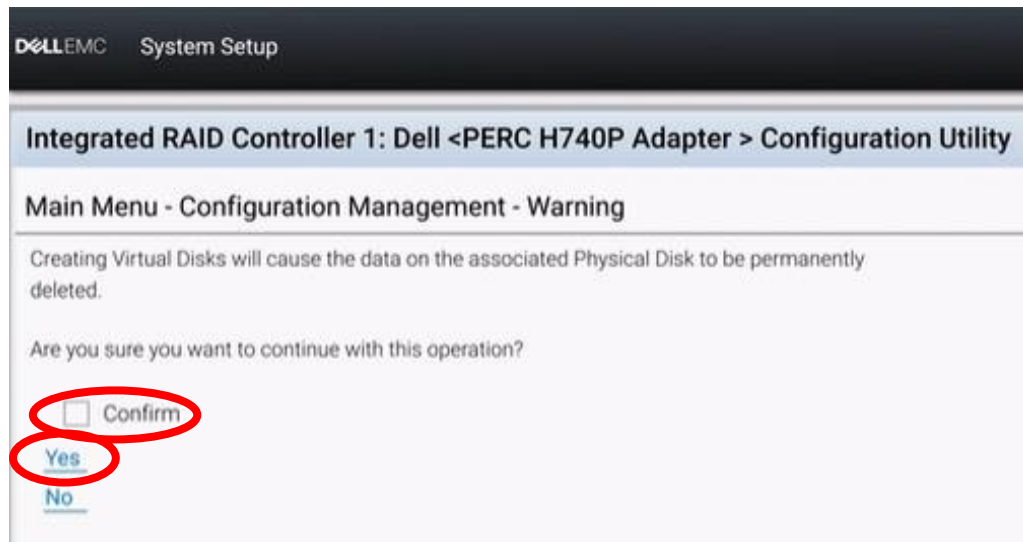
Then you will see a “Success” page which means the process was successful.



Then click on “Create Virtual Disk”



Then click the following options to confirm creating the “Virtual Disk”



This should conclude the process. Then click into “Virtual Disk Management” to confirm the Virtual Disk was created.

Hardware Requirements for SPY-6 Linux Station

What type of System is required?

What kind of processors, which processors have you found don't work because they aren't compatible, I know this was an issue some time ago.

The type of processors required are processors from Intel that are compatible with Linux distributions.

What kind of NIC- 10G, how do we set up the NIC? Are all 10G NICS compatible?

10GbE (10 Gigabit Ethernet) is a high-speed wired network technology that transmits data frames at a rate of 10 billion bits per second. It offers significant speed improvements over traditional 1G networks. What 10GbE Offers:

Speed: 10GbE provides a transfer rate of 10 Gbps. To put this in perspective, it allows you to transfer a 20 GB file in under 20 seconds.

Setting Up 10GbE:

Direct Host-to-Host: The simplest setup involves directly linking two hosts.

Some 10G NICS aren't compatible. Such as the non-Linux supported NICs (any NIC from manufacturers that do not provide Linux drivers. Another example is the NIC's that use proprietary chipsets for which there are no open-source drivers. Next is the NICs that are designed to work virtually may not work with Linux.

Are there any differences for DREX, DC, or FS line? (TBD)

DREX line demands high-performance capabilities to support intensive data center and cloud environments. The DC line requires reliability and minimal latency which is essential for data center servers where consistent and quick data processing is critical. The FS line must be cost-effective and suitable for general purpose networking needs where basic functionality and affordability are prioritized.

Hard Drive Configuration and Requirements

What type/size Hard Drives are required?

A 2.5 inch or a 3.5 inch hard drive size is required. You can install a 2.5 inch hard drive into a 3.5 inch hard drive carrier or a 3.5 inch hard drive that'd fit right in. RAID 500 GB, 1 TB Hard drive

RAID Configuration

How are the RAID configurations set up?

So there are 4 different RAID types available. RAID0, RAID1, RAID5, & RAID10. RAID0 is for data striping across drives for faster read/write performance. RAID1 is for data mirroring (100% redundancy) for data protection (requires 2 drives). RAID5 is for striping (performance) and data protection via parity (requires at least 3 drives). RAID10 is for a "stripe of mirrors" since it combines RAID0 and RAID1.

Operating System Configuration

Where do we get the O/S from?

We can get the O/S from the RedHat website. You will need to have an account to install the O/S so username/password is needed. It is also open source.

[Software & Download Center - Red Hat Customer Portal](https://access.redhat.com/downloads)
(<https://access.redhat.com/downloads>)

Account Number: 7049674

Tech Support Number – (919) -301-3003

As of Jul 1 2024:

Subscriptions

Downloads

Red Hat Console

Get Support

 **Red Hat**
Customer Portal

ProductsKnowledgeSecuritySupport

Search

English

All Red Hat

Overview

Subscriptions

Systems

Cloud Access

Subscription Allocations

Contracts

Errata

Manage

Inventory

Filter by Subscription (Number, Name, SKU) or Contract Number

Active

Ready to Renew

Recently Expired

Future Dated

All

Download CSV

Subscription Number	Subscription Name	Contract Number	Quantity	Start Date	End Date	Status
13720451	Red Hat Developer Subscription for Individuals	Not Available	1	2023-10-22	2024-10-21	Expiring in < 120 days Renewal Options
12583043	Red Hat Satellite Infrastructure Subscription	Not Available	50	2023-03-06	2025-02-11	Active
12583045	Red Hat Beta Access	Not Available	1	2023-03-06	2025-02-11	Active
14102927	Red Hat Enterprise Linux Server Entry Level with Satellite, Self-support	12598889	1	2024-02-12	2025-02-11	Active

Show

100

entries

Showing 1 to 4 of 4 entries

First

Previous

1

Next

Last

https://access.redhat.com/management/subscriptions/product/RH00010

Google CAES Home Page Employee Engagem... San Diego CAES Derek MPulse

Red Hat Enterprise Linux Server Entry Level with Satellite, Self-support

Overview Subscriptions Utilization Systems Content

Entitlement Usage

0 of 1 Entitlements in Use Available 1 0%

0 entitlements used by Systems
0 entitlements used by Subscription Allocations

SKU Details

SKU	RH00010
Support Level ⓘ	Self-Support ↗
Support Type	L1-L3
Subscription Quantity	1 ⓘ Number of active subscriptions
Entitlement Quantity	1 1 subscription provides 1 entitlement
Capacity per Subscription	2 sockets 1 subscription covers 2 sockets
Provided Content ⓘ	dotNET on RHEL (for RHEL Server) dotNET on RHEL Beta (for RHEL Server) Oracle Java (for RHEL Server) Red Hat Ansible Engine Red Hat Beta Red Hat CodeReady Linux Builder for x86_64 Red Hat Enterprise Linux for x86_64 Red Hat Enterprise Linux Server

https://access.redhat.com/management/subscriptions/product/RH00010/content?product_id=491

https://access.redhat.com/management/subscriptions/product/RH00010/content?product_id=69

Google CAES Home Page Employee Engagem... San Diego CAES Derek MPulse

Subscriptions Downloads Red Hat Console Get Support

Red Hat Customer Portal Products Knowledge Security Support

Search English All Red Hat

Overview Subscriptions Systems Cloud Access Subscription Allocations Contracts Errata Manage ▾

Red Hat Enterprise Linux Server Entry Level with Satellite, Self-support

Overview Subscriptions Utilization Systems Content

Content Repositories for Products Provided by this Subscription

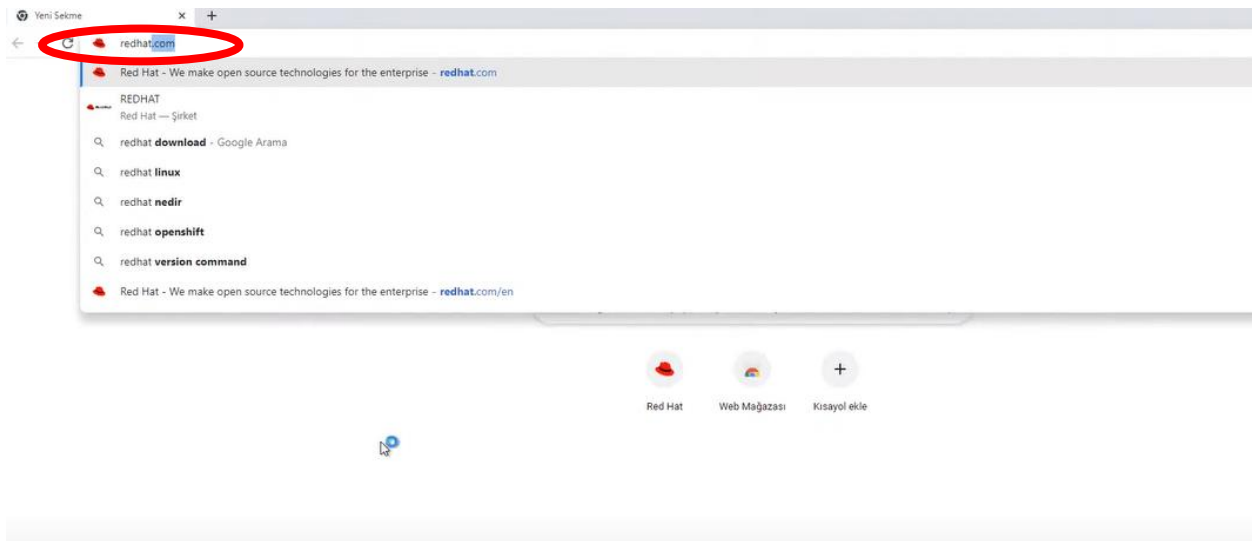
Filter by Repository Name or Label Red Hat Enterprise Linux for x86_64 ▾

Repository Name	Repository Label	Repository Type	Enabled by Default
Red Hat Enterprise Linux 8 for x86_64 - BaseOS (ISOs)	rhel-8-for-x86_64-baseos-isos	file	No
Red Hat Enterprise Linux 8 for x86_64 - BaseOS (Source ISOs)	rhel-8-for-x86_64-baseos-source-isos	file	No
Red Hat Enterprise Linux 8 for x86_64 - Supplementary (ISOs)	rhel-8-for-x86_64-supplementary-isos	file	No
Red Hat Enterprise Linux 9 for x86_64 - BaseOS (ISOs)	rhel-9-for-x86_64-baseos-isos	file	No
Red Hat Enterprise Linux 9 for x86_64 - BaseOS (Source ISOs)	rhel-9-for-x86_64-baseos-source-isos	file	No
Red Hat Enterprise Linux 9 for x86_64 - Supplementary (ISOs)	rhel-9-for-x86_64-supplementary-isos	file	No
Red Hat Enterprise Linux 8 for x86_64 - AppStream (Kickstart)	rhel-8-for-x86_64-appstream-kickstart	kickstart	No
Red Hat Enterprise Linux 8 for x86_64 - BaseOS (Kickstart)	rhel-8-for-x86_64-baseos-kickstart	kickstart	No

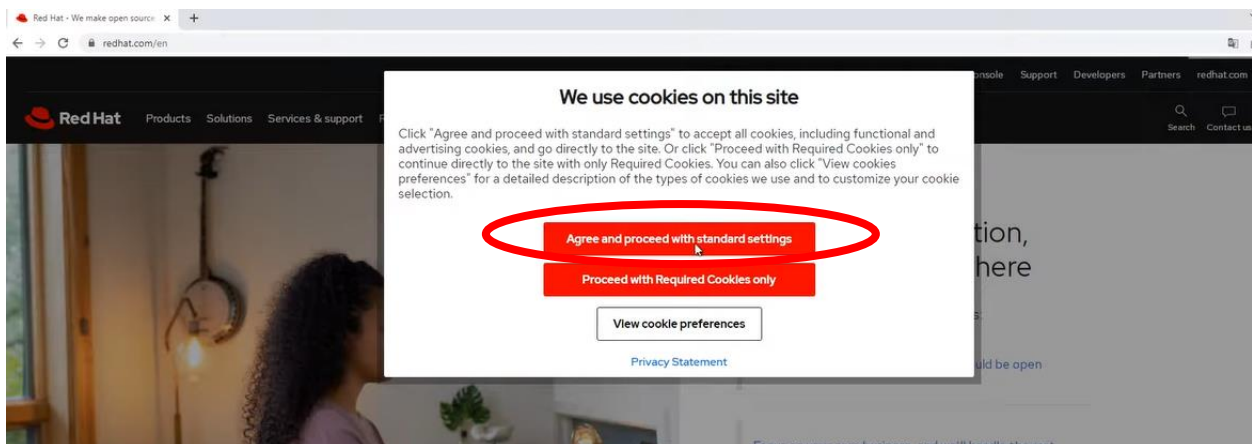
4. Create Red Hat Account and Activate Subscription

Here are the steps –

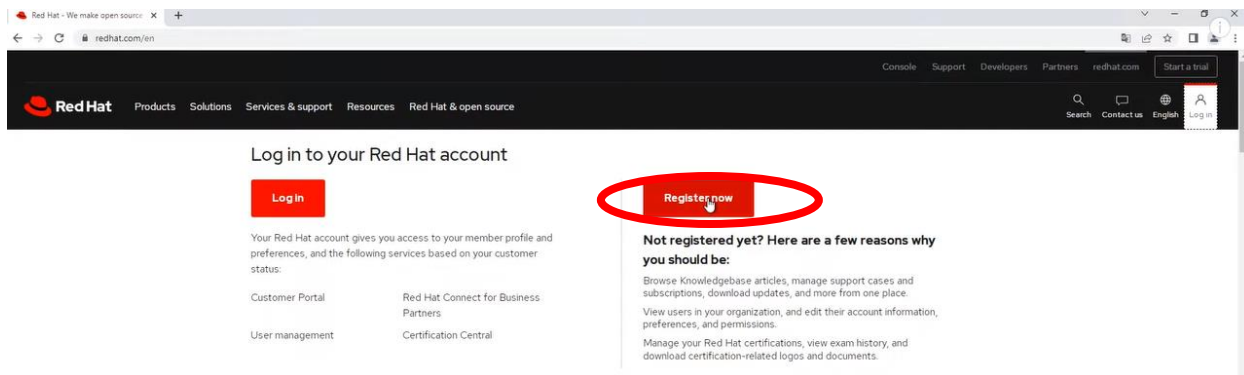
First open the Redhat website



Next, click on this option when entering the website – “Agree and proceed with standard settings”



Then click on “Register now”



Then click into “Corporate” Account type and choose an email and password to login with

contact your organization administrator to join or contact customer service with questions.

* Required fields

Account type

Choose account type *

☒ Corporate

A corporate Red Hat account allows a set of users within your organization to centrally make purchases or administer systems.

☐ Personal

A personal Red Hat account is for purchasing or administering your own personal systems.

Login information

Choose a Red Hat login *

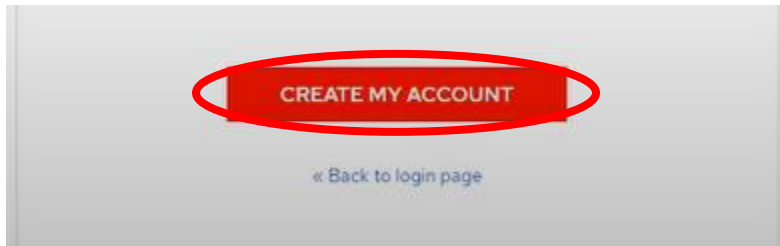
linux I

Your login is a user ID for accessing your account across all Red Hat sites. It must be at least 5 characters and **cannot be changed once created.**

Choose a password *

Strength

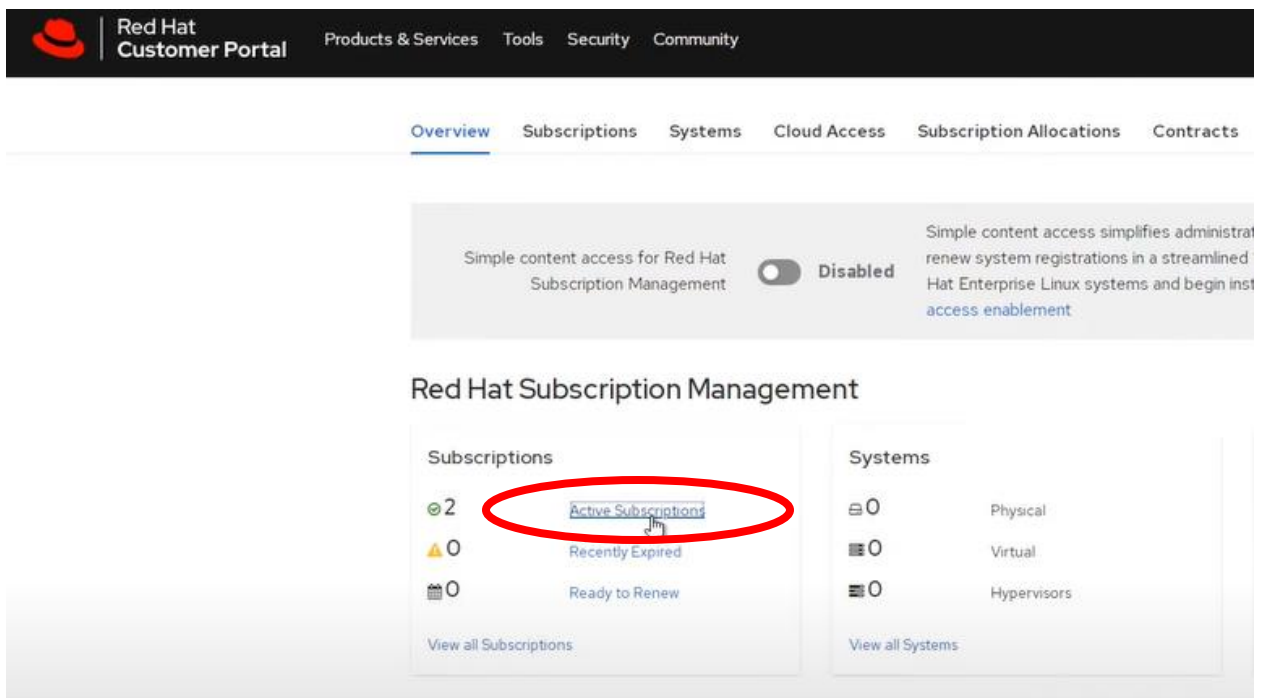
Then when done inputting personal information, click on “Create My Account”



Go ahead and confirm the email



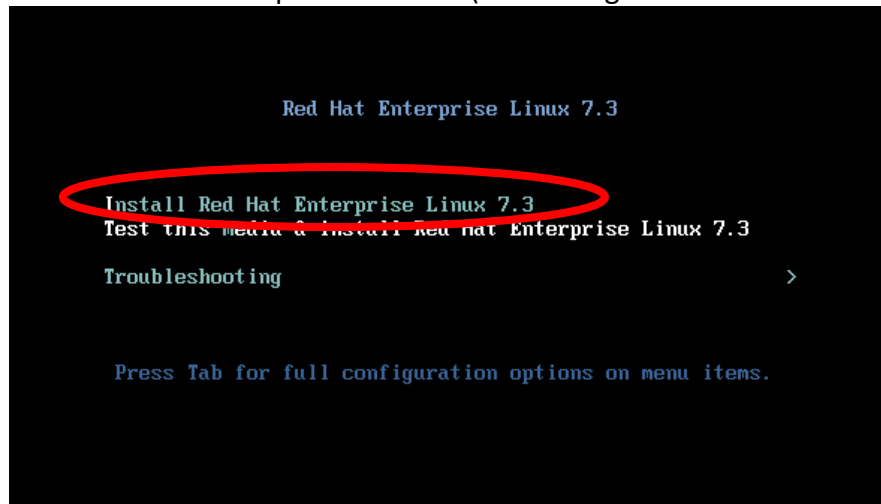
Then go into the Redhat Portal and click on the “Active Subscriptions”



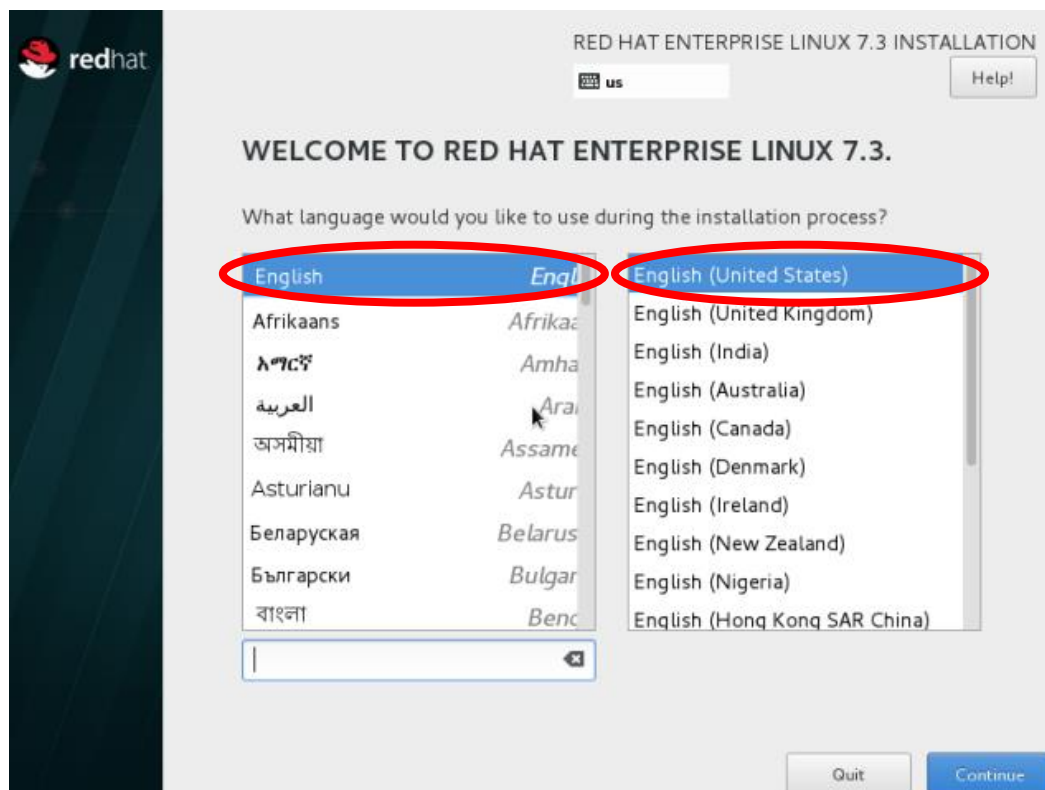
5. Install O/S

Boot the Red Hat Enterprise Server 7.9 installation media, select:

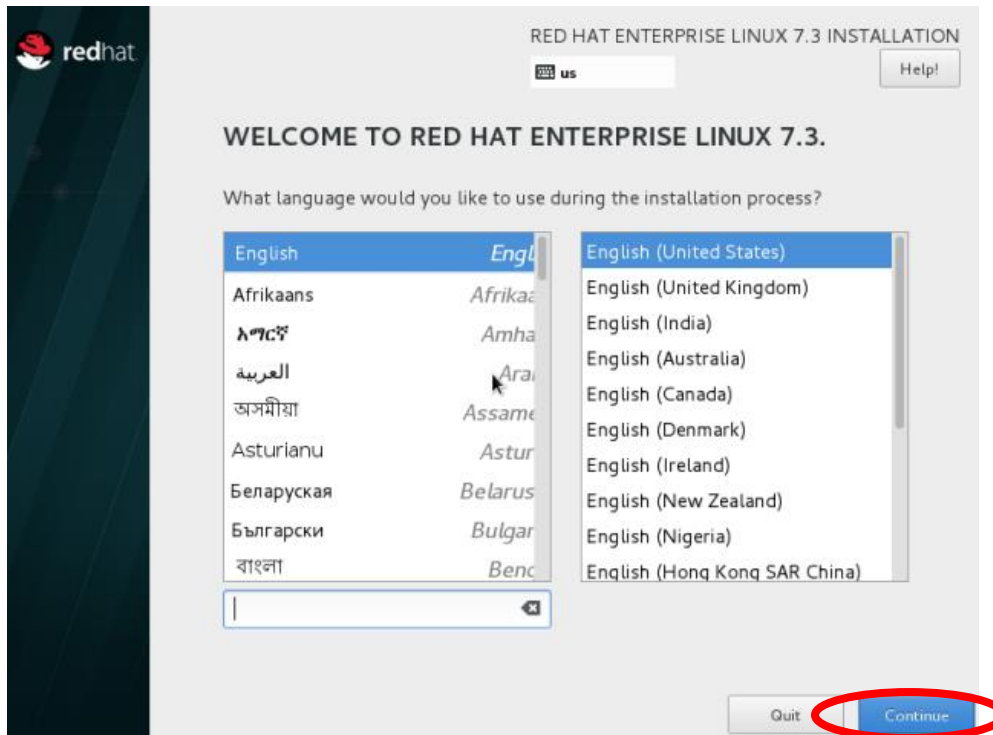
Install Red Hat Enterprise Linux 7.9 (Ex. is using V7.3 but instructions are the same)



Then click on "English" and "English (United States)"

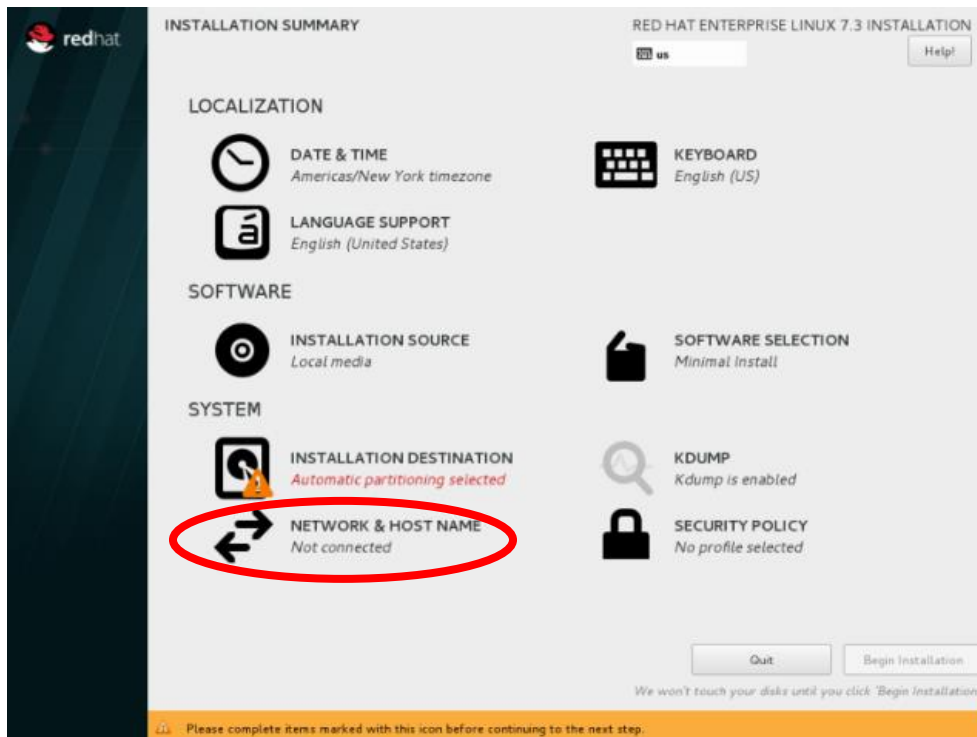


Then click on "Continue"

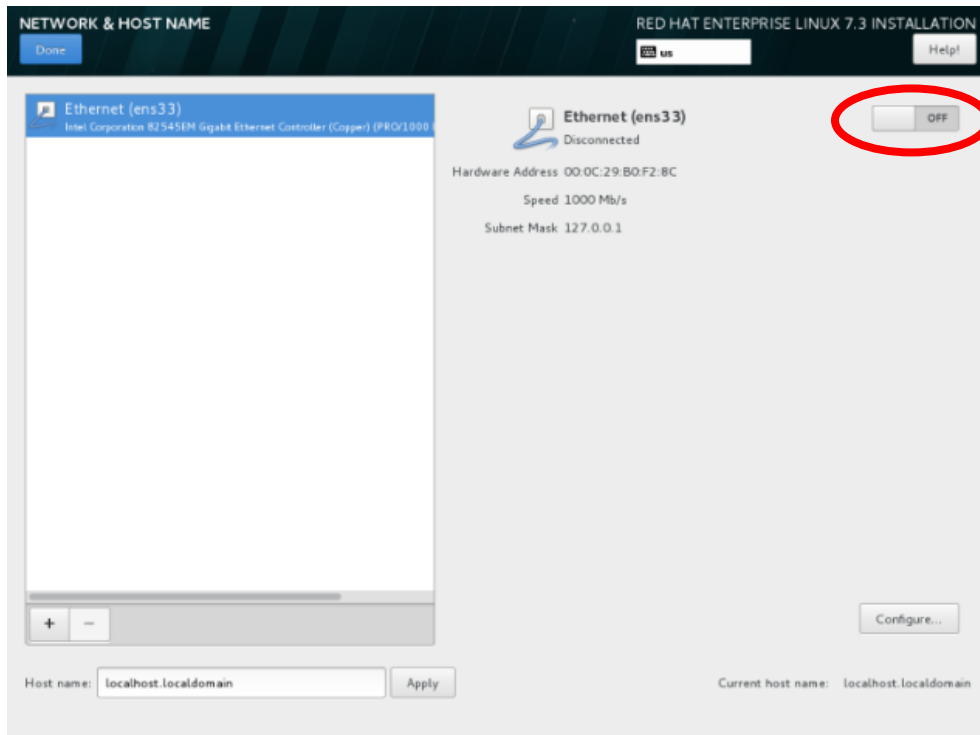


On the Welcome to Red Hat Enterprise Linux 7.9 screen:

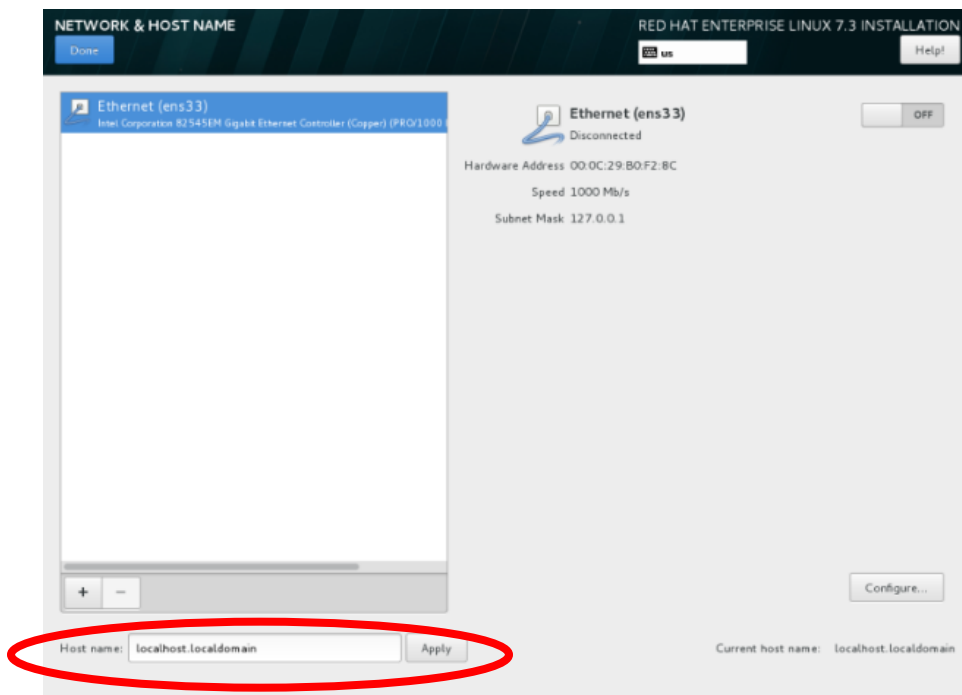
Click on "NETWORK & HOST NAME"



Turn the “primary network connection” ON.



Enter the fully qualified host name in the "Host name" field (include “clan.local” in the host name).



What is the naming scheme we should follow?

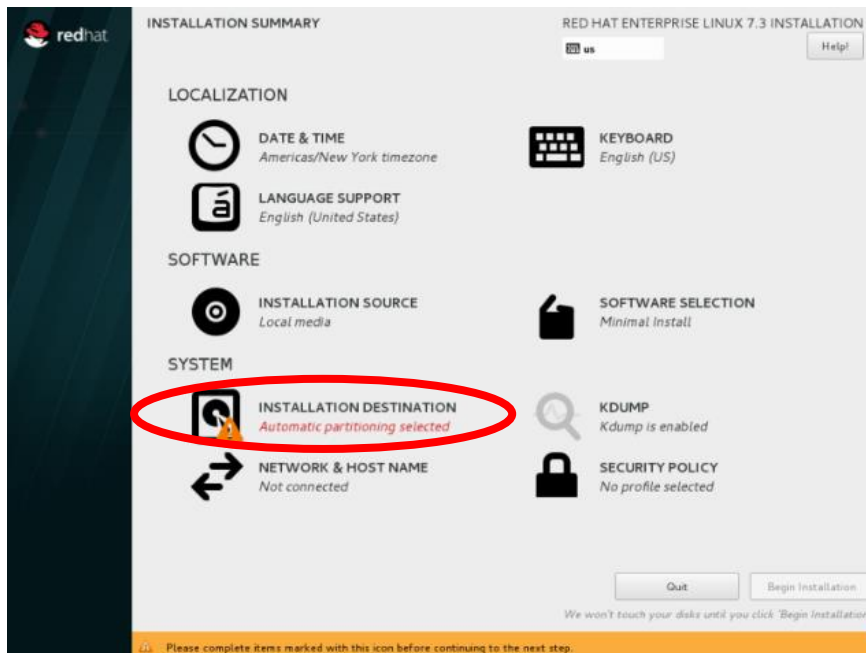
DREX Naming Scheme: CAES Serial number (i.e.: FA105662-16)

DC Naming Scheme: (i.e. LRU-Bench-13)

FS Naming Scheme: (Virtual Server2019 system: (i.e.: SAT01)

Click on "Done" in the top left corner

Click on "INSTALLATION DESTINATION"



Under Other Storage Options select "I will configure partitioning"

The screenshot shows the 'INSTALLATION DESTINATION' window for 'RED HAT ENTERPRISE LINUX 7.3 INSTALLATION'. It includes a 'Done' button in the top left and a 'Help!' button in the top right. The 'Device Selection' section contains a list of 'Local Standard Disks' with one entry: 'VMware, VMware Virtual S' (sda) with 30 GiB free. Below this is a section for 'Specialized & Network Disks' with an 'Add a disk...' button. The 'Other Storage Options' section has two main parts: 'Partitioning' and 'Encryption'. In the 'Partitioning' section, the radio button for 'I will configure partitioning.' is selected and circled in red. The 'Encryption' section has an unchecked checkbox for 'Encrypt my data. You'll set a passphrase next.'

Click on "Done" in the top left corner

This screenshot is identical to the previous one, showing the 'INSTALLATION DESTINATION' window. In this version, the 'Done' button in the top left corner is circled in red, indicating the next step in the process.

(Unable to pull instructions for this section from online)

Set the "New mount points will use the following partitioning scheme" to "Standard Partition"

For each partition listed below, click on the "+"

Fill in the "Mount Point", and "Desired Capacity" and click "Add Mount Point"

Change "File System" to the correct type

Enter the label into the "Label" field

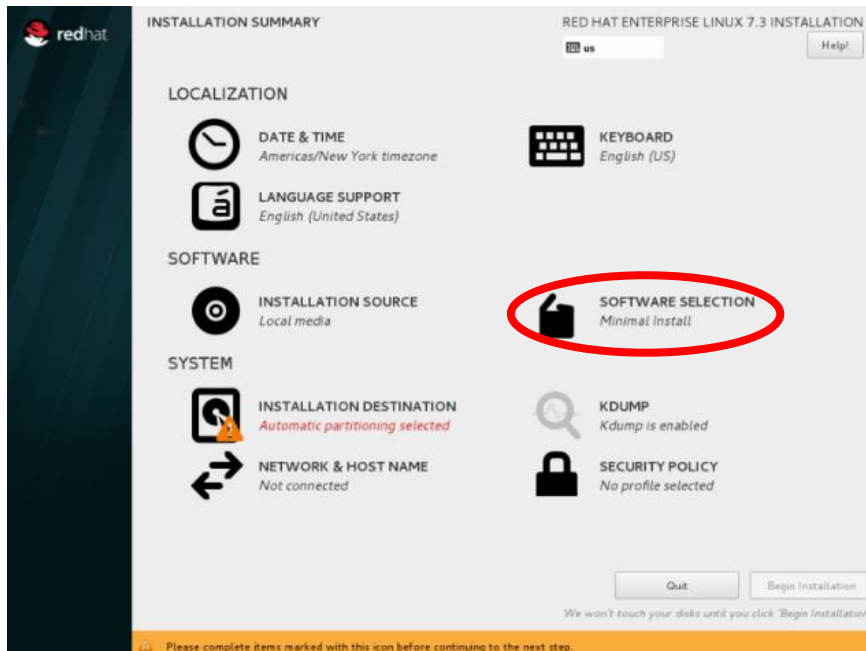
After all partitions have been entered click "Done" in the top left corner

Verify the actions to be taken on the SUMMARY OF CHANGES window and click "Accept Changes" in the bottom right corner

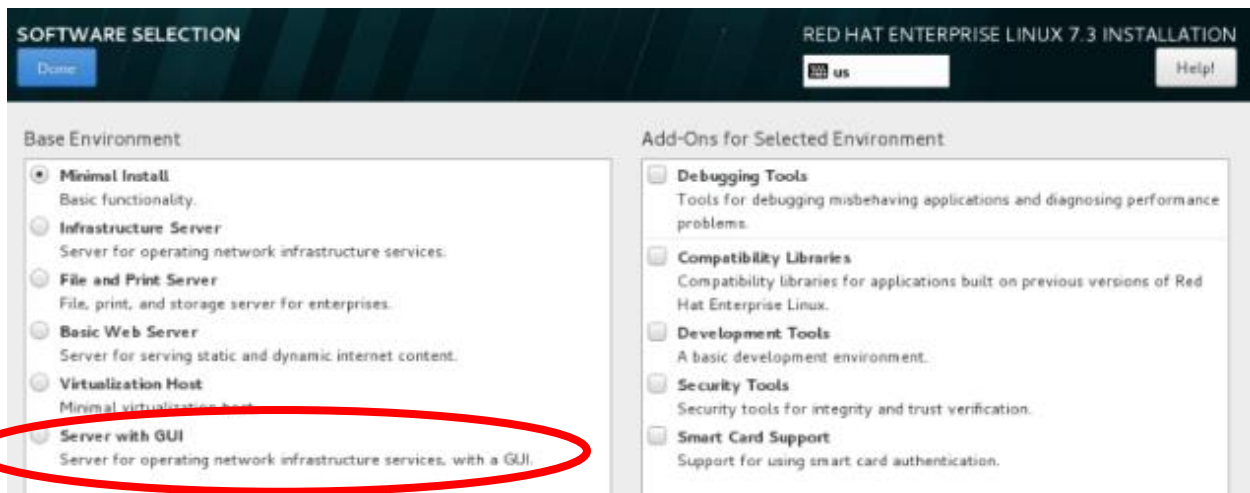
Partitions:

Mount Point: (case sensitive)	Desired Capacity:	File System:	Label: (Case Sensitive)
/boot	1G	xfs	BOOT
/boot/efi	512M	EFI System Partition	BOOTEFI
swap	4G	swap	SWAP
/var	50G	xfs	VAR
/var/tmp	25G	xfs	VARTMP
/var/log	25G	xfs	VARLOG
/var/log/audit	25G	xfs	VARLOGAUDIT
/tmp	25G	xfs	TMP
/	50G	xfs	ROOT
/home	*	xfs	HOME

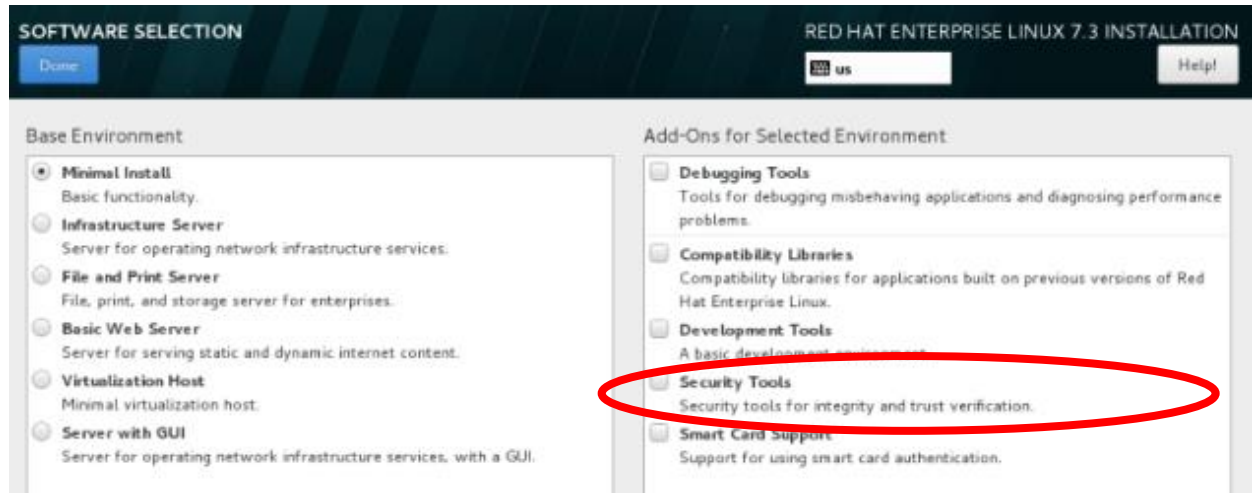
Click on "SOFTWARE SELECTION"



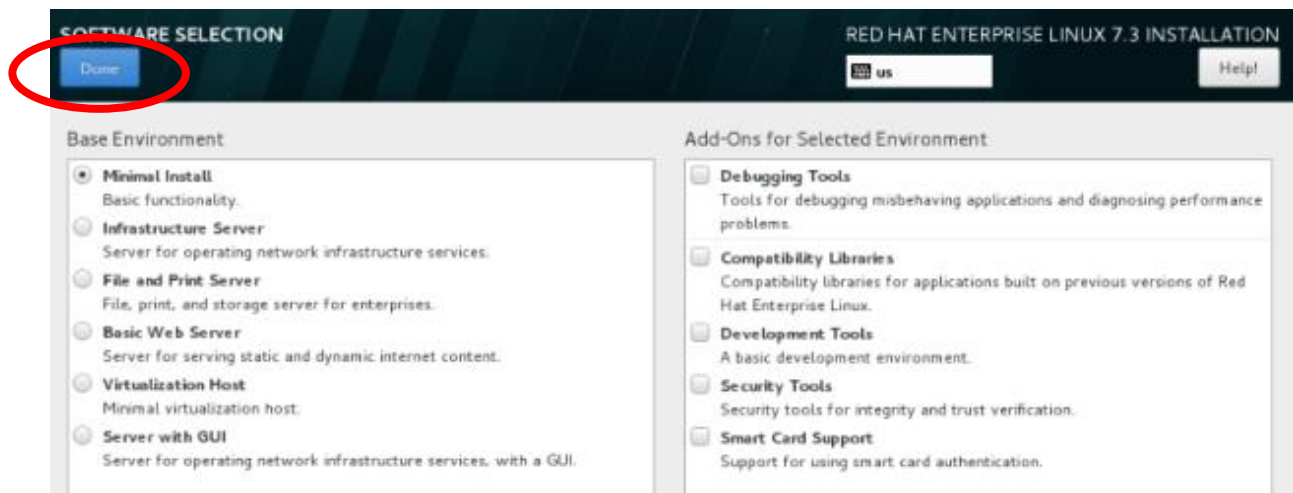
Select "Server with GUI" in the "Base Environment" panel



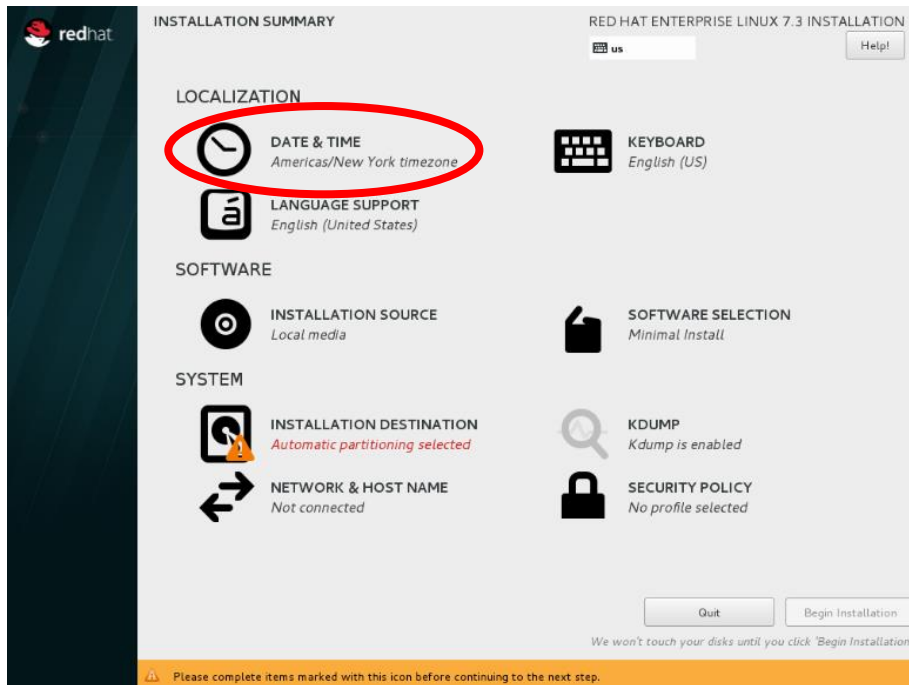
Select ""Security Tools" and "System Administration Tools" in the "Add-Ons for Selected Environment" pane



Click on "Done" in the top left corner of the screen



Click on "DATE & TIME"



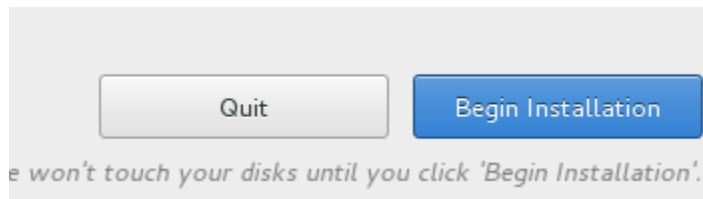
Select the correct time zone (network time)



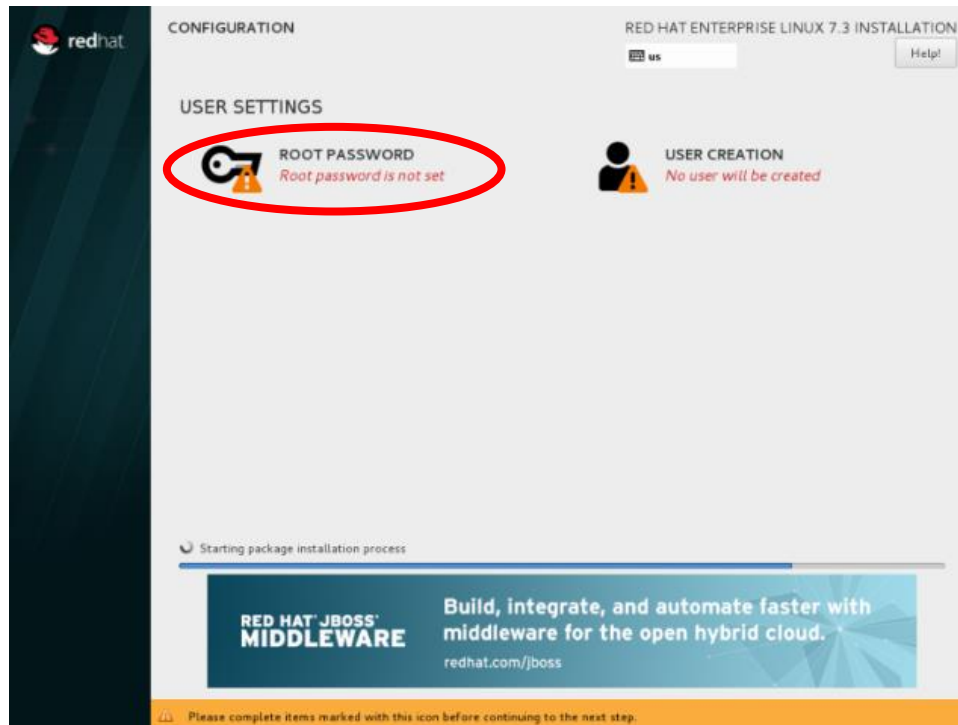
Click "Done" in the top left corner of the screen



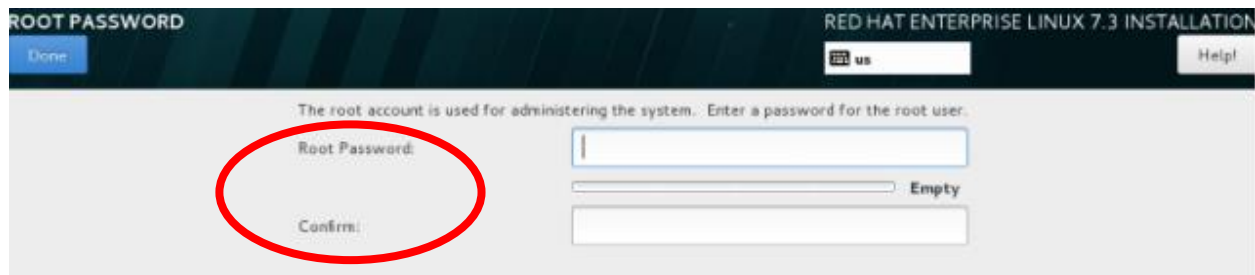
Click "Begin Installation" in the bottom right of the screen



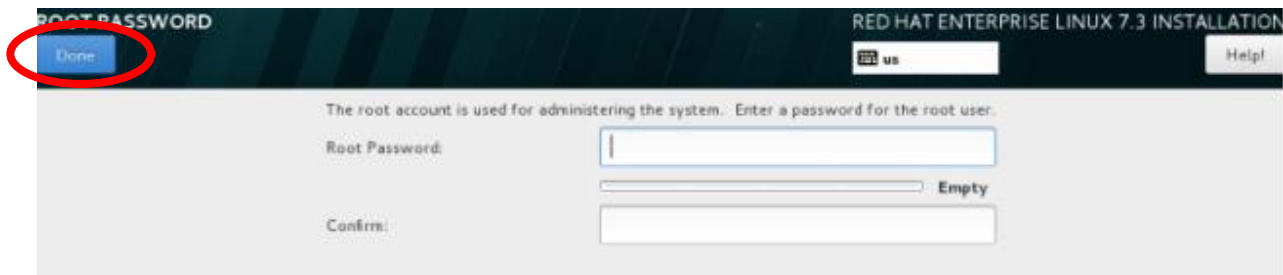
Click on "ROOT PASSWORD"



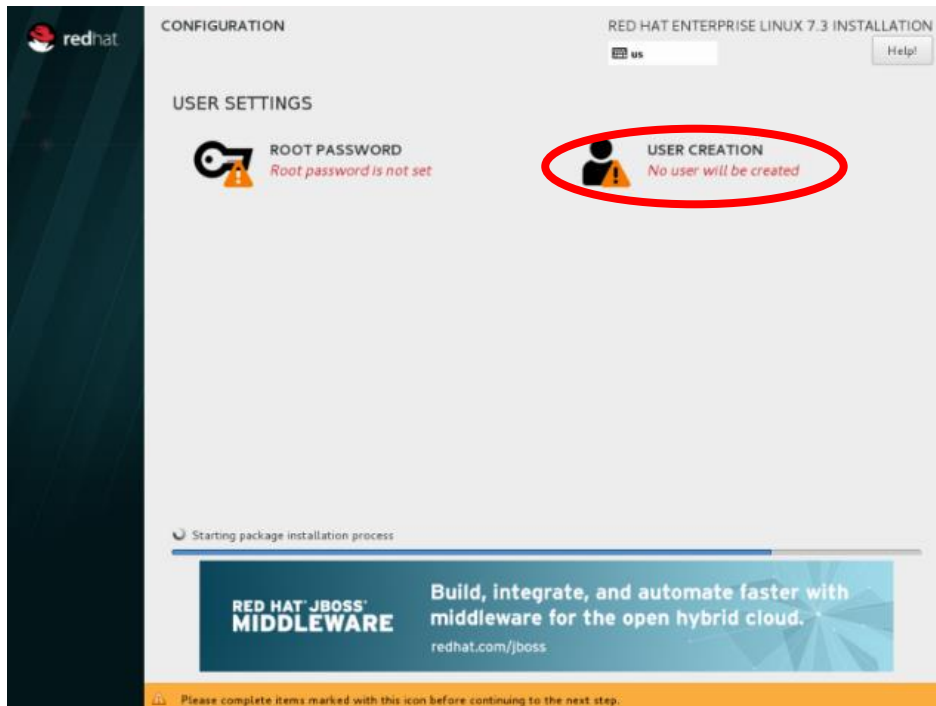
Enter the root password in both fields



Click on "Done" in the top left corner of the screen



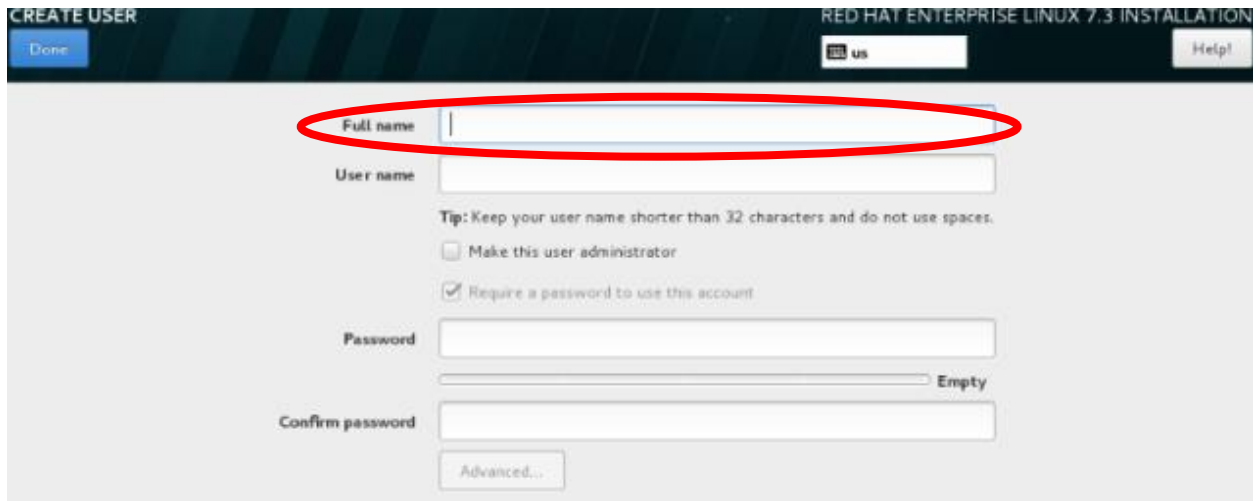
Click on "USER CREATION"



Check "Make this user administrator"

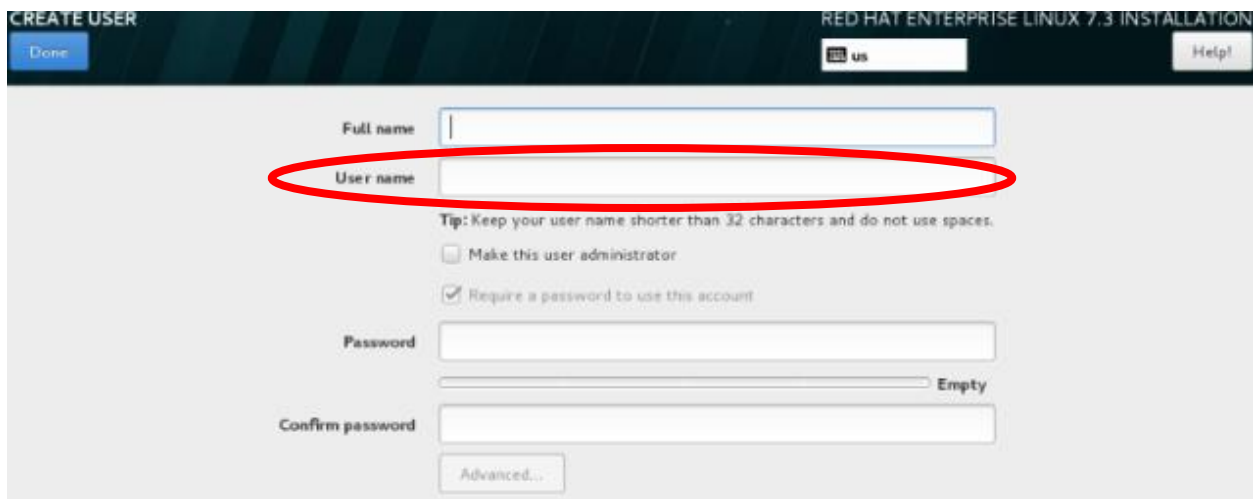
The image shows the 'CREATE USER' window. It has a 'Done' button in the top left. The top right shows 'RED HAT ENTERPRISE LINUX 7.3 INSTALLATION' with a language dropdown set to 'us' and a 'Help!' button. The form contains the following fields and options: 'Full name' (text input), 'User name' (text input), a tip 'Tip: Keep your user name shorter than 32 characters and do not use spaces.', a checked checkbox 'Make this user administrator' (highlighted with a red oval), a checked checkbox 'Require a password to use this account', 'Password' (text input), a strength indicator bar showing 'Empty', 'Confirm password' (text input), and an 'Advanced...' button at the bottom.

Enter the user's full name



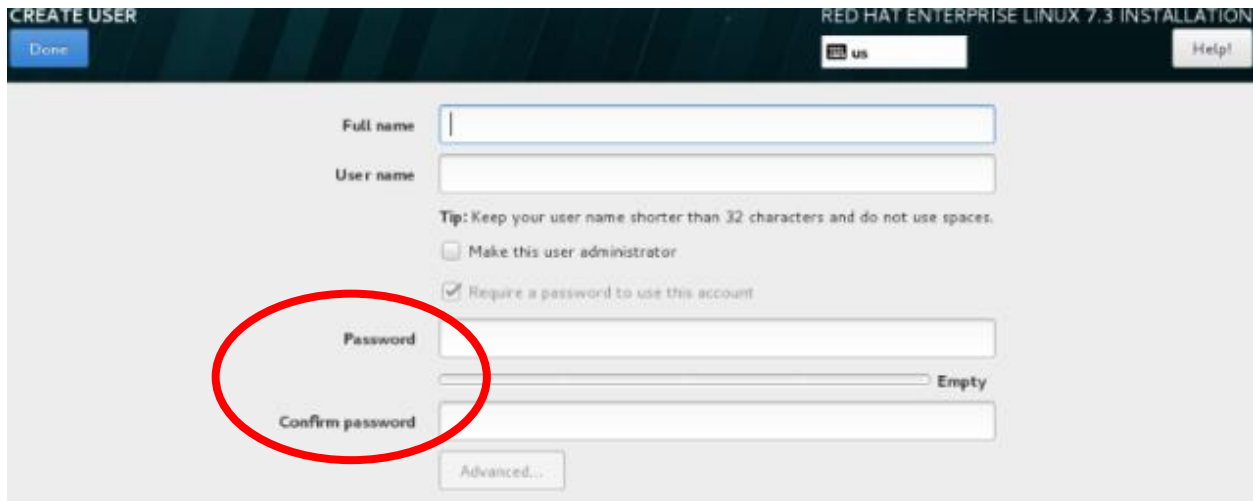
The screenshot shows the 'CREATE USER' window from the Red Hat Enterprise Linux 7.3 installation. The window has a dark header with 'CREATE USER' on the left and 'RED HAT ENTERPRISE LINUX 7.3 INSTALLATION' on the right. Below the header, there is a 'Done' button on the left and a 'Help!' button on the right. The main content area contains several input fields: 'Full name' (highlighted with a red oval), 'User name', 'Password', and 'Confirm password'. There are also checkboxes for 'Make this user administrator' and 'Require a password to use this account'. A 'Tip' message is displayed above the checkboxes. An 'Advanced...' button is located at the bottom right of the form.

Enter the user's User Name



This screenshot is identical to the one above, showing the 'CREATE USER' window. In this instance, the 'User name' field is highlighted with a red oval, indicating the next step in the user creation process.

Set the user's password



CREATE USER RED HAT ENTERPRISE LINUX 7.3 INSTALLATION

Done us Help!

Full name

User name

Tip: Keep your user name shorter than 32 characters and do not use spaces.

☐ Make this user administrator

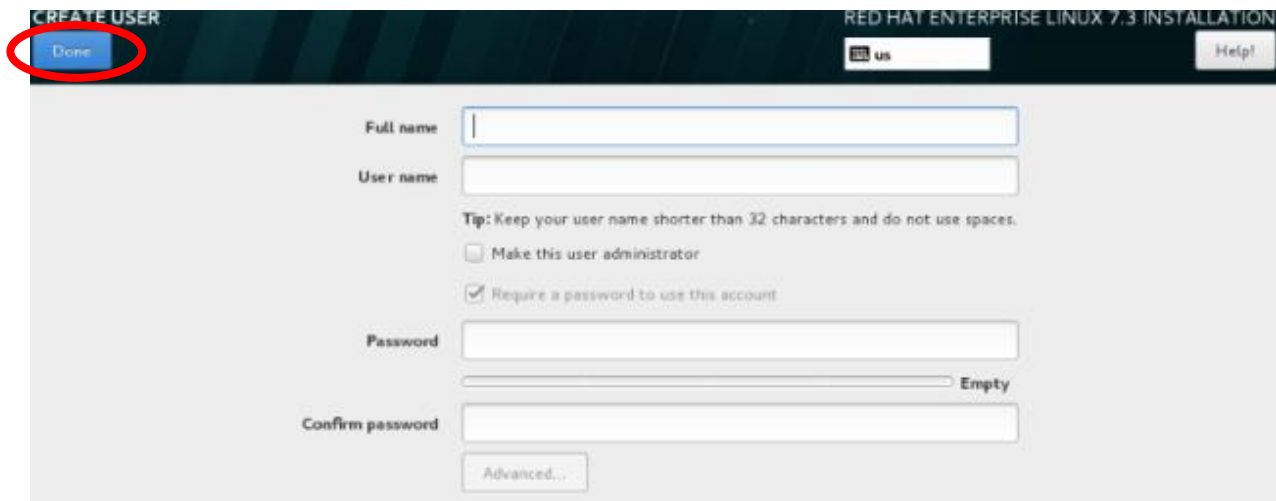
☒ Require a password to use this account

Password

Confirm password

Advanced...

Click on "Done" in the top left corner of the screen



CREATE USER RED HAT ENTERPRISE LINUX 7.3 INSTALLATION

Done us Help!

Full name

User name

Tip: Keep your user name shorter than 32 characters and do not use spaces.

☐ Make this user administrator

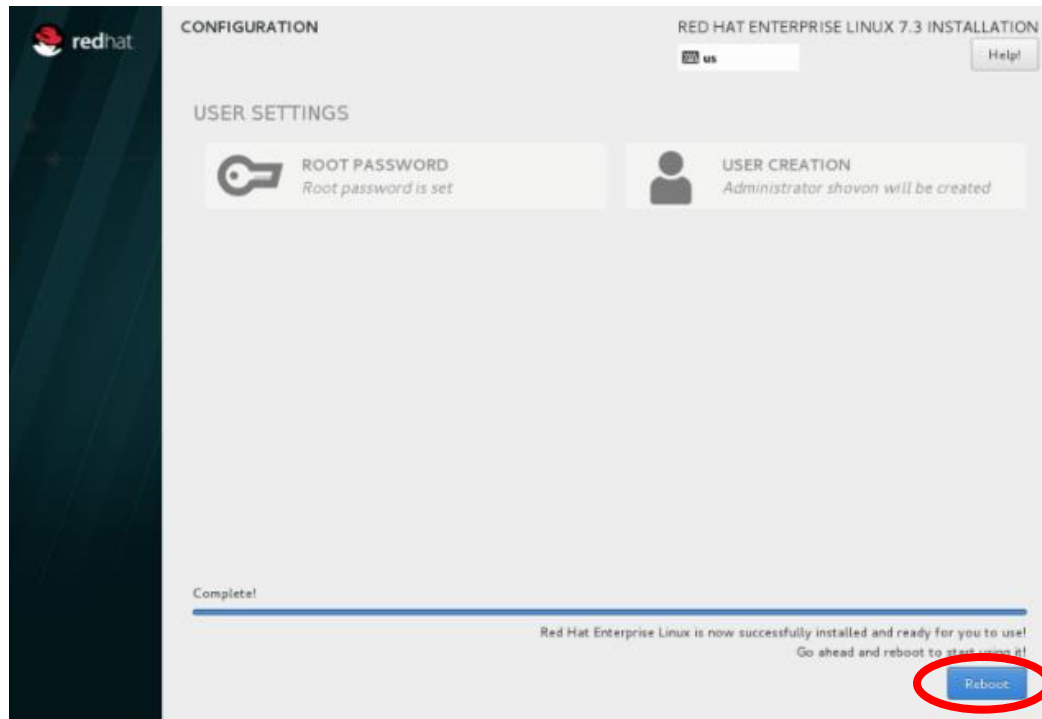
☒ Require a password to use this account

Password

Confirm password

Advanced...

After the install completes click "Reboot" in the bottom right corner of the screen



Automation Configurations

SATELLITE SERVER

How do we push the Automations from the SAT server to the system?

Are there any differences for DREX, DC, or FS line?

Does this include all necessary GD Software installation files?

SAMBA Share

Which systems require a Samba Share Setup?

Are there any differences for DREX, DC, or FS line?

What is required of the Linux Station

What is required of the Windows Station

Installing Trellix

Install the Trellix Agent

How?

Weekly Audits

How to conduct weekly audit?

How?

Where are the Audits stored since they can't be on the system?

How do they get there?

Conducting the Semi-Annual SCAP/STIG Benchmarks

UPLOADING THE SCAP COMPLIANCE TOOL TO THE SATELLITE SERVER

How?

RUNNING THE SCAP COMPLIANCE TOOL ON LINUX STATIONS

How?

RESULTS ARE VIEWED AND STORED

Where?

Conducting the Quarterly Patches

DOWNLOADING PATCHES TO APRICORN DRIVE

How?

UPLOAD PATCHES TO SAT01

How?

ADD PATCHES TO ANSIBLE SCRIPTS

How?

PUSH PATCHES TO TEST SYSTEM

How?

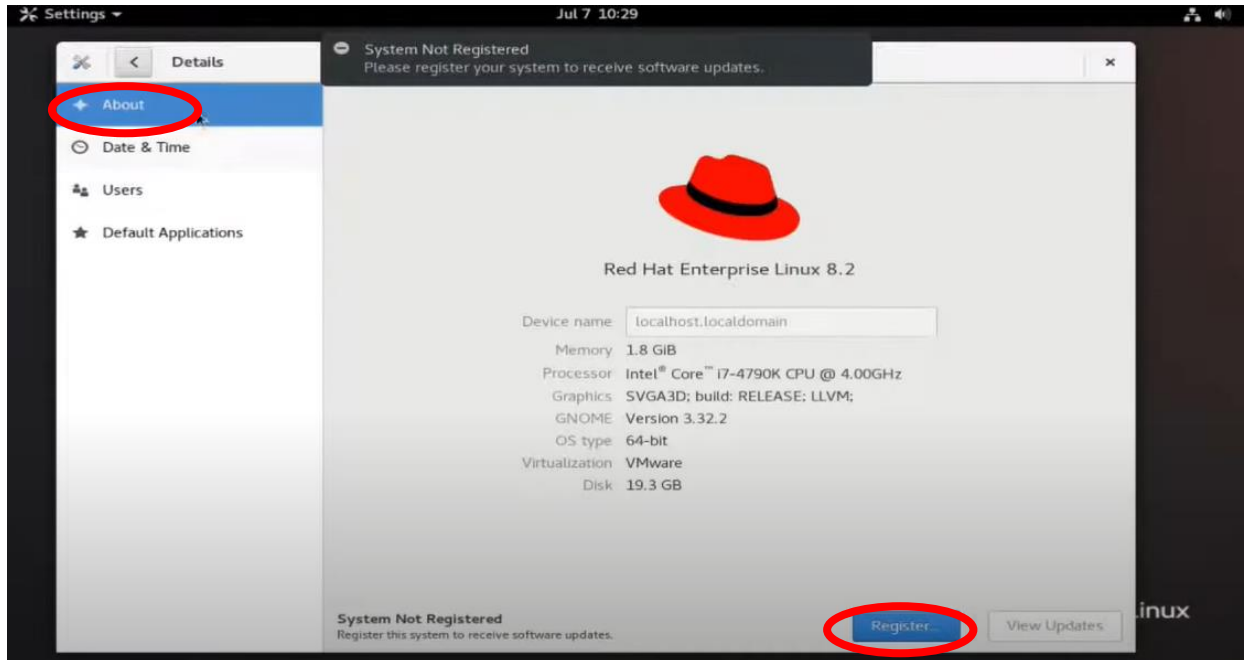
PUSH PATCHES TO REMAINING LRU AND DREX SYSTEMS

How?

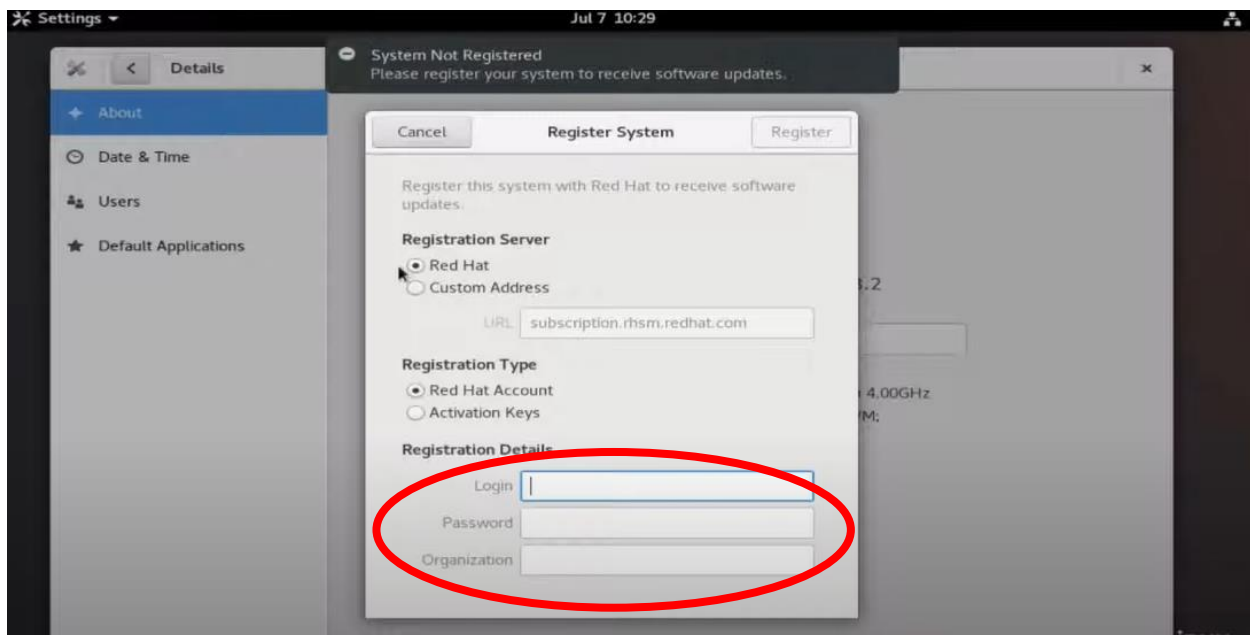
NOTE – The OS is booted through a USB drive and not any other source. Red Hat Linux Version 7.9 is loaded on the USB Drive to get us through this process.

6. Redhat Linux Account Subscription Sign-In - How to Register, Subscribe, and Enable Repos on a RHEL server.

First go into the “Settings”, then “About”, then if you see “System Not Registered”



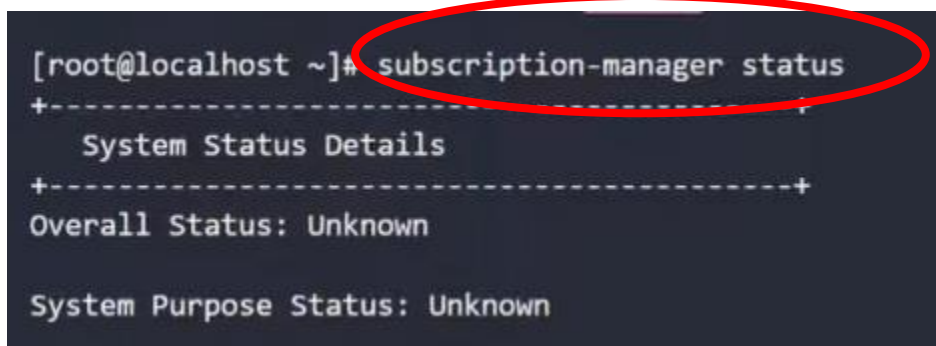
Then go ahead and sign in with the login email and password you used to register with Redhat on a subscription prior



ANOTHER METHOD OF SIGNING IN -

Once logged in, go ahead and open Terminal and get into root by inputting “su” first followed by the admin password

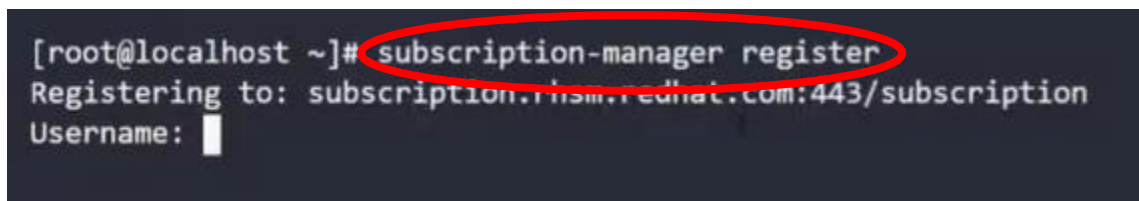
Then input “subscription-manager status” to find out if we are logged in yet.



```
[root@localhost ~]# subscription-manager status
+-----+
      System Status Details
+-----+
Overall Status: Unknown

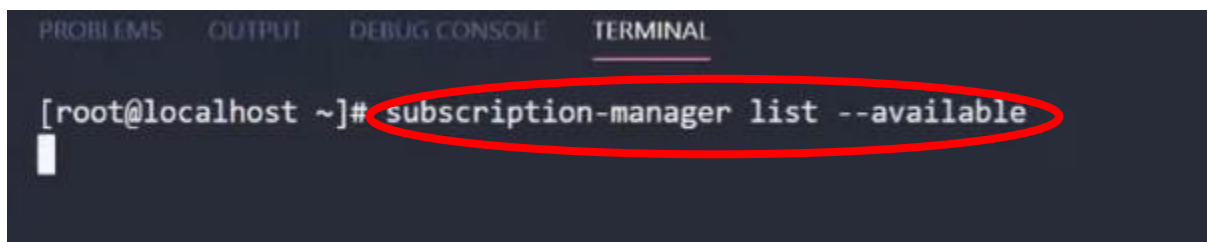
System Purpose Status: Unknown
```

Then input “subscription-manager register” which will then connect you to the Redhat server through terminal to let you sign in through inputting your username/email and password right there inside of terminal.



```
[root@localhost ~]# subscription-manager register
Registering to: subscription.rhsm.redhat.com:443/subscription
Username: █
```

Then input “subscription-manager list --available” to list all the available subscriptions



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL

[root@localhost ~]# subscription-manager list --available
█
```

Next input “subscription-manager subscribe” to subscribe

```
[root@localhost ~]# subscription-manager subscribe
Installed Product Current Status:
Product Name: Red Hat Enterprise Linux for x86_64
Status:      Subscribed
```

Next input “subscription-manager repos –list” in order to list all of the repos available for you to install.

```
[root@localhost ~]# subscription-manager repos --list
```

Next you can go ahead and enable/disable any repos you would like by inputting “subscription-manager repos –enable=” (whichever repo you list)

```
Repo ID:  rhel-8-for-x86_64-baseos-eus-source-rpms
Repo Name: Red Hat Enterprise Linux 8 for x86_64 - BaseOS - Extended Update Support (Source RPMs)
Repo URL:  https://cdn.redhat.com/content/eus/rhel8/$releasever/x86_64/baseos/source/SRPMS
Enabled:   0

Repo ID:  rhel-8-for-x86_64-appstream-e4s-rpms
Repo Name: Red Hat Enterprise Linux 8 for x86_64 - AppStream - Update Services for SAP Solutions (RPMs)
Repo URL:  https://cdn.redhat.com/content/e4s/rhel8/$releasever/x86_64/appstream/os
Enabled:   0

[root@localhost ~]# subscription-manager repos --enable=rhel-8-for-x86_64-appstream-e4s-rpms
```

(We can also enable ALL repos by “–enable=*” input

```
[root@localhost ~]# subscription-manager repos --enable=*
```

7. Install and List Red Hat Enterprise Linux Security Updates and CVEs

First, we'll check how many security notices will be found using "yum updateinfo"

```
[root@ip-172-31-18-181 ~]# yum updateinfo
Last metadata expiration check: 0:16:10 ago on Thu 23 Jul 2020 15:29:43 UTC.
Updates Information Summary: available
  13 Security notice(s)
    9 Important Security notice(s)
    3 Moderate Security notice(s)
    1 Low Security notice(s)
  14 Bugfix notice(s)
    1 Enhancement notice(s)
```

Then search "yum updateinfo info --sec-severity Important" to search up the most important vulnerabilities to take care of

```
[root@ip-172-31-18-181 ~]# yum updateinfo info --sec-severity Important
Last metadata expiration check: 0:17:35 ago on Thu 23 Jul 2020 15:29:43 UTC.
```

Next search up the vulnerability for more information regarding it

```
[root@ip-172-31-18-181 ~]# yum updateinfo info RHSA-2020:2774
Last metadata expiration check: 0:18:14 ago on Thu 23 Jul 2020 15:29:43 UTC.
=====
Important: virt:rhel security update
=====
Update ID: RHSA-2020:2774
Type: security
Updated: 2020-06-30 13:38:53
Bugs: 1798453 - CVE-2020-8608 QEMU: Slirp: potential OOB access due to unsafe snprintf() usages
      1810390 - CVE-2019-20382 QEMU: vnc: memory leakage upon disconnect
CVEs: CVE-2019-20382
      : CVE-2020-8608
```

Then search up the CVE's by searching "yum update --cves CVE-2019..."

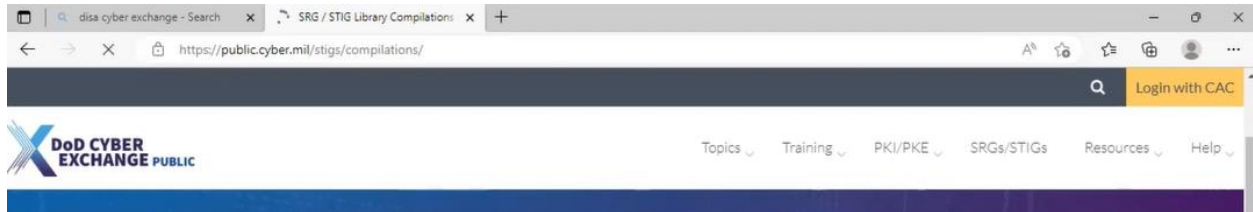
```
[root@ip-172-31-18-181 ~]# yum update --cves CVE-2019-20382,CVE-2020-8608
Last metadata expiration check: 0:19:09 ago on Thu 23 Jul 2020 15:29:43 UTC.
Dependencies resolved.
```

Then we will run upgrades to help solve the CVE by running “yum update --advisory RHSA-2020;2774”

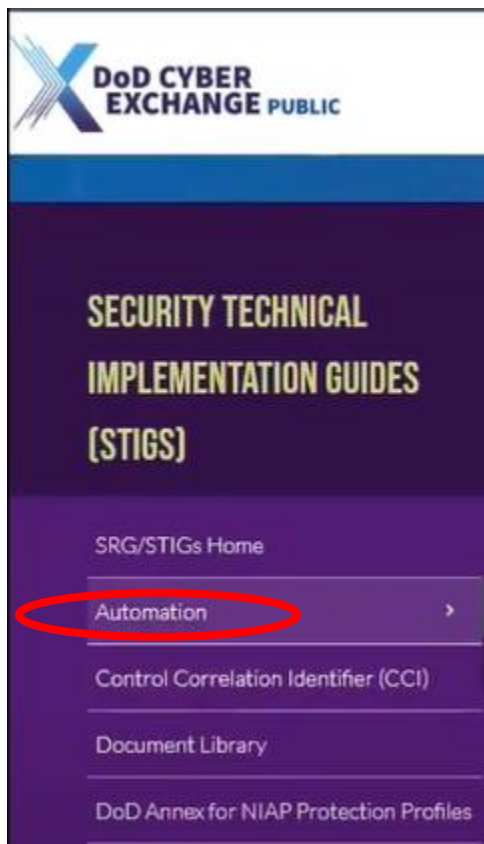
```
[root@ip-172-31-18-181 ~]# yum update --advisory RHSA-2020:2774
Last metadata expiration check: 0:20:06 ago on Thu 25 Jul 2020 15:29:43 UTC.
Dependencies resolved.
=====
Package                Architecture  Version                                           Repository                                Size
=====
Upgrading:
qemu-guest-agent        x86_64        15:2.12.0-99.module+el8.2.0+6870+55b789b4.2    rhel-8-appstream-rhui-rpms              217 k
Transaction Summary
=====
Upgrade 1 Package
```

8. How To Download the SCAP Tool using Cyber.mil

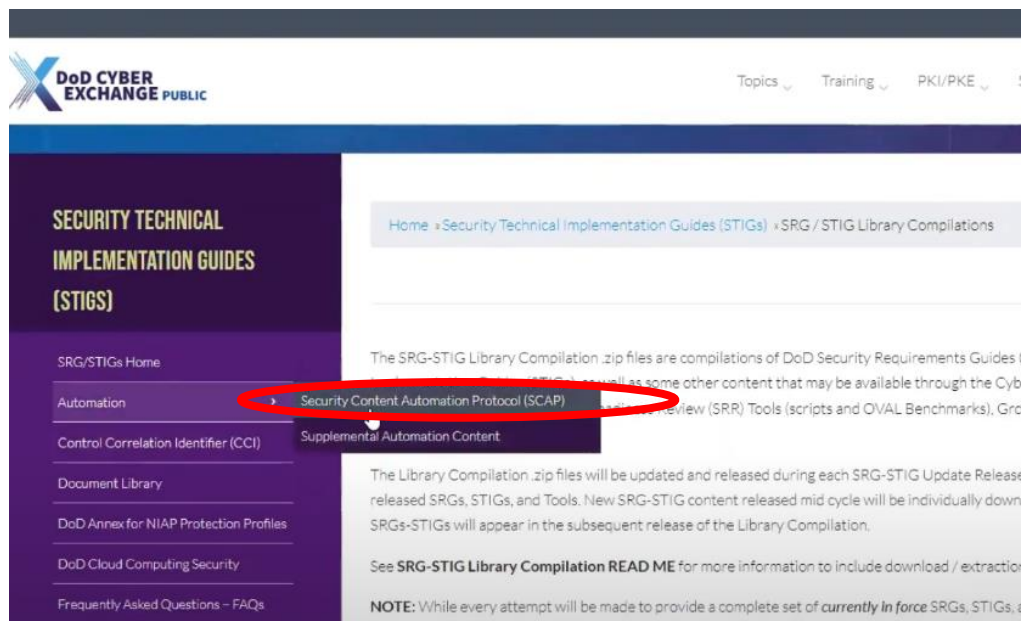
Go to the website public.cyber.mil/stigs/compilations/ to start your search



Next go into the “Automation” settings to find the proper tool



Then click on the “SCAP” menu



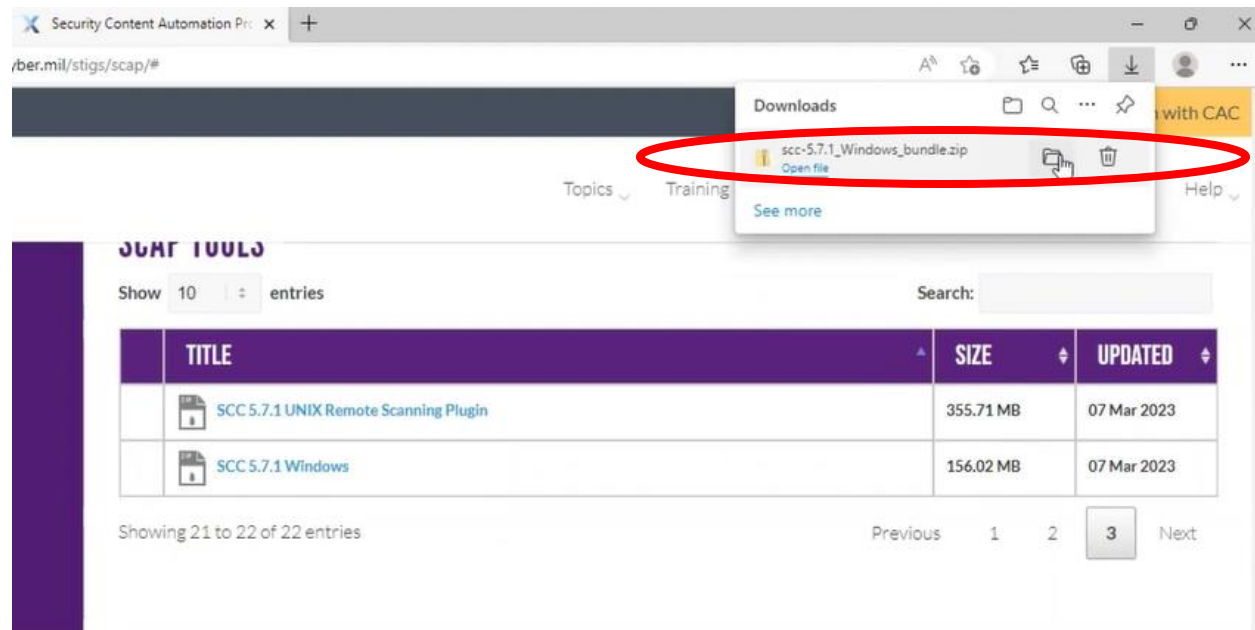
Then click into the latest SCAP tool (SCC) which for RHEL 7.9 we will be using the circled option

SCAP TOOLS

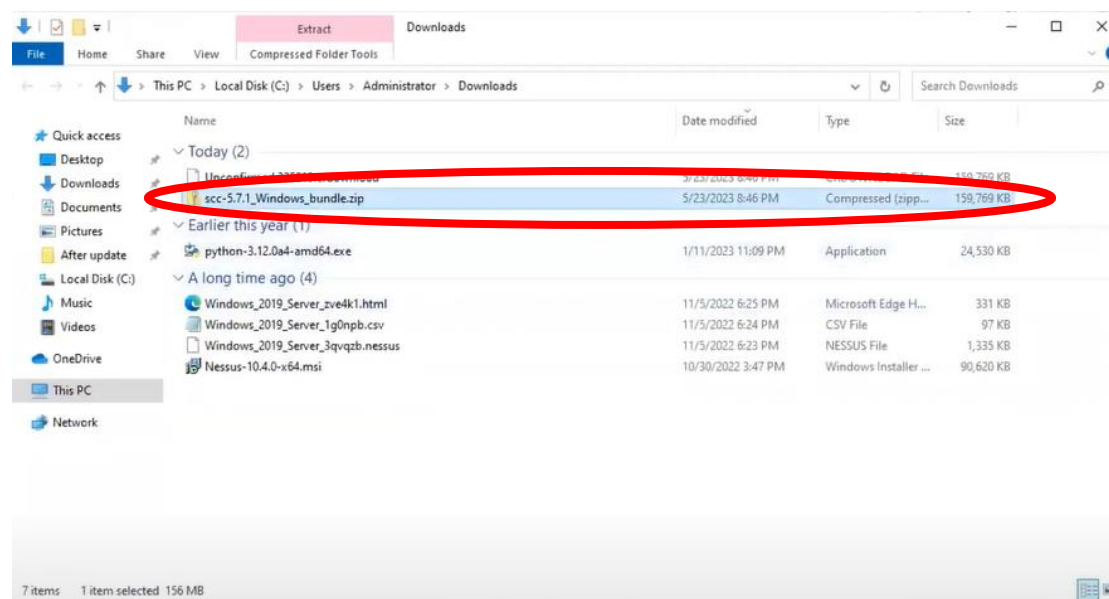
Show 10 entries Search:

	TITLE	SIZE	UPDATED
	SCC 5.7.1 Checksums File	8.74 KB	07 Mar 2023
	SCC 5.7.1 Mac OS X x86 64	41.94 MB	07 Mar 2023
	SCC 5.7.1 Raspbian 11 ARMv7	81.36 MB	07 Mar 2023
	SCC 5.7.1 Readme	2.04 KB	07 Mar 2023
	SCC 5.7.1 Release Notes	11.45 KB	07 Mar 2023
	SCC 5.7.1 RHEL 6 i686	87.05 MB	07 Mar 2023
	SCC 5.7.1 RHEL 6 x86 64	88.19 MB	07 Mar 2023
	SCC 5.7.1 RHEL 7/Oracle Linux 7/SLES12/SLES 15 x86 64	84.97 MB	07 Mar 2023
	SCC 5.7.1 RHEL 8/Oracle Linux 8 Aarch64	85.13 MB	07 Mar 2023
	SCC 5.7.1 RHEL 8/Oracle Linux 8 x86 64	81.61 MB	07 Mar 2023

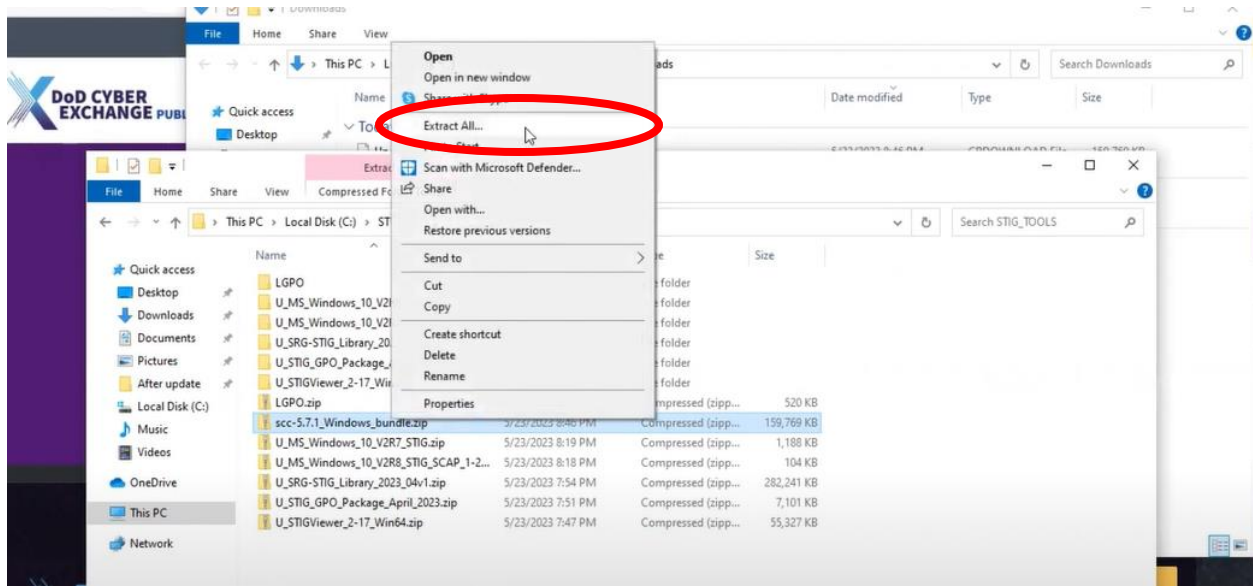
Then it should be downloading -



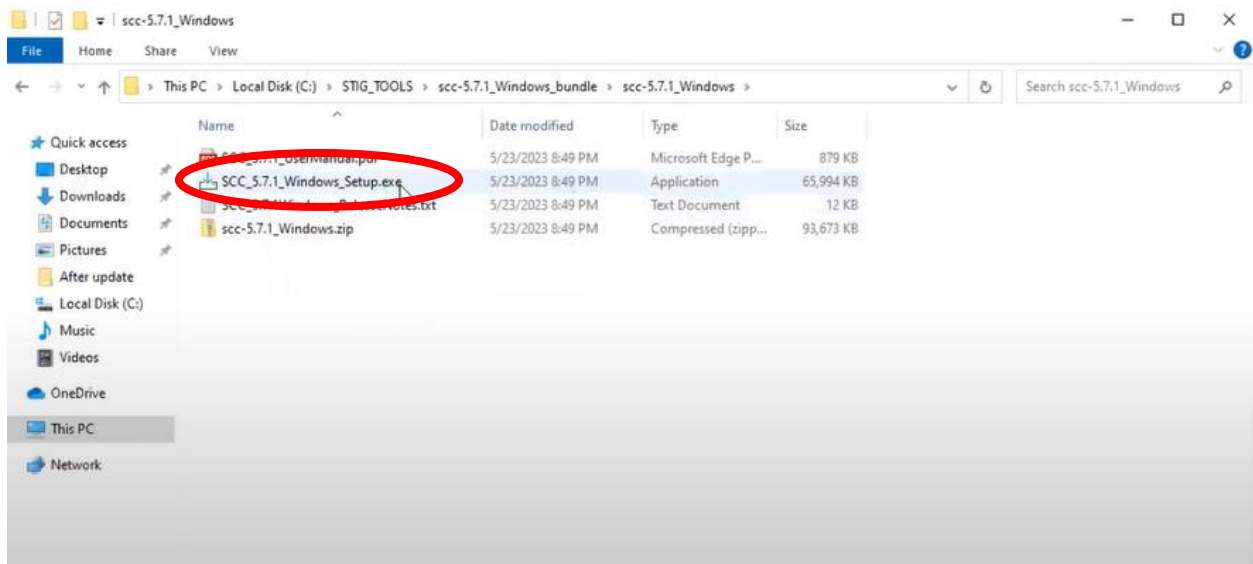
Once the file is loaded click into it to access your downloads' folder



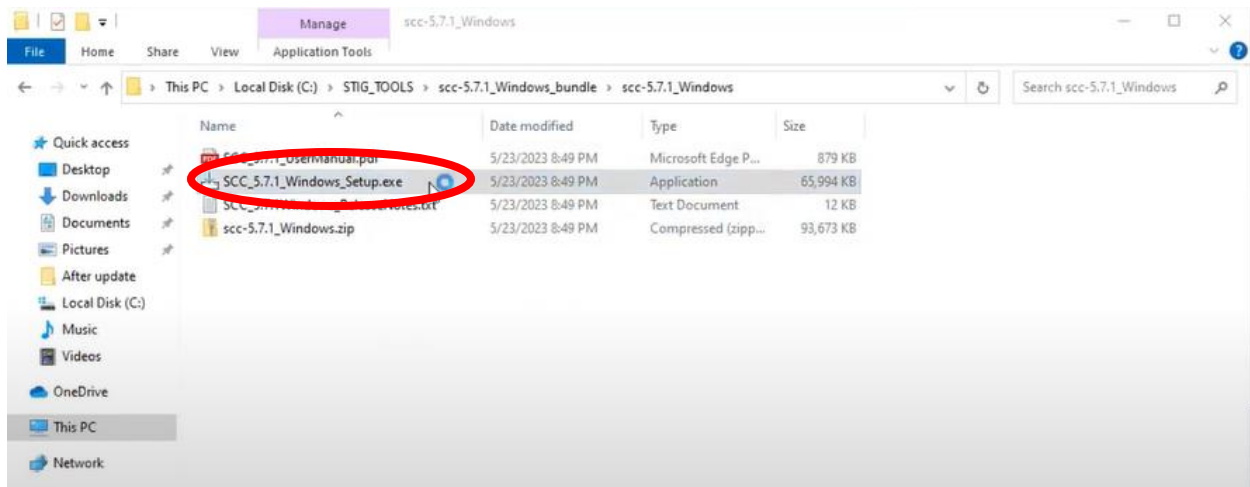
Then go ahead and proceed to unzip the folder



Then run the setup and click on the app.



Once it is unzipped, run the .exe file and run the setup program



9. How To Download the SCAP Tool Using Terminal

First, go into `df -h` to get the different routes

```
[nsahial@nademclan ~]$ df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
devtmpfs	7.7G	0	7.7G	0%	/dev
tmpfs	7.7G	16K	7.7G	1%	/dev/shm
tmpfs	7.7G	27M	7.7G	1%	/run
tmpfs	7.7G	0	7.7G	0%	/sys/fs/cgroup
/dev/nvme0n1p3	50G	4.0G	46G	8%	/
/dev/nvme0n1p5	33G	4.6G	29G	14%	/home
/dev/nvme0n1p2	1014M	166M	849M	17%	/boot
/dev/nvme0n1p6	25G	33M	25G	1%	/tmp
/dev/nvme0n1p1	512M	11M	502M	2%	/boot/efi
/dev/nvme0n1p4	50G	1.4G	49G	3%	/var
/dev/nvme0n1p7	25G	48M	25G	1%	/var/log
/dev/nvme0n1p8	25G	40M	25G	1%	/var/log/audit
/dev/nvme0n1p9	25G	40M	25G	1%	/var/tmp
tmpfs	1.6G	64K	1.6G	1%	/run/user/1000
/dev/sde1	30G	7.2G	23G	25%	/run/media/nsahial/RHEL-7_9

Next, go into the “RHEL-7_9” and then run `ls -lh`

```
[nsahial@nademclan ~]$ cd /run/media/nsahial/RHEL-7_9
[nsahial@nademclan RHEL-7_9]$ ls -lh
```

After running `ls -lh`, we will then run “`cd Nadem`”

```
[nsahial@nademclan RHEL-7_9]$ ls -lh
total 1.1M
drwxr-xr-x. 4 nsahial nsahial 16K Jan 24 2023 addons
-rw-r--r--. 1 nsahial nsahial 34K Jan 24 2023 autorun.ico
-rw-r--r--. 1 nsahial nsahial 224 Jan 24 2023 autorun.inf
drwxr-xr-x. 3 nsahial nsahial 16K Jan 24 2023 EFI
-rw-r--r--. 1 nsahial nsahial 8.1K Jan 24 2023 EULA
-rw-r--r--. 1 nsahial nsahial 1.5K Jan 24 2023 extra_files.json
-rw-r--r--. 1 nsahial nsahial 18K Jan 24 2023 GPL
drwxr-xr-x. 3 nsahial nsahial 16K Jan 24 2023 images
drwxr-xr-x. 2 nsahial nsahial 16K Jan 24 2023 isolinux
-r--r--r--. 1 nsahial nsahial 37K Jan 24 2023 ldlinux.sys
drwxr-xr-x. 2 nsahial nsahial 16K Jan 24 2023 LiveOS
-rw-r--r--. 1 nsahial nsahial 114 Jan 24 2023 media.repo
drwxr-xr-x. 6 nsahial nsahial 16K Jul 15 19:58 Nadem
drwxr-xr-x. 2 nsahial nsahial 720K Jan 24 2023 Packages
drwxr-xr-x. 2 nsahial nsahial 16K Jan 24 2023 repodata
-rw-r--r--. 1 nsahial nsahial 3.3K Jan 24 2023 RPM-GPG-KEY-redhat-beta
-rw-r--r--. 1 nsahial nsahial 3.2K Jan 24 2023 RPM-GPG-KEY-redhat-release
-rw-r--r--. 1 nsahial nsahial 94 Jan 24 2023 syslinux.cfg
drwxr-xr-x. 2 nsahial nsahial 16K Jan 24 2023 System Volume Information
-rw-r--r--. 1 nsahial nsahial 1.8K Jan 24 2023 TRANS.TBL
[nsahial@nademclan RHEL-7_9]$ cd Nadem
```

After running “`CD Nadem`” go ahead and run “`ls -lh`” again

```
[nsahial@nademclan RHEL-7_9]$ cd Nadem
[nsahial@nademclan Nadem]$ ls -lh
```

We will get these results after running “`ls -lh`”

```
total 1.8G
-rw-r--r--. 1 nsahial nsahial 1.8G Jul 10 14:32 All Stig Viewer DL's.zip
drwxr-xr-x. 3 nsahial nsahial 16K Jul 11 14:22 Linux SCAP
drwxr-xr-x. 3 nsahial nsahial 16K Jul 9 16:21 RHEL7.9 STIG Jul24
-rw-r--r--. 1 nsahial nsahial 58K Jul 15 19:07 scc
drwxr-xr-x. 2 nsahial nsahial 16K Jul 15 14:37 SS JEFF
drwxr-xr-x. 3 nsahial nsahial 16K Jul 15 20:00 STIG Viewer
[nsahial@nademclan Nadem]$ cd Linux\ SCAP/
```

Then run “`CD Linux SCAP`” to get into that folder, and then run “`ls`”

```
[nsahial@nademclan Nadem]$ cd Linux\ SCAP/
[nsahial@nademclan Linux SCAP]$ ls
```

Then run the first option which is in blue

```
scc-5.9_rhel7_sles12-15_oracle-linux7_x86_64_bundle
scc-5.9_rhel7_sles12-15_oracle-linux7_x86_64_bundle.zip
[nsahial@nademclan Linux SCAP]$ cd scc-5.9_rhel7_sles12-15_oracle-linux7_x86_64_bundle/
```

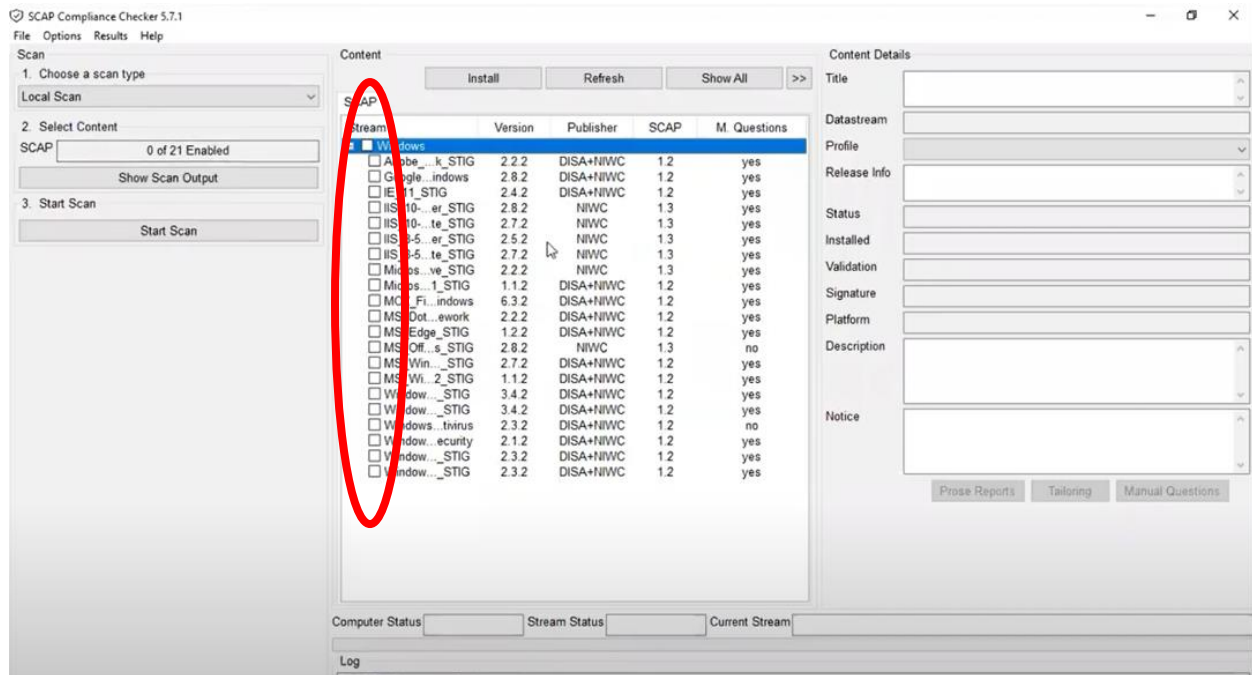
Then after entering that folder, click “ls -lh” then run the first option which is “.x86_64” then “ls -lh” one more time and run the scc rpm file using “sudo”.

```
[nsahial@nademclan Linux SCAP]$ cd scc-5.9_rhel7_sles12-15_oracle-linux7_x86_64_bundle/
[nsahial@nademclan scc-5.9_rhel7_sles12-15_oracle-linux7_x86_64_bundle]$ ls -lh
total 32K
drwxr-xr-x. 3 nsahial nsahial 16K Jul 15 19:07 scc-5.9_rhel7_x86_64
drwxr-xr-x. 2 nsahial nsahial 16K Jul 12 16:17 scc-5.9_Windows
[nsahial@nademclan scc-5.9_rhel7_sles12-15_oracle-linux7_x86_64_bundle]$ cd scc-5.9_rhel7_x86_64/
[nsahial@nademclan scc-5.9_rhel7_x86_64]$ ls -lh
total 89M
drwxr-xr-x. 6 nsahial nsahial 16K Jul 15 19:08 scc_5.9
-rw-r--r--. 1 nsahial nsahial 41M Mar 28 13:16 scc-5.9.rhel7.x86_64.rpm
-rw-r--r--. 1 nsahial nsahial 47M Mar 28 13:15 scc-5.9_rhel7_x86_64.tar.gz
-rw-r--r--. 1 nsahial nsahial 7.5K Mar 28 13:12 SCC_5.9_rhel_x86_64_ReleaseNotes.txt
-rw-r--r--. 1 nsahial nsahial 824K Mar 28 13:12 SCC_5.9_rhel_x86_64_UserManual.pdf
[nsahial@nademclan scc-5.9_rhel7_x86_64]$ sudo rpm -i scc-5.9.rhel7.x86_64.rpm
[sudo] password for nsahial:
warning: scc-5.9.rhel7.x86_64.rpm: Header V4 RSA/SHA256 Signature, key ID fe37a6bc: NOKEY
***** NOTICE *****
*
* By installing this package, you agree to the terms of this license agreement. *
*
***** NOTICE *****
```

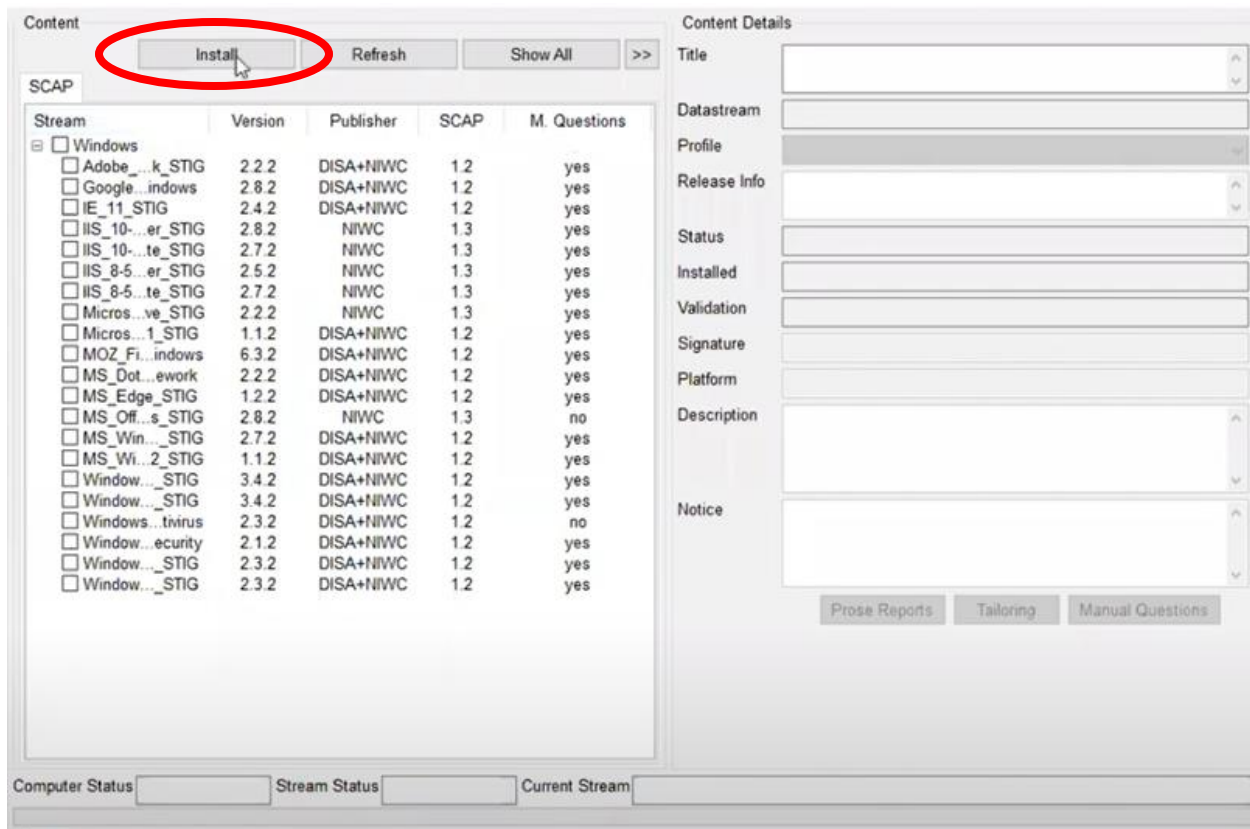
The SCAP Tool should now be successfully installed using Terminal!

10. Loading STIGS using SCAP Tool

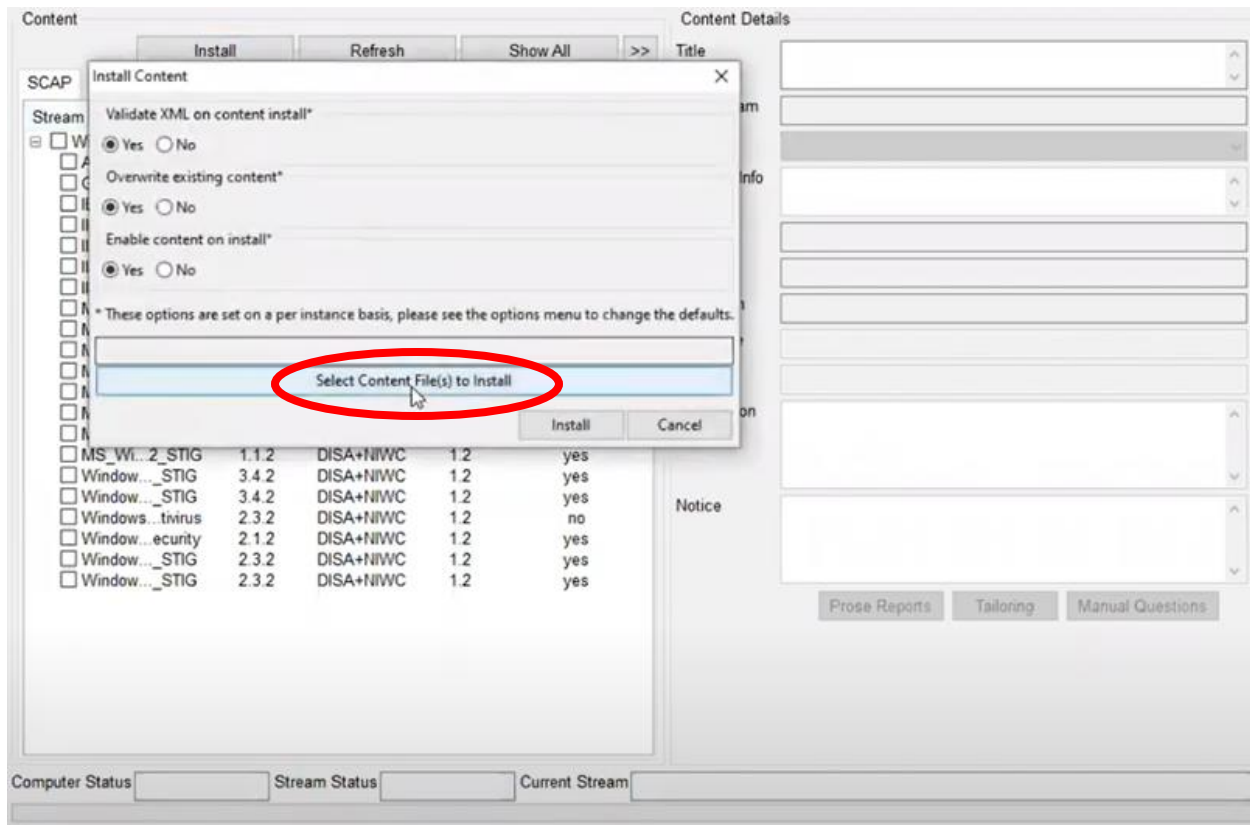
After opening the SCAP Tool unselect all of the streams (We will be loading our own)



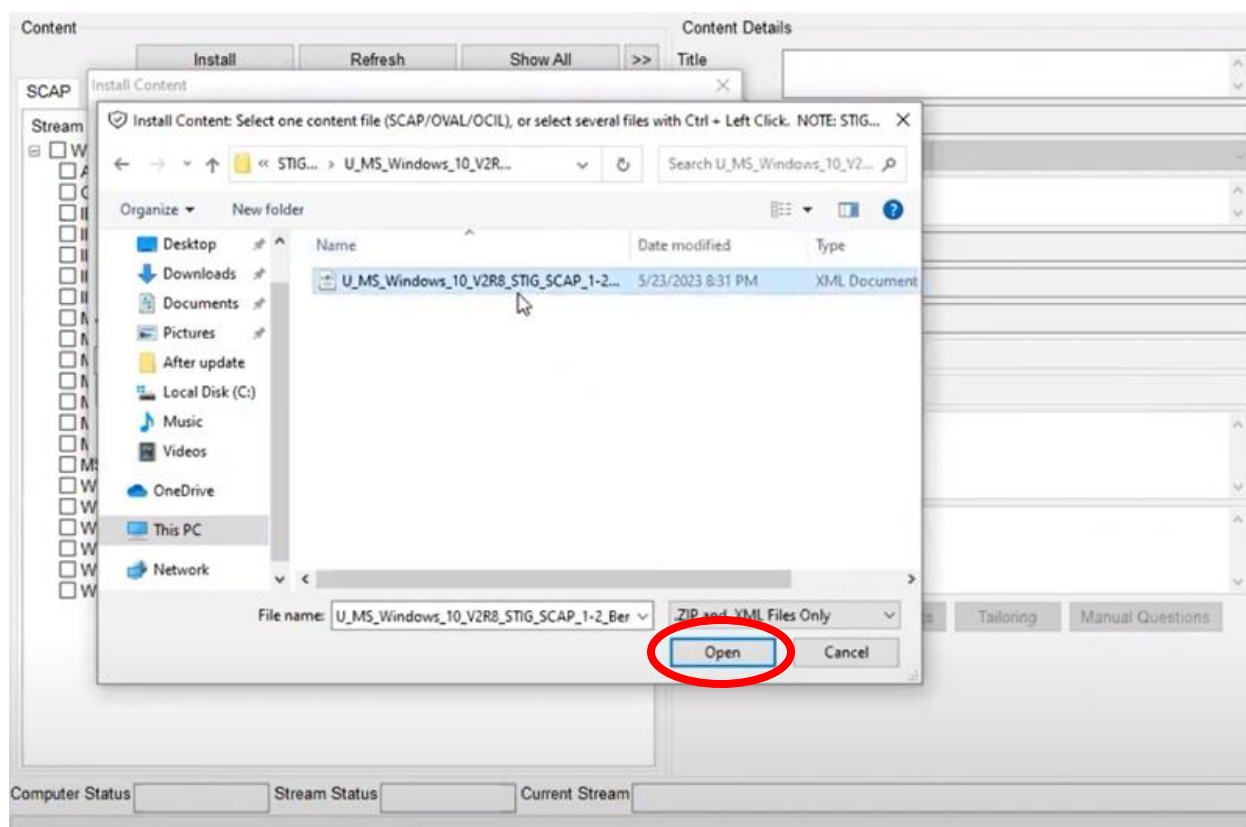
Then click on “Install”



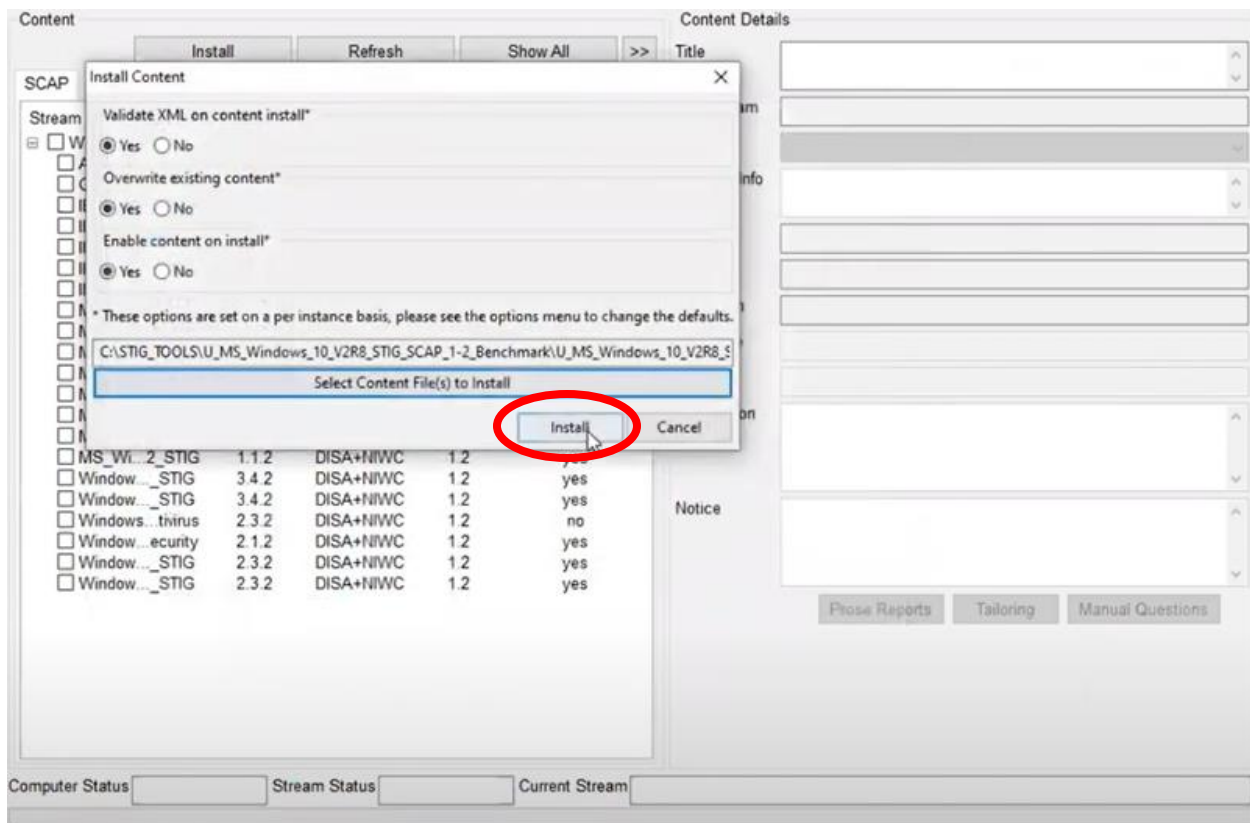
Now click on “Select Content File(s) to Install”



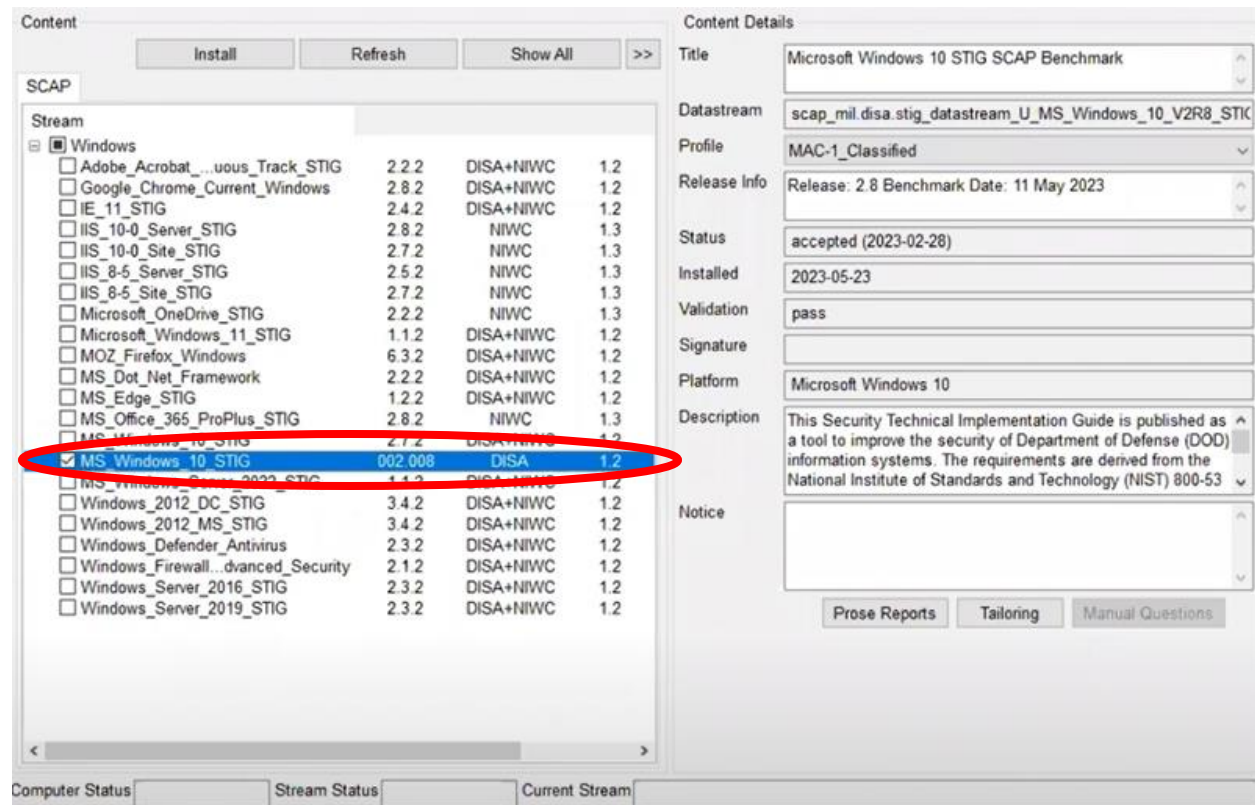
Now locate where the XML file is located and click “Open”



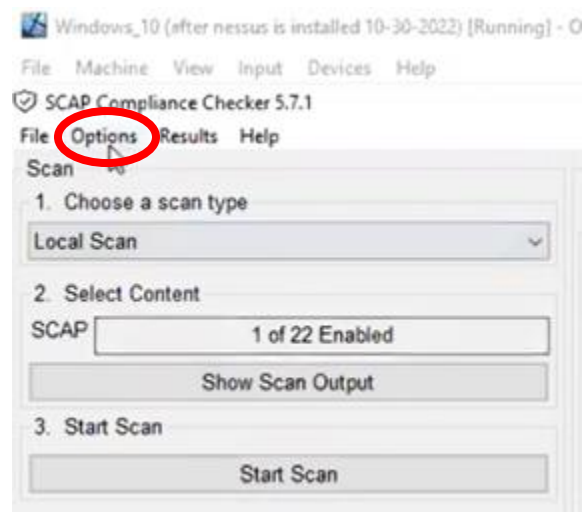
Then click on “Install”



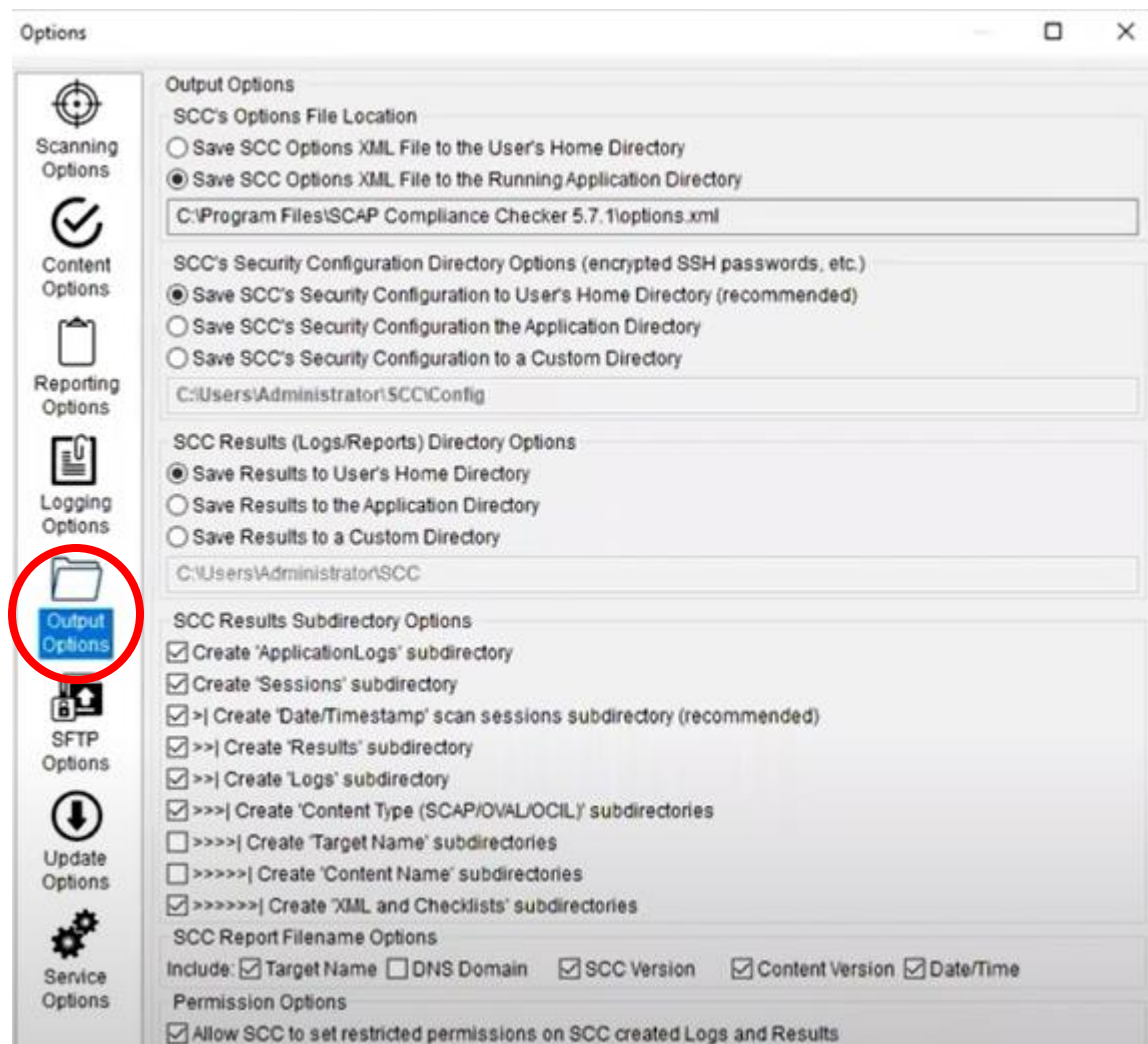
After it loads the STIG file should be selected



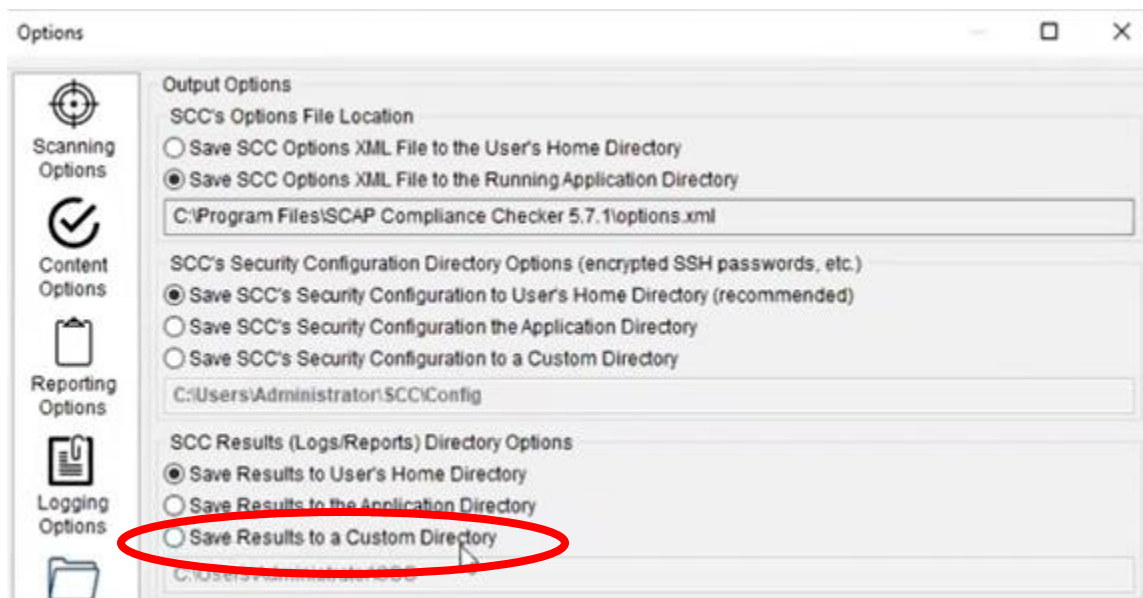
Then go into the "Options"



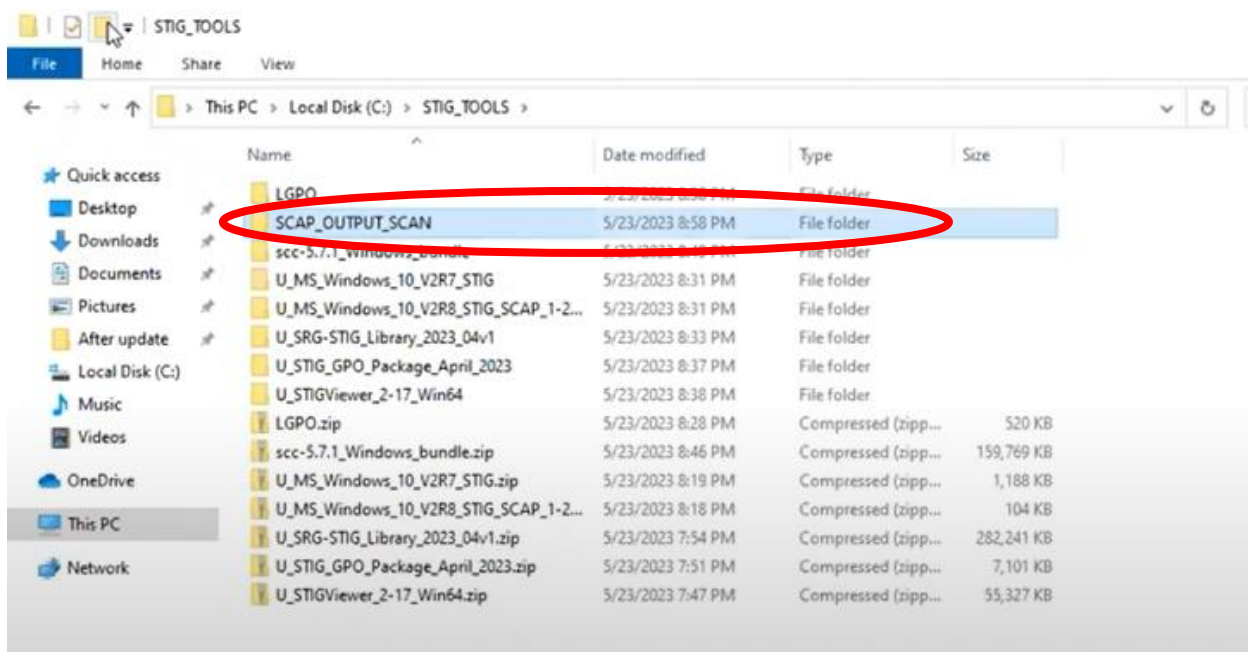
Then click into “Output Options”



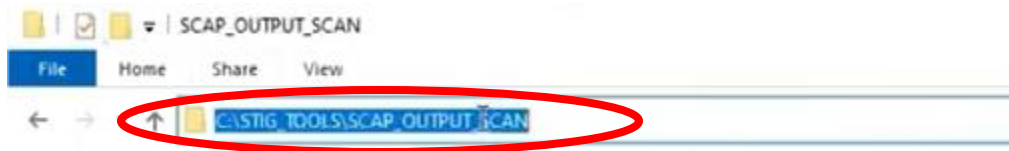
Then click on “Save Results to a Custom Directory”



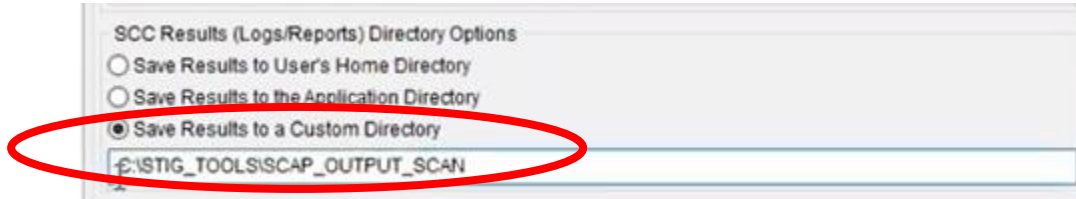
Then go into file explorer and create a folder to load the output scan



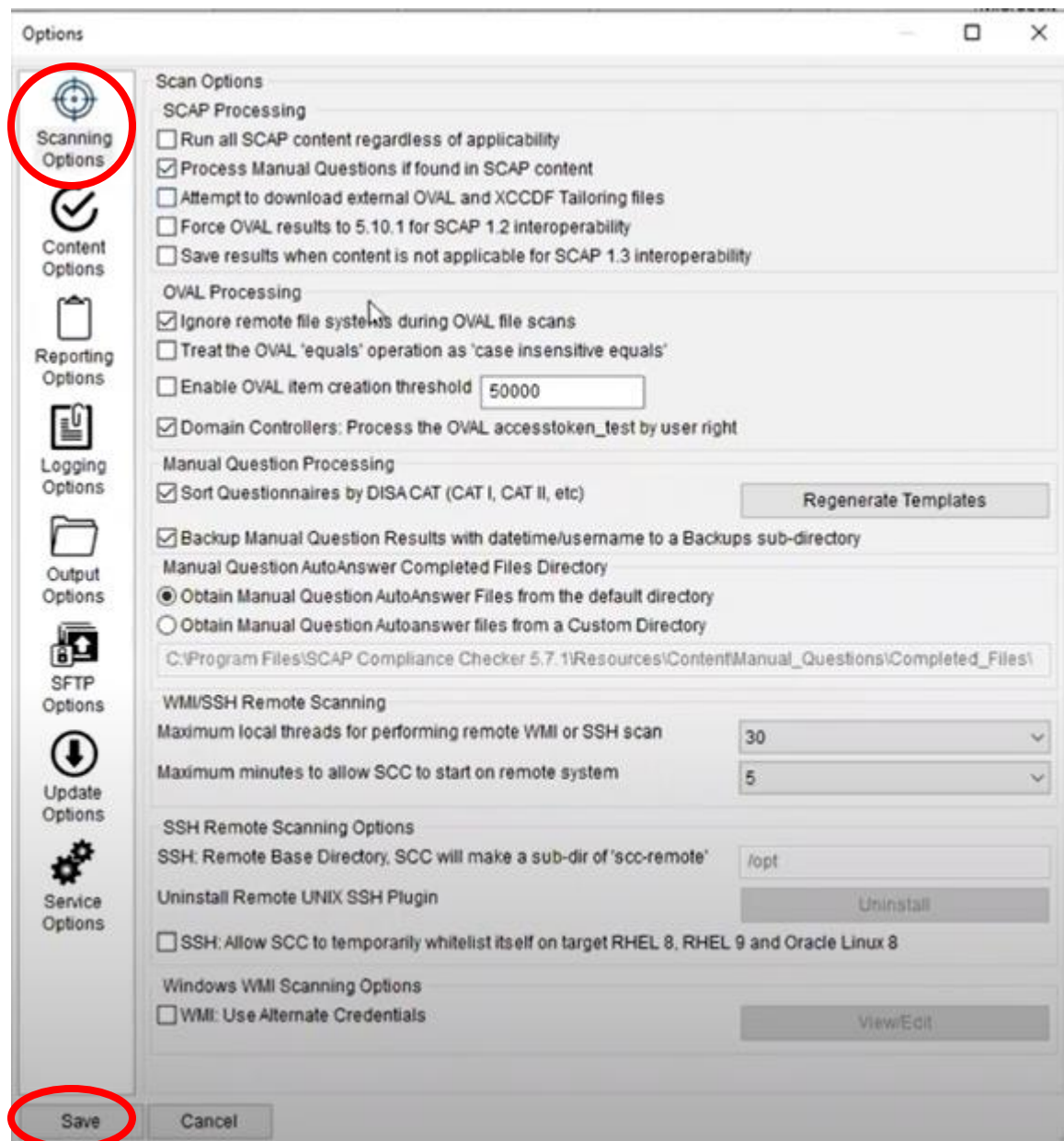
Then go ahead and copy the location of the folder



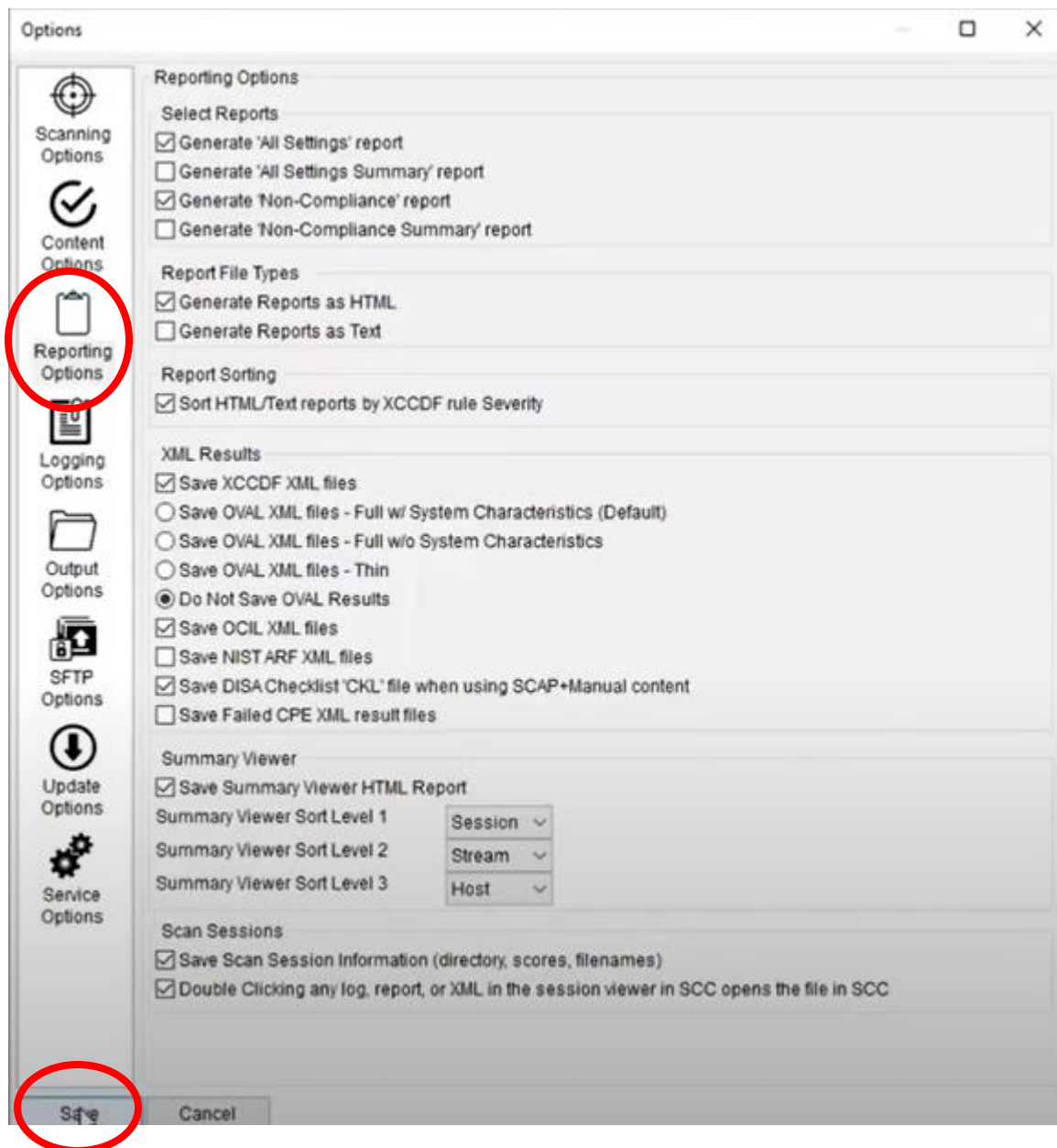
Go ahead and paste the location of the folder here



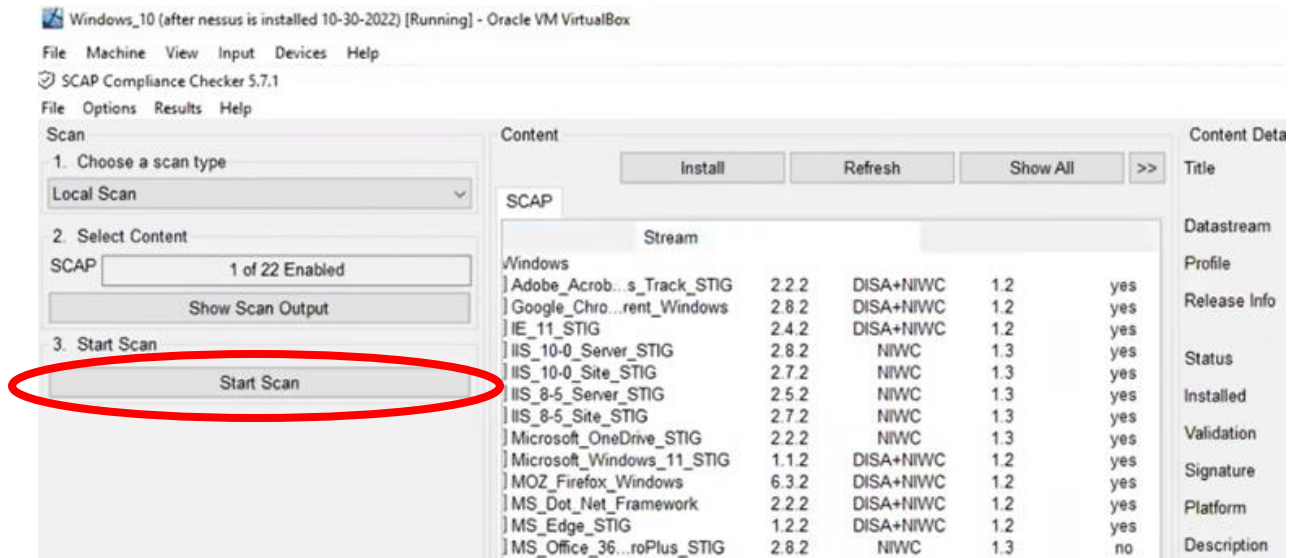
Now head into the “Scanning Options” and configure the settings as such then click “Save”



Then head into the “Reporting Options” and configure the settings as such then click “Save”

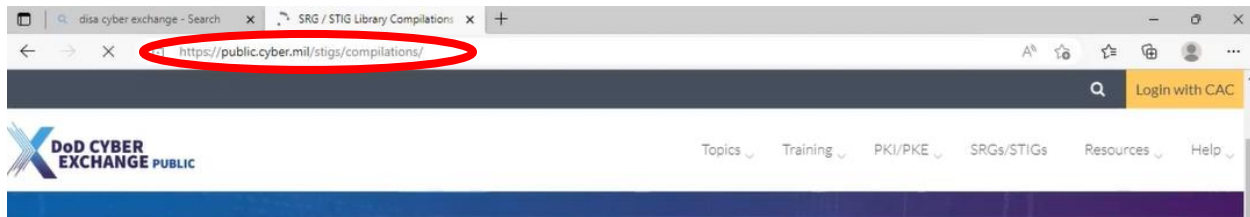


Then you can go ahead and “Start Scan”

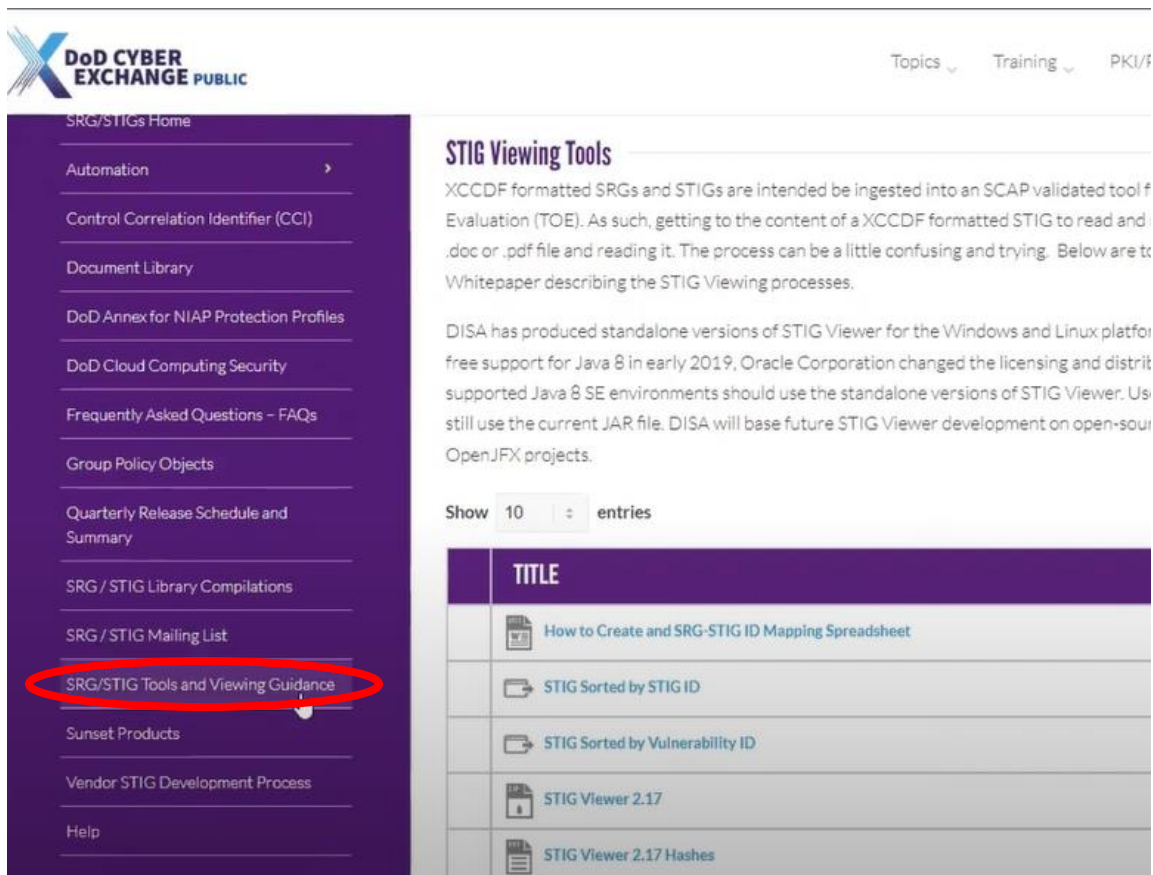


11. How To Download STIG Viewer

Go to the website public.cyber.mil/stigs/compilations to start your search



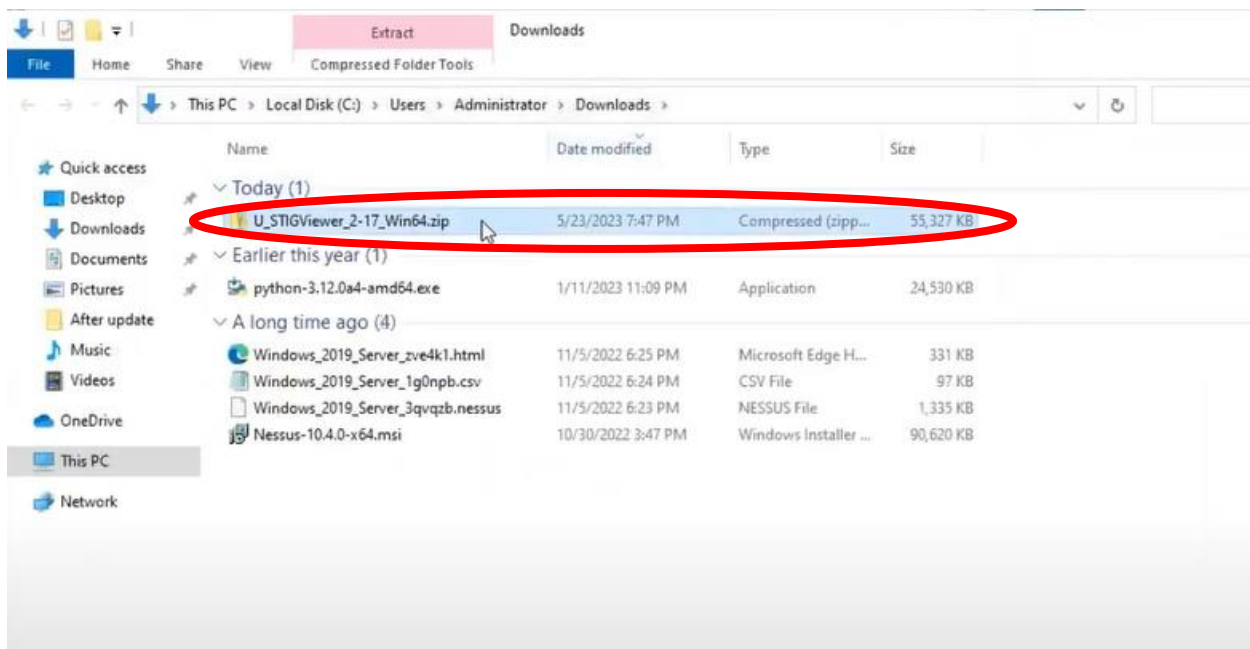
Next go into the “STIG Tools and Viewing Guidance” menu option



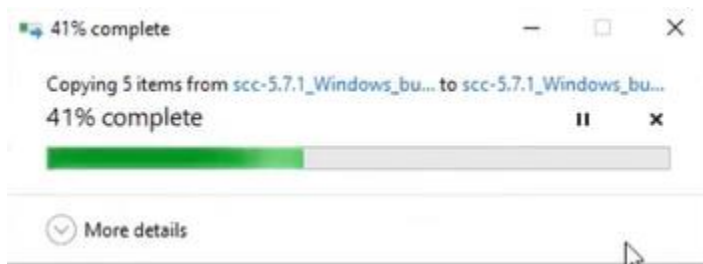
Then find the latest STIG Viewer that's compatible with your O/S

	TITLE	SIZE	UPDATED
	How to Create and SRG-STIG ID Mapping Spreadsheet	298.21 KB	03 Feb 2021
	STIG Sorted by STIG ID	103.46 KB	30 Mar 2015
	STIG Sorted by Vulnerability ID	101.59 KB	30 Mar 2015
	STIG Viewer 2.17	1.14 MB	21 Sep 2022
	STIG Viewer 2.17 Hashes	1.36 KB	21 Sep 2022
	STIG Viewer 2.17-Linux	73.38 MB	21 Sep 2022
	STIG Viewer 2.17-Win64	54.03 MB	21 Sep 2022
	STIG Viewer 2.17-Win64 msi	54.26 MB	21 Sep 2022
	STIG Viewer User Guide (Sep 2022)	1.77 MB	21 Sep 2022
	STIG Viewer Video	--	14 Jun 2018

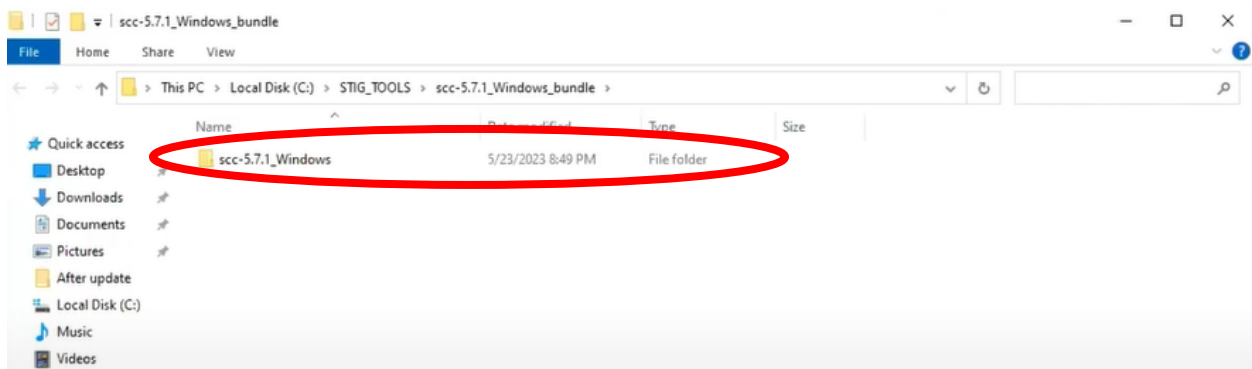
Then go to the Downloads folder to unzip the downloaded folder



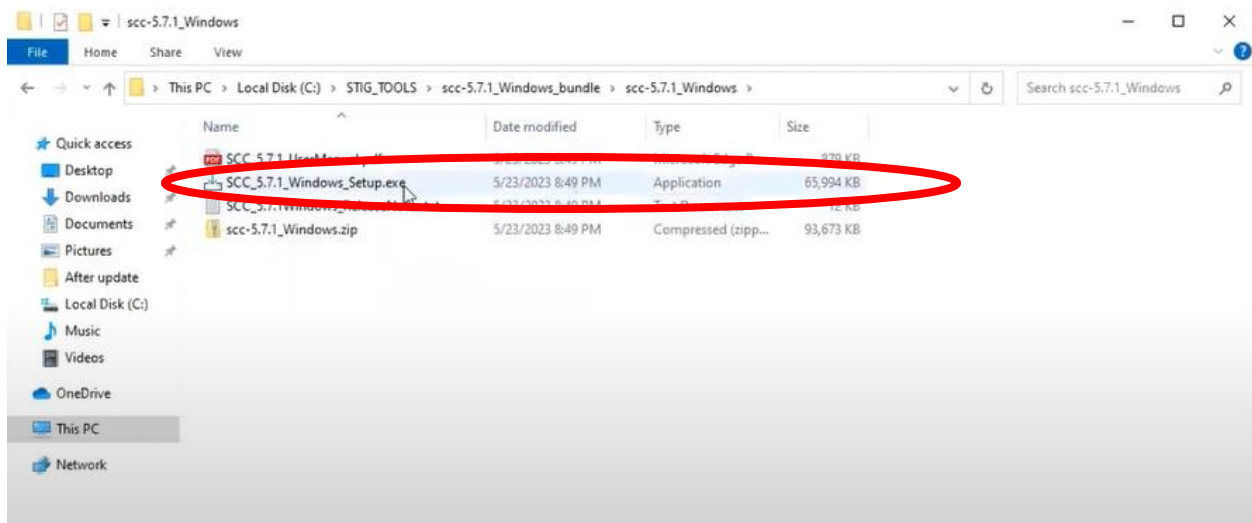
They should be loading and unzipping



Click into the unzipped folder



Then you would have to run the Setup.exe file in order to set up SCAP Tool



12. How To Download STIG Viewer 3 via Terminal

First, make sure to create a folder named “stigviewer” in the “opt” directory

```
[nsahial@nademclan ~]$ cd /opt/
All SCAP Tools DL's/ rh/ stigviewer/
All Stig Viewer DL's/ scc/
```

Then “cd” the app that we located in the “opt” but then it may give you an error for sandbox and root.

```
[nsahial@nademclan ~]$ cd /opt/stigviewer/stig_viewer_3-linux-x64/
[nsahial@nademclan stig_viewer_3-linux-x64]$ ./STIG\ Viewer\ 3
[26029:0806/130758.367839:FATAL:setuid_sandbox_host.cc(158)] The SUID sandbox helper binary was
that /opt/stigviewer/stig_viewer_3-linux-x64/chrome-sandbox is owned by root and has mode 4755.
Trace/breakpoint trap (core dumped)
```

Then “cd” your way back a few folders and also get into root access by “su”

```
[nsahial@nademclan stig_viewer_3-linux-x64]$ cd ../
[nsahial@nademclan stigviewer]$ su
Password:
```

Then add “chmod 4755” to the files as such to eliminate the errors we receive

```
[root@nademclan stigviewer]# chmod 4755 /opt/stigviewer/
stig_viewer_3-linux-x64/ U_STIGViewer-linux_x64-3-3-0.zip
[root@nademclan stigviewer]# chmod 4755 /opt/stigviewer/stig_viewer_3-linux-x64/
stig_viewer_3-linux-x64/ U_STIGViewer-linux_x64-3-3-0.zip
[root@nademclan stigviewer]# chmod 4755 /opt/stigviewer/stig_viewer_3-linux-x64/ -rwx
stig_viewer_3-linux-x64/ U_STIGViewer-linux_x64-3-3-0.zip
[root@nademclan stigviewer]# chmod 4755 /opt/stigviewer/stig_viewer_3-linux-x64/ -rwx
stig_viewer_3-linux-x64/ U_STIGViewer-linux_x64-3-3-0.zip
[root@nademclan stigviewer]# chmod 4755 /opt/stigviewer/stig_viewer_3-linux-x64/
```

Then run “ls” again and go into the Stig viewer folder

```
[root@nademclan stigviewer]# ls
stig_viewer_3-linux-x64 U_STIGViewer-linux_x64-3-3-0.zip
[root@nademclan stigviewer]# cd stig_viewer_3-linux-x64/
[root@nademclan stig_viewer_3-linux-x64]# ls
```

Then run the app, if an error occurs mentioning “Sandbox”, the following step will solve that issue.

```
[root@nademclan stig_viewer_3-linux-x64]# ./STIG\ Viewer\ 3
[0806/131857.790592:FATAL:electron_main_delegate.cc(294)] Running as root without --no-sandbox is not supported. See https://crbug.com/638180.
Trace/breakpoint trap (core dumped)
```

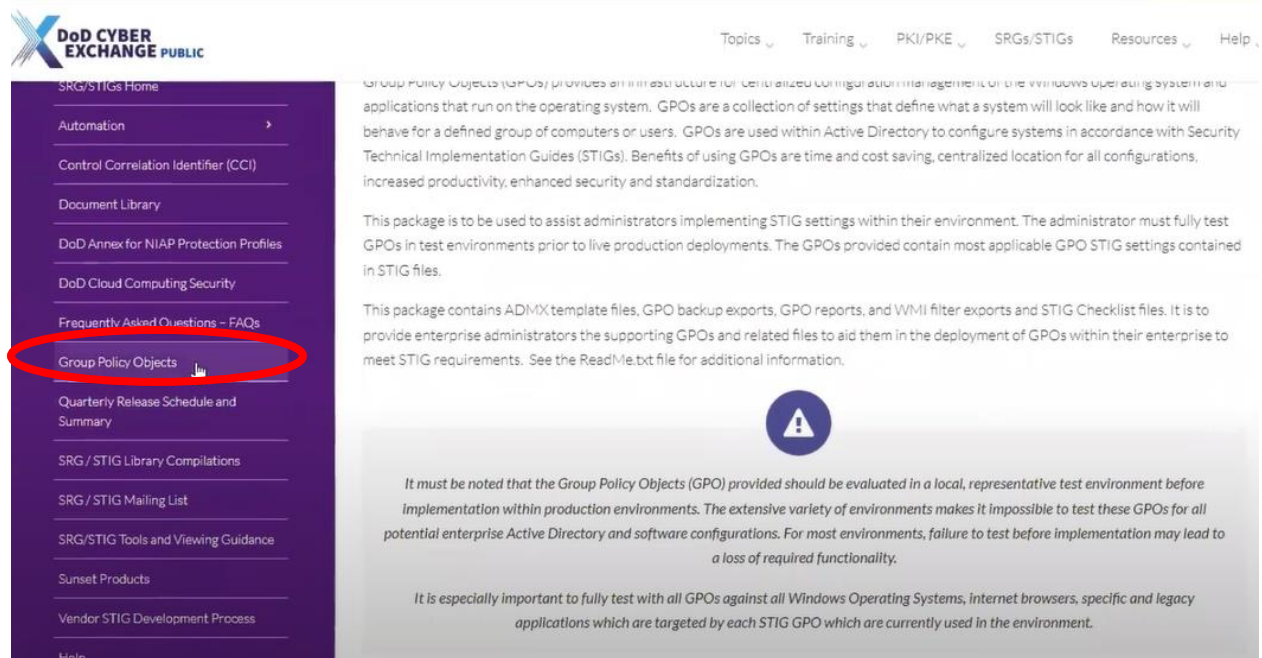
In order to fix that issue with Sandbox, run “--no sandbox” and that should solve it.

```
[root@nademclan stig_viewer_3-linux-x64]# sudo ./STIG\ Viewer\ 3 --no-sandbox
VERSION: 3.3.0
```

The STIG Viewer Tool should now be successfully installed using Terminal!

13. How To Download STIGs

First off on the same website, click onto “Group Policy Objects”



The screenshot shows the DoD CYBER EXCHANGE PUBLIC website. The left sidebar contains a list of links, with 'Group Policy Objects' highlighted by a red circle. The main content area provides information about Group Policy Objects (GPOs) and includes a warning icon and text stating that the GPOs should be evaluated in a local test environment before implementation.

DoD CYBER EXCHANGE PUBLIC

Topics Training PKI/PKE SRGs/STIGs Resources Help

SRGs/STIGs Home

Automation

Control Correlation Identifier (CCI)

Document Library

DoD Annex for NIAP Protection Profiles

DoD Cloud Computing Security

Frequently Asked Questions - FAQs

Group Policy Objects

Quarterly Release Schedule and Summary

SRG / STIG Library Compilations

SRG / STIG Mailing List

SRG/STIG Tools and Viewing Guidance

Sunset Products

Vendor STIG Development Process

Help

Group Policy Objects (GPOs) provides an infrastructure for centralized configuration management on the Windows operating system and applications that run on the operating system. GPOs are a collection of settings that define what a system will look like and how it will behave for a defined group of computers or users. GPOs are used within Active Directory to configure systems in accordance with Security Technical Implementation Guides (STIGs). Benefits of using GPOs are time and cost saving, centralized location for all configurations, increased productivity, enhanced security and standardization.

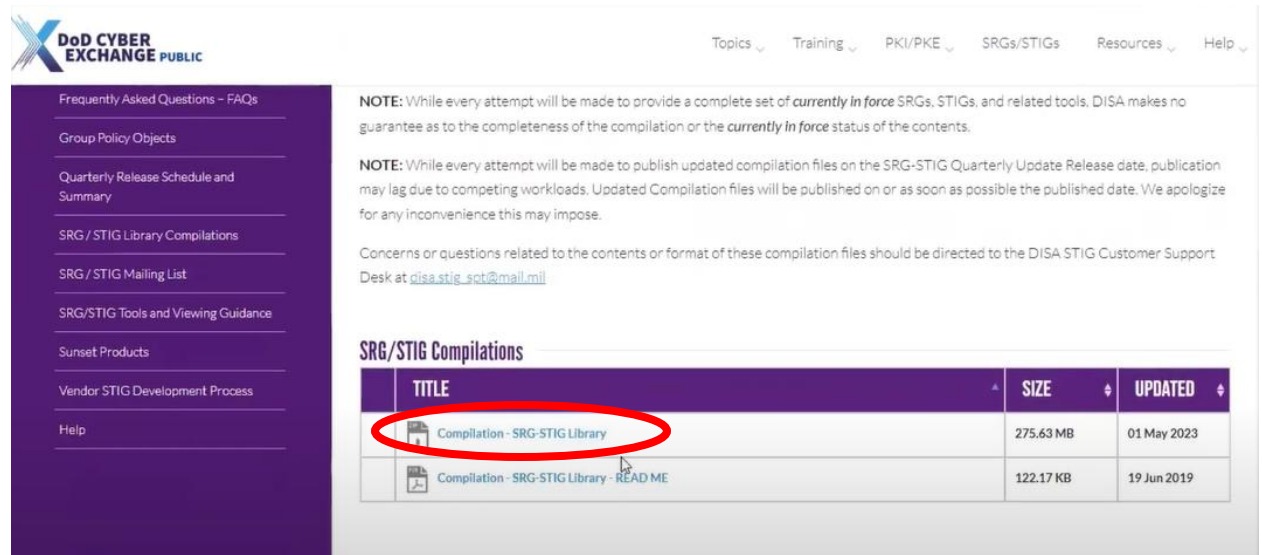
This package is to be used to assist administrators implementing STIG settings within their environment. The administrator must fully test GPOs in test environments prior to live production deployments. The GPOs provided contain most applicable GPO STIG settings contained in STIG files.

This package contains ADMX template files, GPO backup exports, GPO reports, and WMI filter exports and STIG Checklist files. It is to provide enterprise administrators the supporting GPOs and related files to aid them in the deployment of GPOs within their enterprise to meet STIG requirements. See the ReadMe.txt file for additional information.

Warning: It must be noted that the Group Policy Objects (GPO) provided should be evaluated in a local, representative test environment before implementation within production environments. The extensive variety of environments makes it impossible to test these GPOs for all potential enterprise Active Directory and software configurations. For most environments, failure to test before implementation may lead to a loss of required functionality.

It is especially important to fully test with all GPOs against all Windows Operating Systems, internet browsers, specific and legacy applications which are targeted by each STIG GPO which are currently used in the environment.

Then go ahead and click on STIG Library needed to load up in SCAP tool or STIG Viewer



The screenshot shows the DoD CYBER EXCHANGE PUBLIC website. The left sidebar contains a list of links, with 'Group Policy Objects' highlighted by a red circle. The main content area provides information about SRG/STIG Library Compilations and includes a table listing the compilations.

DoD CYBER EXCHANGE PUBLIC

Topics Training PKI/PKE SRGs/STIGs Resources Help

Frequently Asked Questions - FAQs

Group Policy Objects

Quarterly Release Schedule and Summary

SRG / STIG Library Compilations

SRG / STIG Mailing List

SRG/STIG Tools and Viewing Guidance

Sunset Products

Vendor STIG Development Process

Help

NOTE: While every attempt will be made to provide a complete set of *currently in force* SRGs, STIGs, and related tools, DISA makes no guarantee as to the completeness of the compilation or the *currently in force* status of the contents.

NOTE: While every attempt will be made to publish updated compilation files on the SRG-STIG Quarterly Update Release date, publication may lag due to competing workloads. Updated Compilation files will be published on or as soon as possible the published date. We apologize for any inconvenience this may impose.

Concerns or questions related to the contents or format of these compilation files should be directed to the DISA STIG Customer Support Desk at disa.stig_sqt@mail.mil

SRG/STIG Compilations

TITLE	SIZE	UPDATED
Compilation - SRG-STIG Library	275.63 MB	01 May 2023
Compilation - SRG-STIG Library - READ ME	122.17 KB	19 Jun 2019

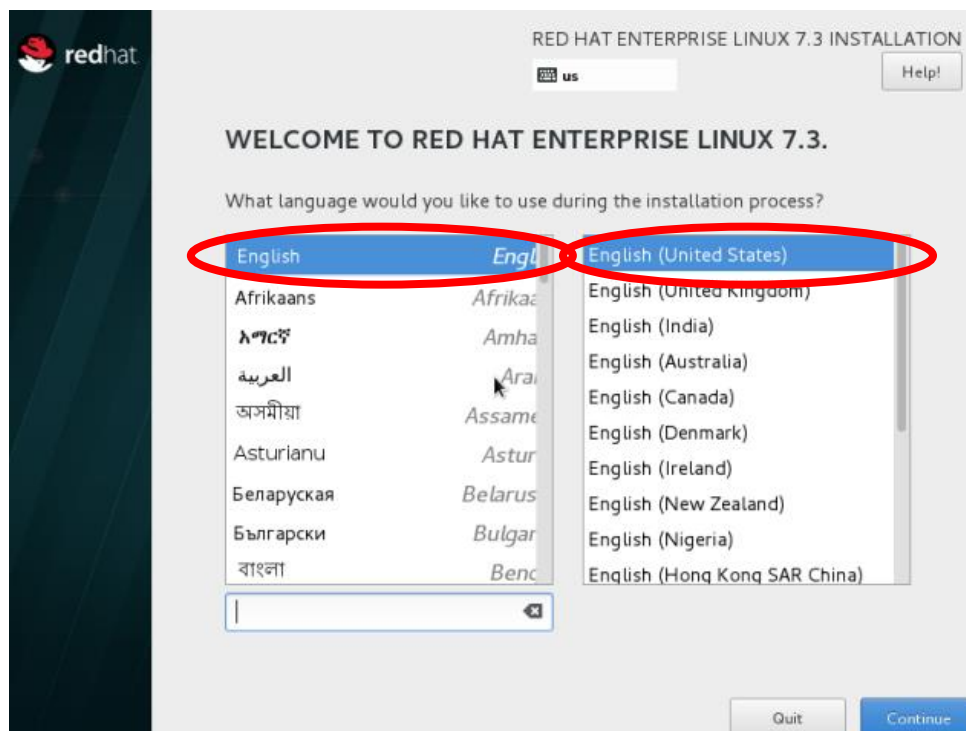
Now you'll be able to load up the STIG's after locating them in the folder at the time of loading the STIG's in SCAP tool or STIG Viewer.

14. SECURITY POLICY – STIG

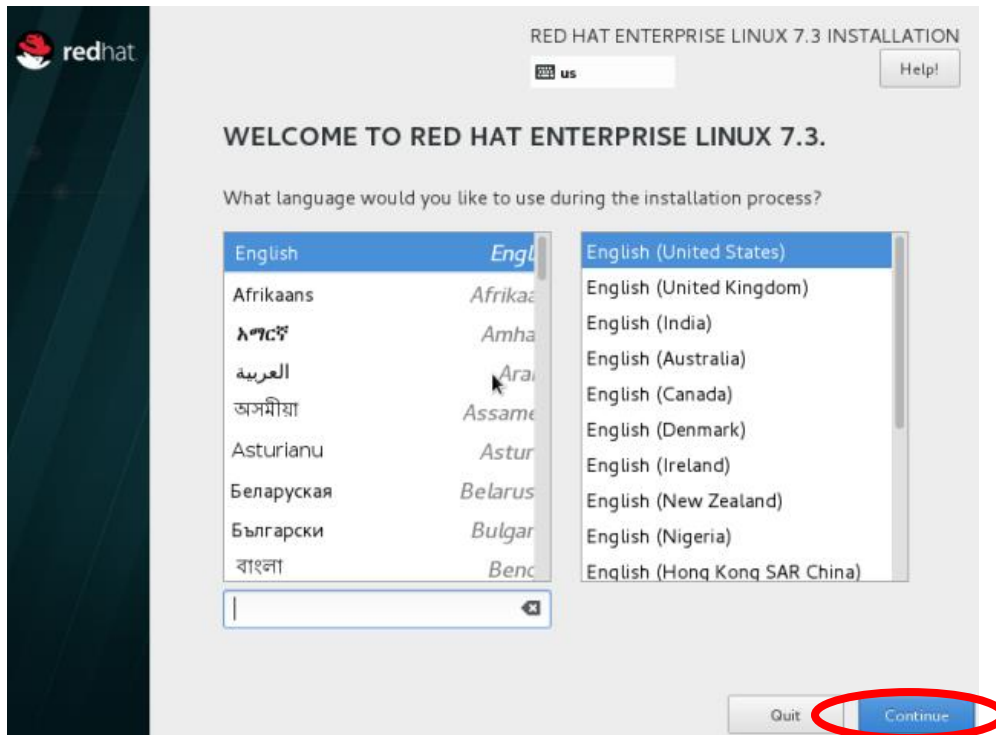
A STIG is a document published by the DoD. It contains guidance on how to configure systems to defend against potential threats. These threats mainly include cyberattacks, but they can also be problems caused by the use of misconfigured systems.

These are the steps to enable STIG on your machine.

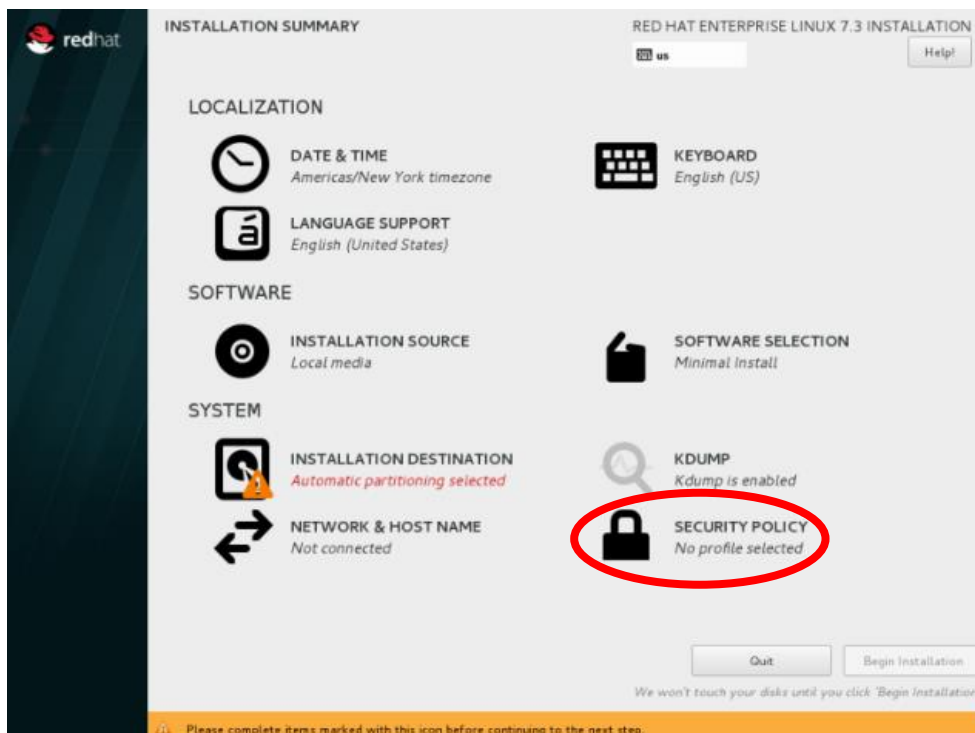
First click on "English" and "English (United States)"



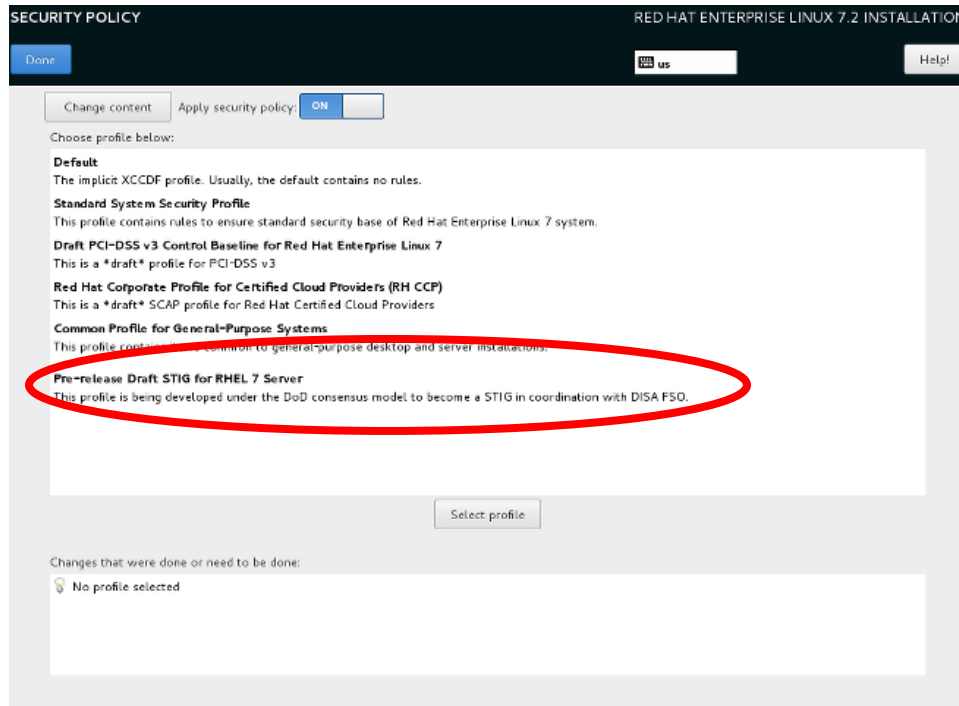
Then click on "Continue"



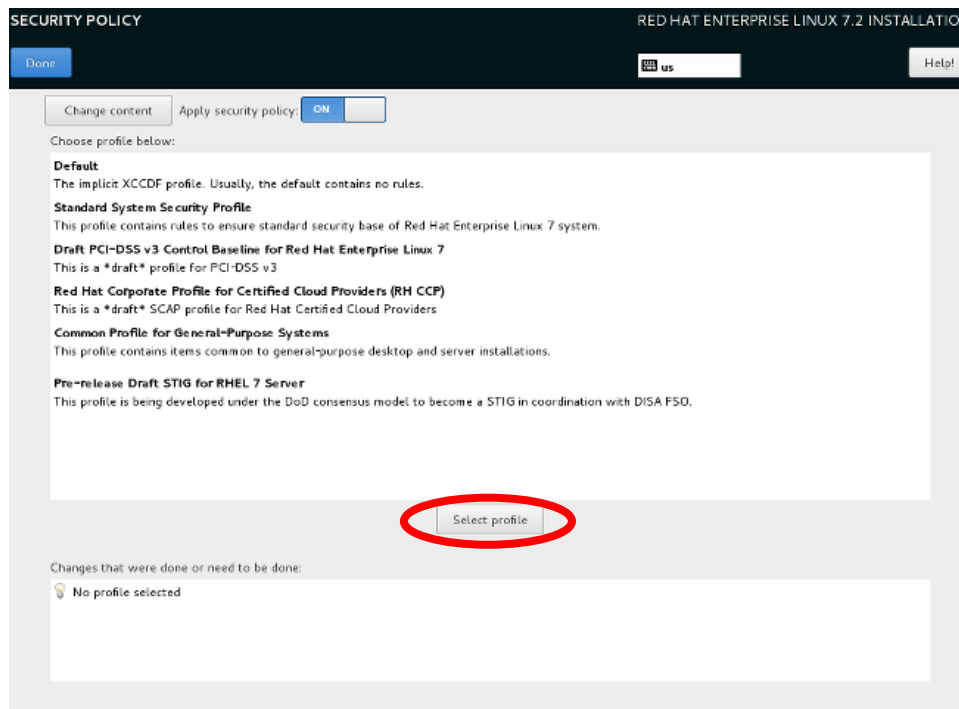
Then click on "Security Policy"



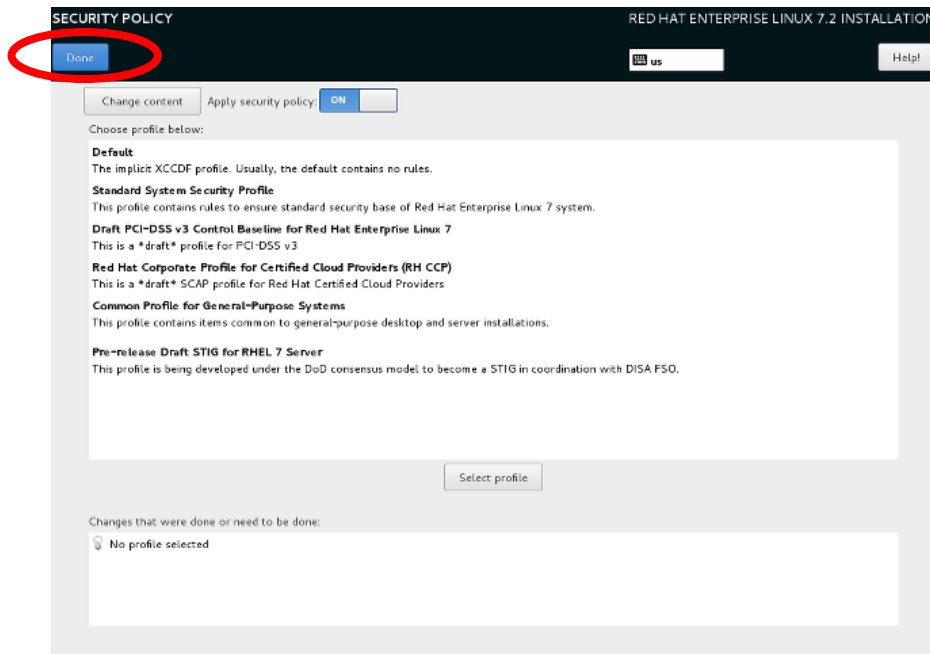
Then scroll down and click on “[DRAFT] DISA STIG for Red Hat Enterprise Linux Virtualization Host (RHELH)”



Then click on “Select Profile”



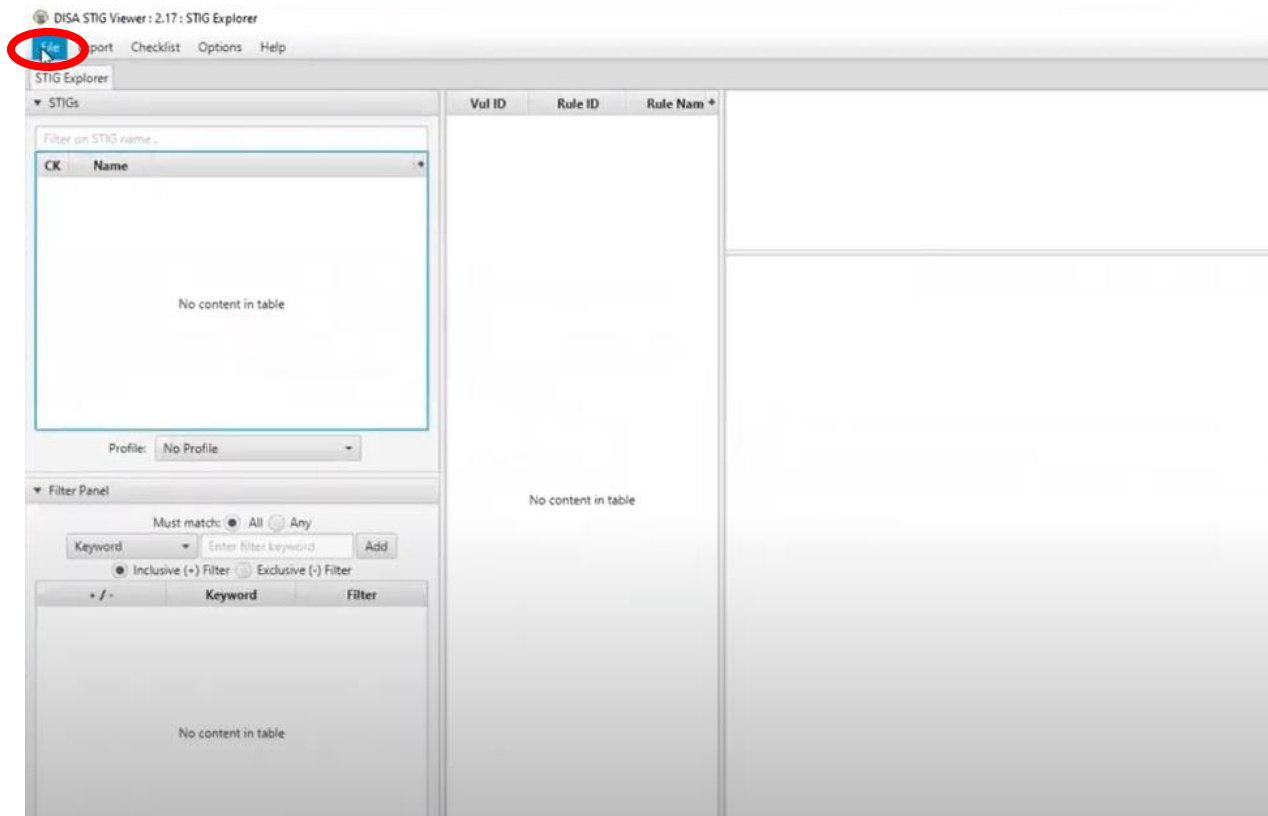
Lastly, click on “Done”



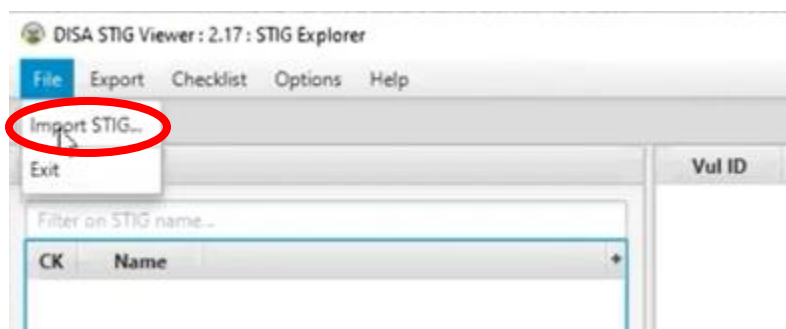
If changes have to be done to the partitions, usually reinputting the ten partitions separately one more time will solve the issues. After the new partitions have been added, remove the old ones. Most of the changes should be solved at this point.

15. Loading STIGs Using STIG Viewer 3

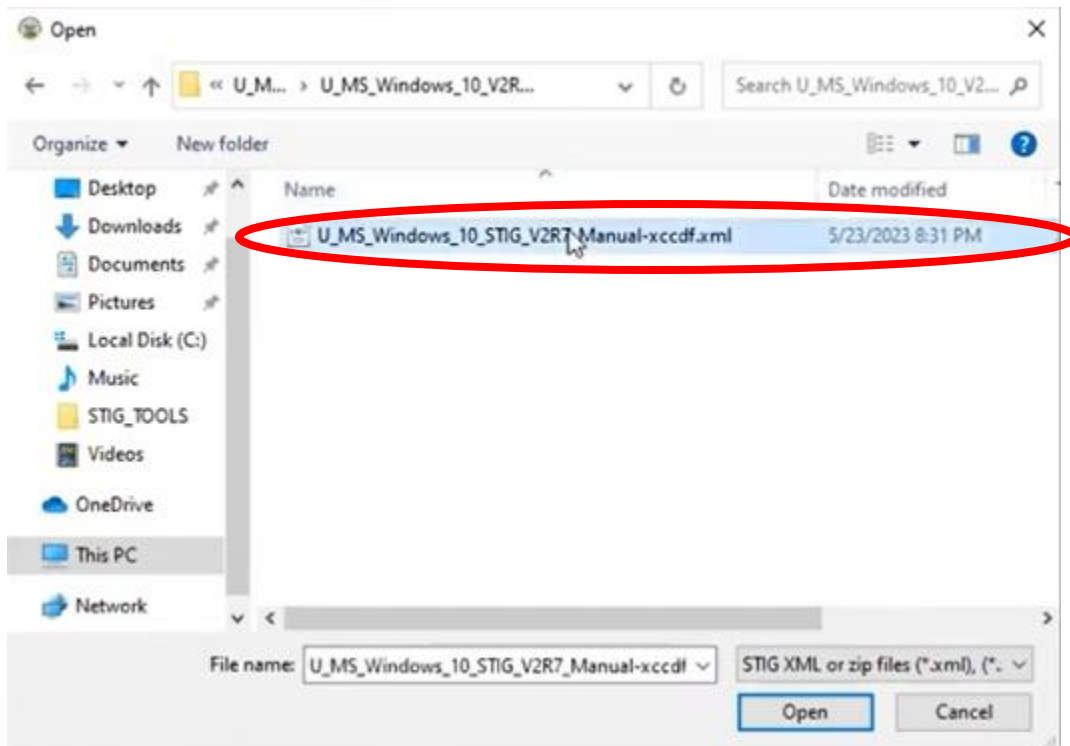
First open up STIG Viewer and click on “File”



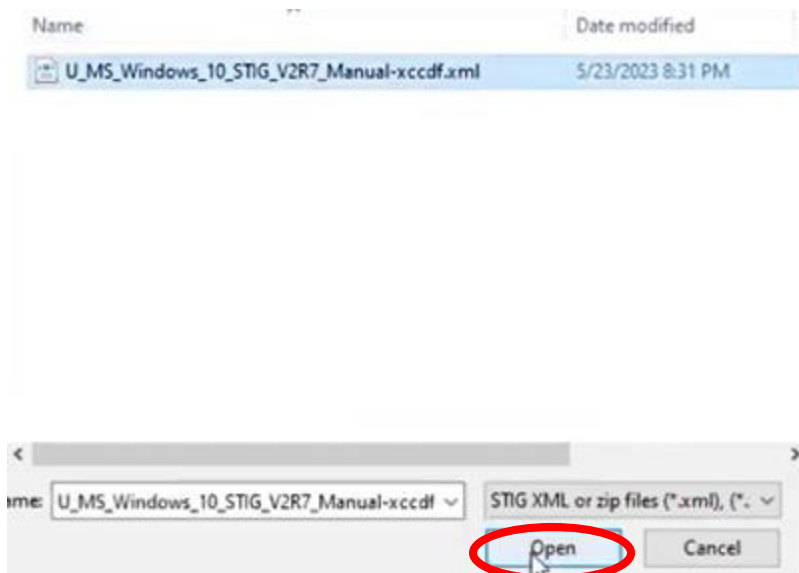
Then click on “Import STIG..”



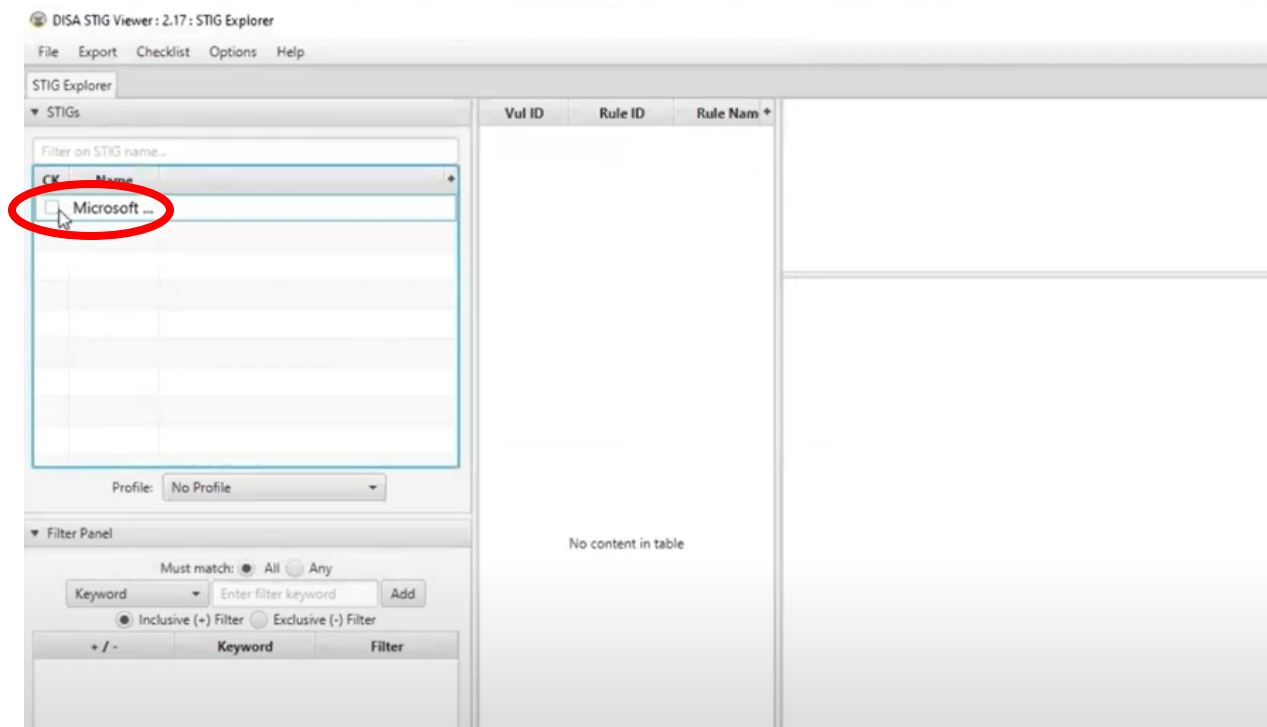
Then after this window opens, locate the XML file to then open it



Now click on “Open”



Now click on the file



Now it will give you a list of vulnerabilities to take a look at

Microsoft Windows 10 Security Technical Implementation Guide :: Version 2, Release: 7 Benchmark Date: 07 Jun 2023

Vul ID: V-220700 **Rule ID:** SV-220700r569187_rule **STIG ID:** WN10-00-000020

Severity: CAT III **Classification:** Unclass **Legacy IDs:** V-77085; SV-91781

Group Title: SRG-OS-000480-GPOS-00227

Rule Title: Secure Boot must be enabled on Windows 10 systems.

Discussion: Secure Boot is a standard that ensures systems boot only to a trusted operating system. Secure Boot is required to support additional security features in Windows 10, including Virtualization Based Security and Credential Guard. If Secure Boot is turned off, these security features will not function.

Check Text: Some older systems may not have UEFI firmware. This is currently a CAT III; it will be raised in severity at a future date when broad support of Windows 10 hardware and firmware requirements are expected to be met. Devices that have UEFI firmware must have Secure Boot enabled.

For virtual desktop implementations (VDIs) where the virtual desktop instance is deleted or refreshed upon logoff, this is NA.

Run "System Information".

Under "System Summary", if "Secure Boot State" does not display "On", this is finding.

Fix Text: Enable Secure Boot in the system firmware.

References

CCI: CCI-000366: The organization implements the security configuration settings.
 NIST SP 800-53 :: CM-6 b
 NIST SP 800-53A :: CM-6.1 (iv)
 NIST SP 800-53 Revision 4 :: CM-6 b

Vul ID	Rule ID	Rule Name
V-220697	SV-220697r857178_rule	SRG-OS-000480-...
V-220698	SV-220698r857181_rule	SRG-OS-000480-...
V-220699	SV-220699r569187_rule	SRG-OS-000480-...
V-220700	SV-220700r569187_rule	SRG-OS-000480-...
V-220701	SV-220701r793197_rule	SRG-OS-000191-...
V-220702	SV-220702r859296_rule	SRG-OS-000185-...
V-220703	SV-220703r859157_rule	SRG-OS-000185-...
V-220704	SV-220704r859297_rule	SRG-OS-000185-...
V-220705	SV-220705r890420_rule	SRG-OS-000370-...
V-220706	SV-220706r857183_rule	SRG-OS-000480-...
V-220707	SV-220707r793194_rule	SRG-OS-000480-...
V-220708	SV-220708r569187_rule	SRG-OS-000080-...
V-220709	SV-220709r569187_rule	SRG-OS-000480-...
V-220710	SV-220710r569187_rule	SRG-OS-000138-...
V-220711	SV-220711r569187_rule	SRG-OS-000118-...
V-220712	SV-220712r877392_rule	SRG-OS-000324-...
V-220713	SV-220713r569187_rule	SRG-OS-000480-...
V-220714	SV-220714r569187_rule	SRG-OS-000095-...
V-220715	SV-220715r890423_rule	SRG-OS-000480-...
V-220716	SV-220716r569187_rule	SRG-OS-000076-...
V-220717	SV-220717r851965_rule	SRG-OS-000312-...
V-220718	SV-220718r569187_rule	SRG-OS-000095-...
V-220719	SV-220719r569187_rule	SRG-OS-000096-...
V-220720	SV-220720r569187_rule	SRG-OS-000095-...
V-220721	SV-220721r569187_rule	SRG-OS-000096-...
V-220722	SV-220722r569187_rule	SRG-OS-000096-...
V-220723	SV-220723r569187_rule	SRG-OS-000480-...
V-220724	SV-220724r569187_rule	SRG-OS-000480-...

16. **APPENDIX:**

When using CMD remember to press “TAB” when inputting a previously used command. ~
More convenience and saves time.

When trying to go into previous folder input “cd ../” into the command.

Download new files into the “opt” folder to keep organized.